

PHYS1112 - Electricity and Magnetism

Lecture Notes

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Chapter 1

Vector Algebra

1.1 Definitions

A **vector** consists of two components: *magnitude* and *direction* .
(e.g. force, velocity, pressure)

A **scalar** consists of *magnitude* only.
(e.g. mass, charge, density)

1.2 Vector Algebra

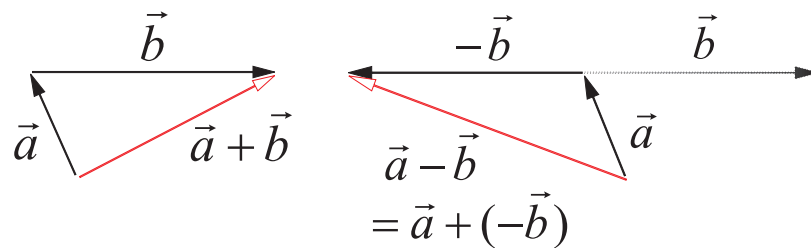


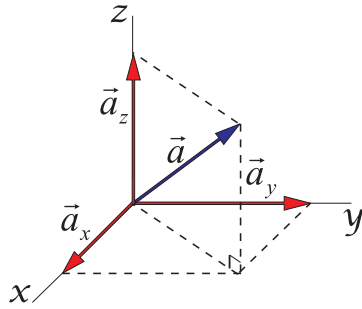
Figure 1.1: Vector algebra

$$\begin{aligned}\vec{a} + \vec{b} &= \vec{b} + \vec{a} \\ \vec{a} + (\vec{c} + \vec{d}) &= (\vec{a} + \vec{c}) + \vec{d}\end{aligned}$$

1.3 Components of Vectors

Usually vectors are expressed according to **coordinate system**. Each vector can be expressed in terms of *components*.

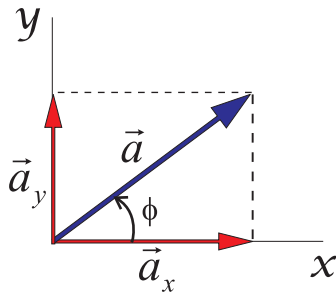
The most common coordinate system: **Cartesian**



$$\vec{a} = \vec{a}_x + \vec{a}_y + \vec{a}_z$$

Magnitude of $\vec{a} = |\vec{a}| = a$,

$$a = \sqrt{a_x^2 + a_y^2 + a_z^2}$$



$$\vec{a} = \vec{a}_x + \vec{a}_y$$

$$a = \sqrt{a_x^2 + a_y^2}$$

$$a_x = a \cos \phi; \quad a_y = a \sin \phi$$

$$\tan \phi = \frac{a_y}{a_x}$$

Figure 1.2: ϕ measured anti-clockwise from position x -axis

Unit vectors have magnitude of 1

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \text{unit vector along } \vec{a} \text{ direction}$$

$$\begin{array}{ccc} \hat{i} & \hat{j} & \hat{k} \\ \updownarrow & \updownarrow & \updownarrow \\ x & y & z \end{array} \text{ are unit vectors along directions}$$

$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

Other coordinate systems:

1. Polar Coordinate:

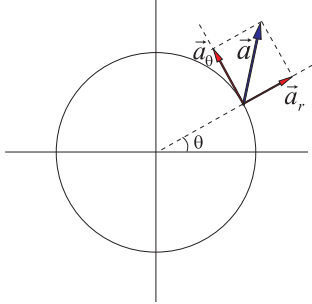


Figure 1.3: Polar Coordinates

$$\vec{a} = a_r \hat{r} + a_\theta \hat{\theta}$$

2. Cylindrical Coordinates:

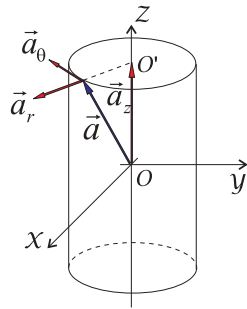


Figure 1.4: Cylindrical Coordinates

$$\vec{a} = a_r \hat{r} + a_\theta \hat{\theta} + a_z \hat{z}$$

$\hat{r}$ originated from nearest point on z-axis (Point O')

3. Spherical Coordinates:

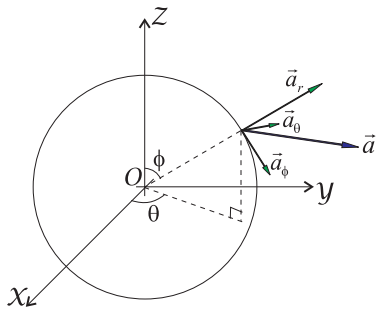


Figure 1.5: Spherical Coordinates

$$\vec{a} = a_r \hat{r} + a_\theta \hat{\theta} + a_\phi \hat{\phi}$$

$\hat{r}$ originated from Origin O

1.4 Multiplication of Vectors

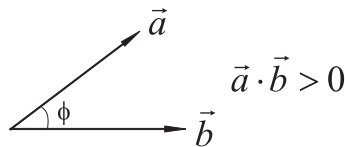
1. Scalar multiplication:

If $\vec{b} = m \vec{a}$ $\vec{b}, \vec{a}$ are vectors; m is a scalar
 then $b = m a$ (Relation between magnitude)
 $\left. \begin{array}{l} b_x = m a_x \\ b_y = m a_y \end{array} \right\}$ Components also follow relation

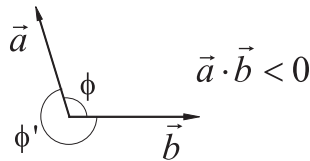
i.e.

$$\begin{aligned} \vec{a} &= a_x \hat{i} + a_y \hat{j} + a_z \hat{k} \\ m\vec{a} &= ma_x \hat{i} + ma_y \hat{j} + ma_z \hat{k} \end{aligned}$$

2. Dot Product (Scalar Product):

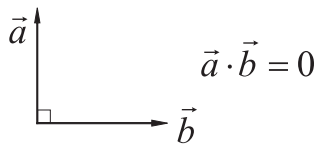


$$\vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| \cos \phi$$



Result is **always** a scalar. It can be positive or negative depending on ϕ .

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$



Notice: $\vec{a} \cdot \vec{b} = ab \cos \phi = ab \cos \phi'$
 i.e. Doesn't matter how you measure angle ϕ between vectors.

Figure 1.6: Dot Product

$$\begin{aligned} \hat{i} \cdot \hat{i} &= |\hat{i}| |\hat{i}| \cos 0^\circ = 1 \cdot 1 \cdot 1 = 1 \\ \hat{i} \cdot \hat{j} &= |\hat{i}| |\hat{j}| \cos 90^\circ = 1 \cdot 1 \cdot 0 = 0 \end{aligned}$$

$$\begin{aligned} \hat{i} \cdot \hat{i} &= \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1 \\ \hat{i} \cdot \hat{j} &= \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0 \end{aligned}$$

$$\begin{aligned} \text{If } \vec{a} &= a_x \hat{i} + a_y \hat{j} + a_z \hat{k} \\ \vec{b} &= b_x \hat{i} + b_y \hat{j} + b_z \hat{k} \\ \text{then } \vec{a} \cdot \vec{b} &= a_x b_x + a_y b_y + a_z b_z \\ \vec{a} \cdot \vec{a} &= |\vec{a}| \cdot |\vec{a}| \cos 0^\circ = a \cdot a = a^2 \end{aligned}$$

3. Cross Product (Vector Product):

If $\vec{c} = \vec{a} \times \vec{b}$,
 then $c = |\vec{c}| = ab \sin \phi$

$$\vec{a} \times \vec{b} \neq \vec{b} \times \vec{a} !!!$$

$$\boxed{\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}}$$

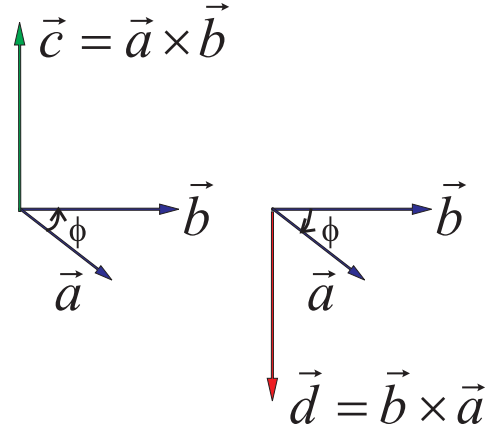


Figure 1.7: Note: How angle ϕ is measured

- Direction of cross product determined from *right hand rule*.
- Also, $\vec{a} \times \vec{b}$ is $\perp$ to $\vec{a}$ and $\vec{b}$, i.e.

$$\begin{aligned}\vec{a} \cdot (\vec{a} \times \vec{b}) &= 0 \\ \vec{b} \cdot (\vec{a} \times \vec{b}) &= 0\end{aligned}$$

- IMPORTANT:

$$\boxed{\vec{a} \times \vec{a} = a \cdot a \sin 0^\circ = 0}$$

$$\begin{aligned}|\hat{i} \times \hat{i}| &= |\hat{i}| |\hat{i}| \sin 0^\circ = 1 \cdot 1 \cdot 0 = 0 \\ |\hat{i} \times \hat{j}| &= |\hat{i}| |\hat{j}| \sin 90^\circ = 1 \cdot 1 \cdot 1 = 1\end{aligned}$$

$$\boxed{\begin{aligned}\hat{i} \times \hat{i} &= \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0 \\ \hat{i} \times \hat{j} &= \hat{k}; \hat{j} \times \hat{k} = \hat{i}; \hat{k} \times \hat{i} = \hat{j}\end{aligned}}$$



$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = \begin{aligned} &(a_y b_z - a_z b_y) \hat{i} \\ &+ (a_z b_x - a_x b_z) \hat{j} \\ &+ (a_x b_y - a_y b_x) \hat{k} \end{aligned}$$

4. Vector identities:

$$\begin{aligned}\vec{a} \times (\vec{b} + \vec{c}) &= \vec{a} \times \vec{b} + \vec{a} \times \vec{c} \\ \vec{a} \cdot (\vec{b} \times \vec{c}) &= \vec{b} \cdot (\vec{c} \times \vec{a}) = \vec{c} \cdot (\vec{a} \times \vec{b}) \\ \vec{a} \times (\vec{b} \times \vec{c}) &= (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}\end{aligned}$$

1.5 Vector Field (Physics Point of View)

A **vector field** $\vec{\mathcal{F}}(x, y, z)$ is a mathematical function which has a *vector* output for a *position* input.

(Scalar field $\vec{\mathcal{U}}(x, y, z)$)

1.6 Other Topics

Tangential Vector

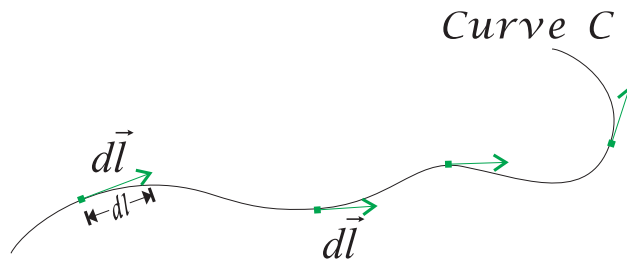


Figure 1.8: $d\vec{l}$ is a vector that is always tangential to the curve C with infinitesimal length dl

Surface Vector

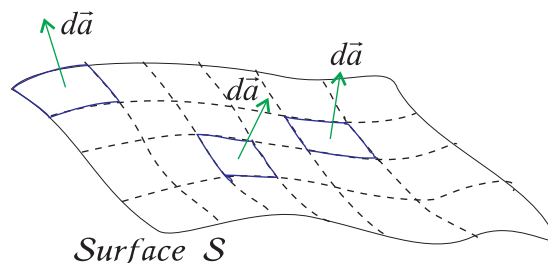


Figure 1.9: $d\vec{a}$ is a vector that is always perpendicular to the surface S with infinitesimal area da

Some uncertainty! ($d\vec{a}$ versus $-d\vec{a}$)

Two conventions:

- Area formed from a closed curve

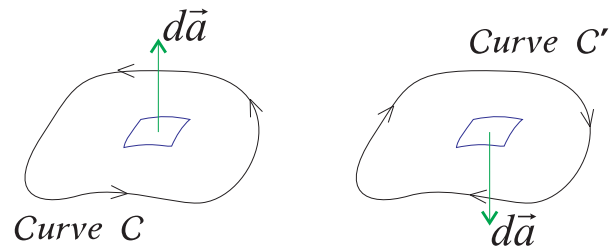


Figure 1.10: Direction of $d\vec{a}$ determined from right-hand rule

- Closed surface enclosing a volume

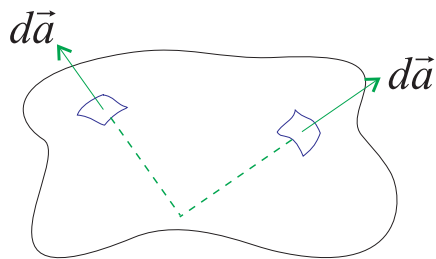


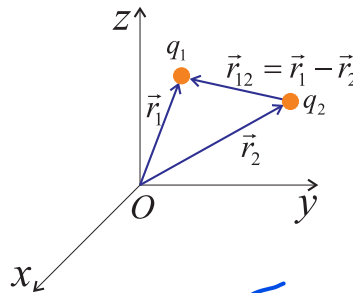
Figure 1.11: Direction of $d\vec{a}$ going from inside to outside

Chapter 2

Electric Force & Electric Field

2.1 Electric Force

The electric force between two **charges** q_1 and q_2 can be described by **Coulomb's Law**.



$\vec{F}_{12}$ = Force on q_1 exerted by q_2

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r_{12}^2} \cdot \hat{r}_{12}$$

$$E = \frac{F}{q}$$

where $\hat{r}_{12} = \frac{\vec{r}_{12}}{|\vec{r}_{12}|}$ is the *unit vector* which locates particle 1 relative to particle 2.

i.e. $\vec{r}_{12} = \vec{r}_1 - \vec{r}_2$

- q_1, q_2 are electrical charges in units of *Coulomb*(C)
- Charge is *quantized*
Recall 1 electron carries $1.602 \times 10^{-19}C$
- ϵ_0 = Permittivity of free space = $8.85 \times 10^{-12}C^2/Nm^2$

COULOMB'S LAW:

- (1) q_1, q_2 can be either positive or negative.

- (2) If q_1, q_2 are of same sign, then the force experienced by q_1 is in direction away from q_2 , that is, *repulsive*.
- (3) Force on q_2 exerted by q_1 :

$$\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_2 q_1}{r_{21}^2} \cdot \hat{r}_{21}$$

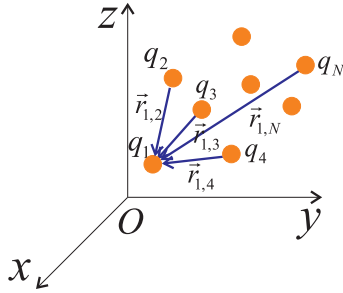
BUT:

$$r_{12} = r_{21} = \text{distance between } q_1, q_2$$

$$\hat{r}_{21} = \frac{\vec{r}_{21}}{r_{21}} = \frac{\vec{r}_2 - \vec{r}_1}{r_{21}} = \frac{-\vec{r}_{12}}{r_{12}} = -\hat{r}_{12}$$

$$\therefore \boxed{\vec{F}_{21} = -\vec{F}_{12} \text{ Newton's 3rd Law}}$$

SYSTEM WITH MANY CHARGES:



The total force experienced by charge q_1 is the *vector sum* of the forces on q_1 exerted by other charges.

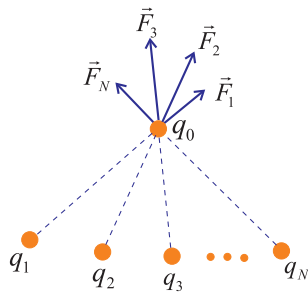
$$\begin{aligned} \vec{F}_1 &= \text{Force experienced by } q_1 \\ &= \vec{F}_{1,2} + \vec{F}_{1,3} + \vec{F}_{1,4} + \cdots + \vec{F}_{1,N} \end{aligned}$$

PRINCIPLE OF SUPERPOSITION:

$$\vec{F}_1 = \sum_{j=2}^N \vec{F}_{1,j}$$

2.2 The Electric Field

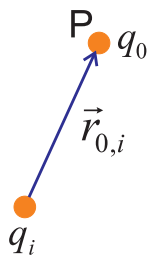
While we need two charges to quantify the **electric force**, we define the **electric field** for any single charge distribution to describe its effect on other charges.



Total force $\vec{F} = \vec{F}_1 + \vec{F}_2 + \cdots + \vec{F}_N$
 The **electric field** is defined as

$$\lim_{q_0 \rightarrow 0} \frac{\vec{F}}{q_0} = \vec{E}$$

(a) E-field due to a single charge q_i :



From the definitions of **Coulomb's Law**, the force experienced at location of q_0 (point P)

$$\vec{F}_{0,i} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_0 q_i}{r_{0,i}^2} \cdot \hat{r}_{0,i}$$

where $\hat{r}_{0,i}$ is the unit vector along the direction *from charge q_i to q_0* ,

$$\begin{aligned} \hat{r}_{0,i} &= \text{Unit vector from charge } q_i \text{ to point P} \\ &= \hat{r}_i \text{ (radical unit vector from } q_i) \end{aligned}$$

Recall $\vec{E} = \lim_{q_0 \rightarrow 0} \frac{\vec{F}}{q_0}$

$\therefore$ E-field due to q_i at point P:

$$\vec{E}_i = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_i}{r_i^2} \cdot \hat{r}_i$$

where $\vec{r}_i$ = Vector pointing from q_i to point P,
 thus $\hat{r}_i$ = Unit vector pointing from q_i to point P
 Note:

- (1) E-field is a **vector**.
 - (2) Direction of E-field depends on **both** position of P and sign of q_i .
- (b) E-field due to system of charges:

Principle of Superposition:

In a system with N charges, the **total** E-field due to all charges is the **vector sum** of E-field due to individual charges.

i.e.
$$\vec{E} = \sum_i \vec{E}_i = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i^2} \hat{r}_i$$

(c) Electric Dipole

System of *equal and opposite* charges separated by a distance d .

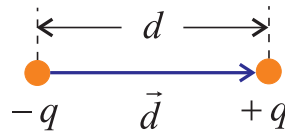


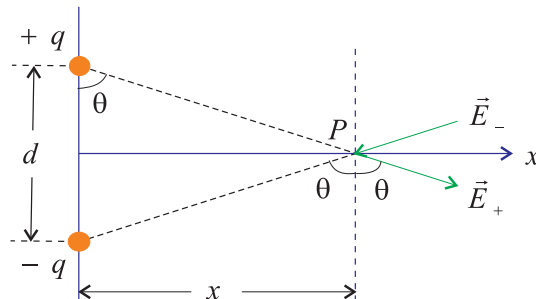
Figure 2.1: An electric dipole. (Direction of $\vec{d}$ from negative to positive charge)

Electric Dipole Moment

$$\vec{p} = q\vec{d} = qd\hat{d}$$

$$p = qd$$

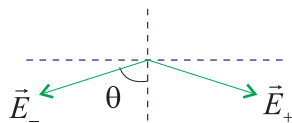
Example: $\vec{E}$ due to dipole along x -axis



Consider point P at distance x along the perpendicular axis of the dipole $\vec{p}$:

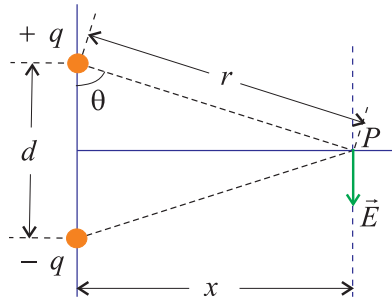
$$\vec{E} = \begin{matrix} \vec{E}_+ & + & \vec{E}_- \\ \uparrow & & \uparrow \\ \text{(E-field} & & \text{(E-field} \\ \text{due to } +q) & & \text{due to } -q) \end{matrix}$$

Notice: Horizontal E-field components of $\vec{E}_+$ and $\vec{E}_-$ cancel out.



$\therefore$ Net E-field points along the axis opposite to the dipole moment vector.

Magnitude of E-field = $2E_+ \cos \theta$



$$E_+ \text{ or } E_- \text{ magnitude!}$$

$$\therefore E = 2 \left(\frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} \right) \cos \theta$$

$$\text{But } r = \sqrt{\left(\frac{d}{2}\right)^2 + x^2}$$

$$\cos \theta = \frac{d/2}{r}$$

$$\therefore E = \frac{1}{4\pi\epsilon_0} \cdot \frac{p}{[x^2 + (\frac{d}{2})^2]^{\frac{3}{2}}}$$

$$(p = qd)$$

Special case: When $x \gg d$

$$[x^2 + (\frac{d}{2})^2]^{\frac{3}{2}} = x^3 [1 + (\frac{d}{2x})^2]^{\frac{3}{2}}$$

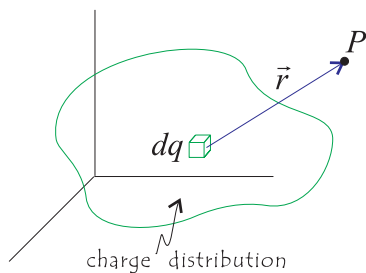
- Binomial Approximation:

$$(1 + y)^n \approx 1 + ny \quad \text{if } y \ll 1$$

$$\text{E-field of dipole} \doteq \frac{1}{4\pi\epsilon_0} \cdot \frac{p}{x^3} \sim \frac{1}{x^3}$$

- Compare with $\frac{1}{r^2}$ E-field for single charge
- Result also valid for point P along any axis with respect to dipole

2.3 Continuous Charge Distribution



E-field at point P due to dq :

$$d\vec{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r^2} \cdot \hat{r}$$

$\therefore$ E-field due to charge distribution:

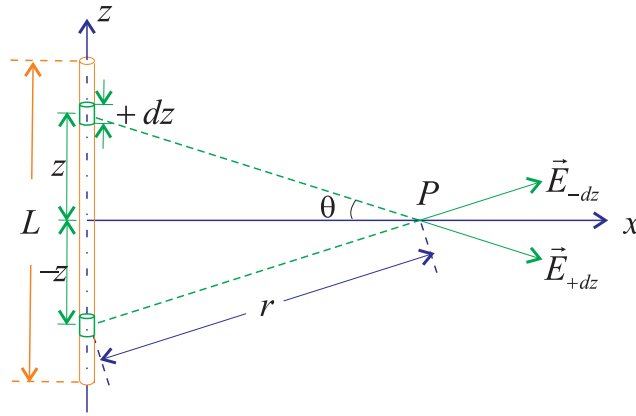
$$\vec{E} = \int_{\text{Volume}} d\vec{E} = \int_{\text{Volume}} \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r^2} \cdot \hat{r}$$

(1) In many cases, we can take advantage of the *symmetry* of the system to simplify the integral.

(2) To write down the small charge element dq :

1-D	$dq = \lambda ds$	$\lambda = \text{linear charge density}$	$ds = \text{small length element}$
2-D	$dq = \sigma dA$	$\sigma = \text{surface charge density}$	$dA = \text{small area element}$
3-D	$dq = \rho dV$	$\rho = \text{volume charge density}$	$dV = \text{small volume element}$

Example 1: Uniform line of charge



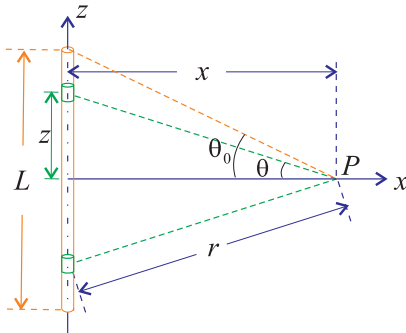
charge per
unit length
 $= \lambda$

(1) Symmetry considered: The E-field from $+z$ and $-z$ directions *cancel along z-direction*, $\therefore$ Only horizontal E-field components need to be considered.

(2) For each element of length dz , charge $dq = \lambda dz$

$$\therefore \text{Horizontal E-field at point P due to element } dz = \underbrace{|d\vec{E}| \cos \theta}_{dE_{dz}} = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda dz}{r^2} \cos \theta$$

$\therefore$ E-field due to entire line charge at point P



$$\begin{aligned} E &= \int_{-L/2}^{L/2} \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda dz}{r^2} \cos \theta \\ &= 2 \int_0^{L/2} \frac{\lambda}{4\pi\epsilon_0} \cdot \frac{dz}{r^2} \cos \theta \end{aligned}$$

To calculate this integral:

- First, notice that x is fixed, but z , r , θ all varies.
- Change of variable (from z to θ)

$$(1) \quad \begin{aligned} z &= x \tan \theta & \therefore dz &= x \sec^2 \theta d\theta \\ x &= r \cos \theta & \therefore r^2 &= x^2 \sec^2 \theta \end{aligned}$$

$$(2) \quad \begin{aligned} &z = 0, \quad \theta = 0^\circ \\ \text{When } &z = L/2 \quad \theta = \theta_0 \quad \text{where } \tan \theta_0 = \frac{L/2}{x} \end{aligned}$$

$$\begin{aligned} E &= 2 \cdot \frac{\lambda}{4\pi\epsilon_0} \int_0^{\theta_0} \frac{x \sec^2 \theta d\theta}{x^2 \sec^2 \theta} \cdot \cos \theta \\ &= 2 \cdot \frac{\lambda}{4\pi\epsilon_0} \int_0^{\theta_0} \frac{1}{x} \cdot \cos \theta d\theta \\ &= 2 \cdot \frac{\lambda}{4\pi\epsilon_0} \cdot \frac{1}{x} \cdot (\sin \theta) \Big|_0^{\theta_0} \\ &= 2 \cdot \frac{\lambda}{4\pi\epsilon_0} \cdot \frac{1}{x} \cdot \sin \theta_0 \\ &= 2 \cdot \frac{\lambda}{4\pi\epsilon_0} \cdot \frac{1}{x} \cdot \frac{L/2}{\sqrt{x^2 + (L/2)^2}} \end{aligned}$$

$$\boxed{E = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda L}{x \sqrt{x^2 + (L/2)^2}}} \quad \text{along } x\text{-direction}$$

Important limiting cases:

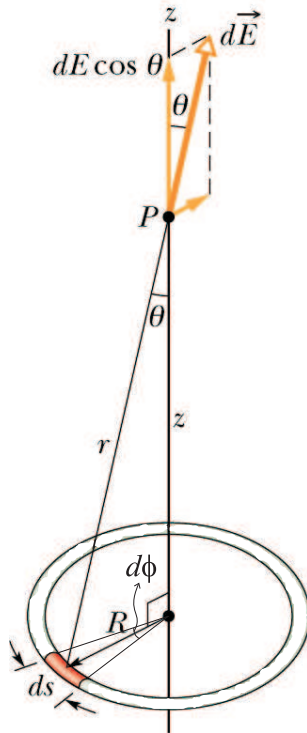
1. $x \gg L$: $E \doteq \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda L}{x^2}$
But $\lambda L = \text{Total charge on rod}$
 $\therefore$ System behave like a point charge

2. $L \gg x$: $E \doteq \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda L}{x \cdot \frac{L}{2}}$

$$\boxed{E_x = \frac{\lambda}{2\pi\epsilon_0 x}}$$

ELECTRIC FIELD DUE TO INFINITELY LONG LINE OF CHARGE

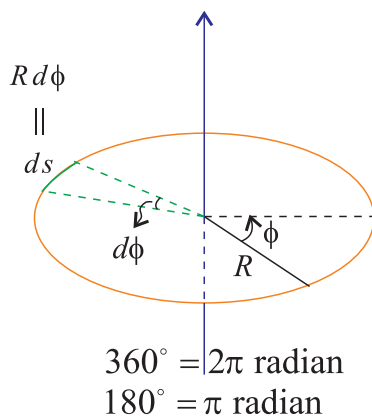
Example 2: Ring of Charge



E-field at a height z above a ring of charge of radius R

- (1) Symmetry considered: For every charge element dq considered, there exists dq' where the horizontal $\vec{E}$ field components cancel.
 $\Rightarrow$ Overall E-field lies along z -direction.

- (2) For each element of length ds , charge



$$dq = \underset{\substack{\uparrow \\ \text{Linear} \\ \text{charge density}}}{\lambda} \cdot \underset{\substack{\uparrow \\ \text{Circular} \\ \text{length element}}}{ds}$$

$dq = \lambda \cdot R d\phi$, where ϕ is the angle measured on the ring plane

$\therefore$ Net E-field along z -axis due to dq :

$$dE = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r^2} \cdot \cos \theta$$

$$\begin{aligned}
 \text{Total E-field} &= \int dE \\
 &= \int_0^{2\pi} \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda R d\phi}{r^2} \cdot \cos\theta \quad \left(\cos\theta = \frac{z}{r}\right)
 \end{aligned}$$

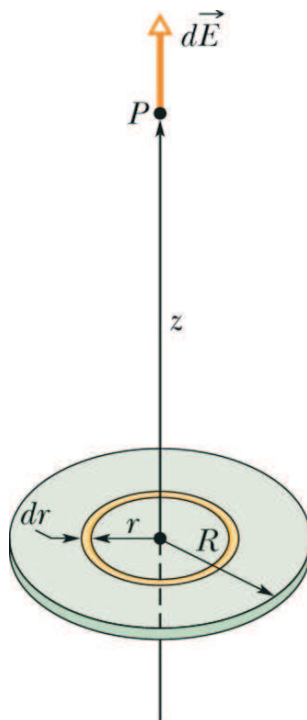
Note: Here in this case, θ, R and r are *fixed* as ϕ varies! BUT we want to convert r, θ to R, z .

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda R z}{r^3} \int_0^{2\pi} d\phi$$

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda(2\pi R)z}{(z^2 + R^2)^{3/2}} \quad \text{along } z\text{-axis}$$

BUT: $\lambda(2\pi R) = \text{total charge on the ring}$

Example 3: E-field from a disk of surface charge density σ

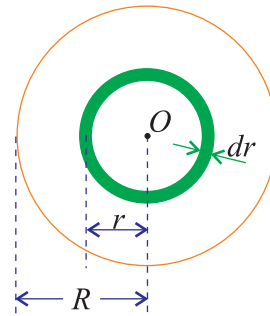


We find the E-field of a disk by integrating concentric rings of charges.

Total charge of ring

$$dq = \sigma \cdot (\underbrace{2\pi r \, dr}_{\text{Area of the ring}})$$

view from the top:



Recall from Example 2:

$$\text{E-field from ring: } dE = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq \, z}{(z^2 + r^2)^{3/2}}$$

$$\begin{aligned} \therefore E &= \frac{1}{4\pi\epsilon_0} \int_0^R \frac{2\pi\sigma r \, dr \cdot z}{(z^2 + r^2)^{3/2}} \\ &= \frac{1}{4\pi\epsilon_0} \int_0^R 2\pi\sigma z \frac{r \, dr}{(z^2 + r^2)^{3/2}} \end{aligned}$$

- Change of variable:

$$\begin{aligned} u &= z^2 + r^2 \quad \Rightarrow \quad (z^2 + r^2)^{3/2} = u^{3/2} \\ \Rightarrow \quad du &= 2r \, dr \quad \Rightarrow \quad r \, dr = \frac{1}{2} du \end{aligned}$$

- Change of integration limit:

$$\begin{cases} r = 0 & , & u = z^2 \\ r = R & , & u = z^2 + R^2 \end{cases}$$

$$\therefore E = \frac{1}{4\pi\epsilon_0} \cdot 2\pi\sigma z \int_{z^2}^{z^2+R^2} \frac{1}{2} u^{-3/2} du$$

BUT: $\int u^{-3/2} du = \frac{u^{-1/2}}{-1/2} = -2u^{-1/2}$

$$\begin{aligned} \therefore E &= \frac{1}{2\epsilon_0} \sigma z \left(-u^{-1/2} \right) \Big|_{z^2}^{z^2+R^2} \\ &= \frac{1}{2\epsilon_0} \sigma z \left(\frac{-1}{\sqrt{z^2 + R^2}} + \frac{1}{z} \right) \end{aligned}$$

$$\boxed{E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{z^2 + R^2}} \right]}$$

VERY IMPORTANT LIMITING CASE:

If $R \gg z$, that is if we have an infinite sheet of charge with charge density σ :

$$\begin{aligned} E &= \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{z^2 + R^2}} \right] \\ &\simeq \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{R} \right] \end{aligned}$$

$$E \approx \frac{\sigma}{2\epsilon_0}$$

E-field is normal to the charged surface

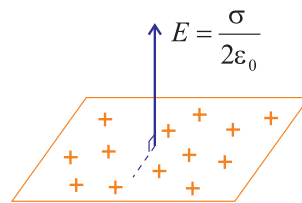
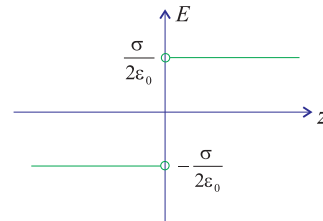


Figure 2.2: E-field due to an infinite sheet of charge, charge density = σ

Q: What's the E-field below the charged sheet?

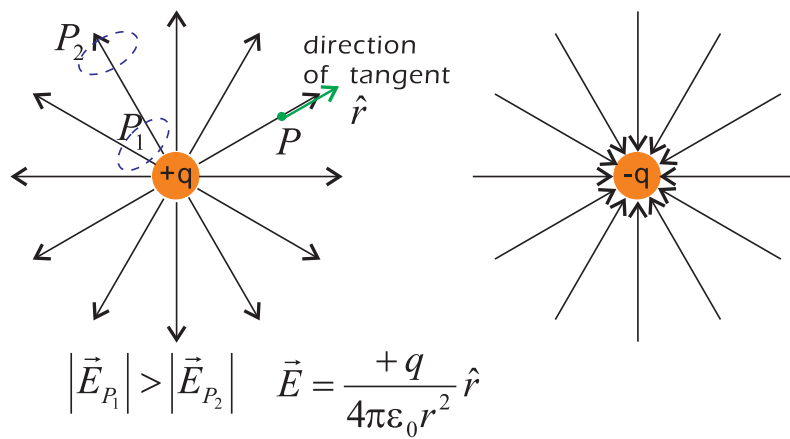
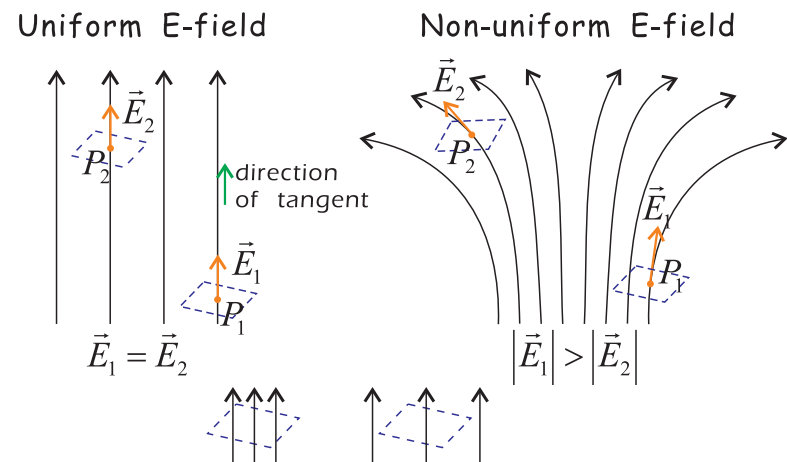


2.4 Electric Field Lines

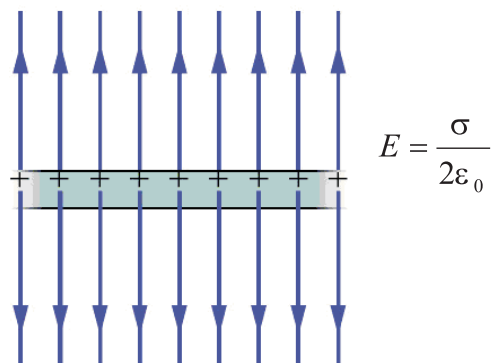
To visualize the electric field, we can use a graphical tool called the **electric field lines**.

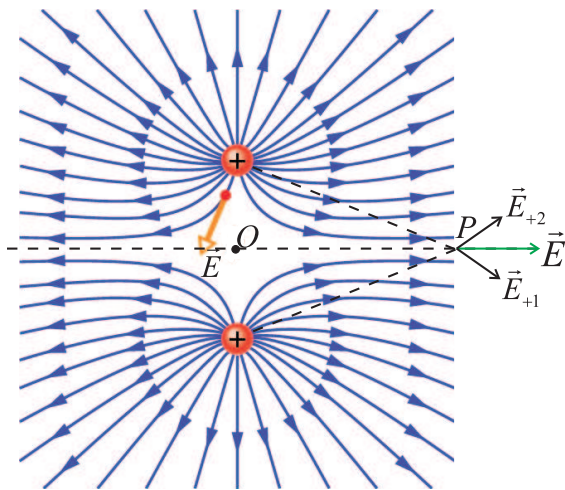
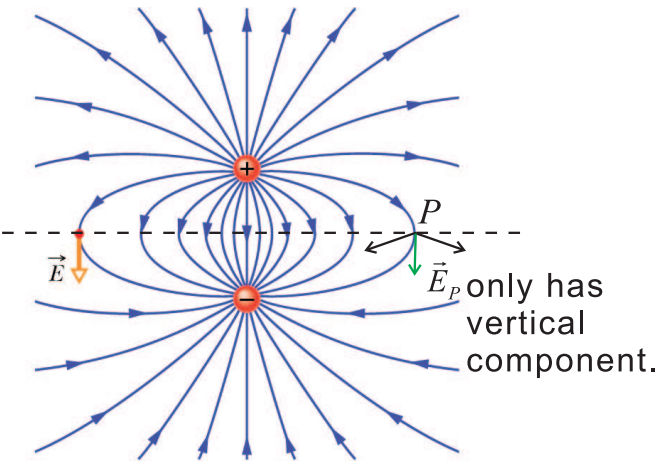
Conventions:

1. The start on position charges and end on negative charges.
2. *Direction* of E-field at any point is given by *tangent* of E-field line.
3. *Magnitude* of E-field at any point is proportional to *number of E-field lines per unit area perpendicular to the lines*.



Infinite sheet of charge





$\vec{E}_{\text{at point } O} = 0$

2.5 Point Charge in E-field

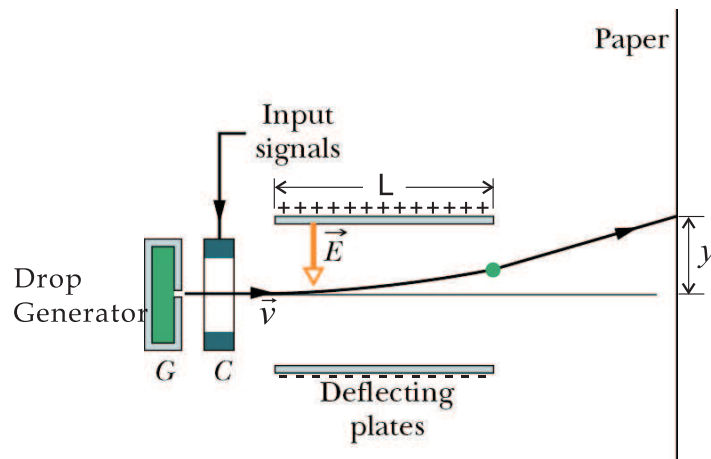
When we place a charge q in an E-field $\vec{E}$, the force experienced by the charge is

$$\vec{F} = q\vec{E} = m\vec{a}$$

Applications: *Ink-jet printer, TV cathode ray tube.*

Example:

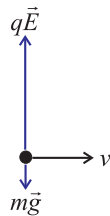
Ink particle has mass m , charge q ($q < 0$ here)



Assume that mass of inkdrop is small, what's the deflection y of the charge?

Solution:

First, the charge carried by the inkdrop is *negative*, i.e. $q < 0$.



Note: $q\vec{E}$ points in opposite direction of $\vec{E}$.

Horizontal motion: Net force = 0

$$\therefore L = vt \quad (2.1)$$

Vertical motion: $|q\vec{E}| \gg |m\vec{g}|$, q is negative,

$\therefore$ Net force $= -qE = ma$ (Newton's 2nd Law)

$$\therefore a = -\frac{qE}{m} \quad (2.2)$$

Vertical distance travelled:

$$y = \frac{1}{2} at^2$$

2.6 Dipole in E-field

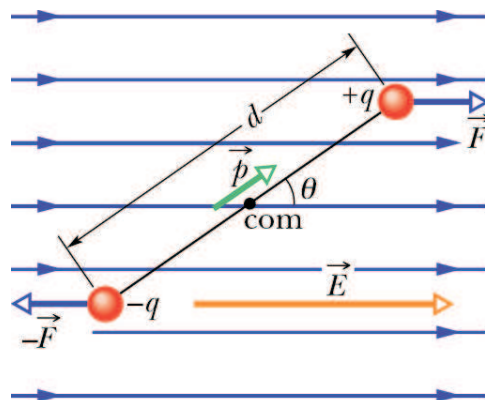
Consider the force exerted on the dipole in an *external* E-field:

Assumption: E-field from dipole doesn't affect the external E-field.

- Dipole moment:

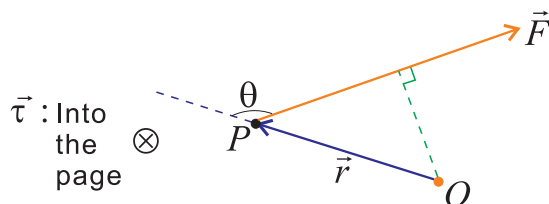
$$\vec{p} = q\vec{d}$$

- Force due to the E-field on $+ve$ and $-ve$ charge are *equal and opposite in direction*. Total external force on dipole $= 0$.



BUT: There is an external **torque** on the center of the dipole.

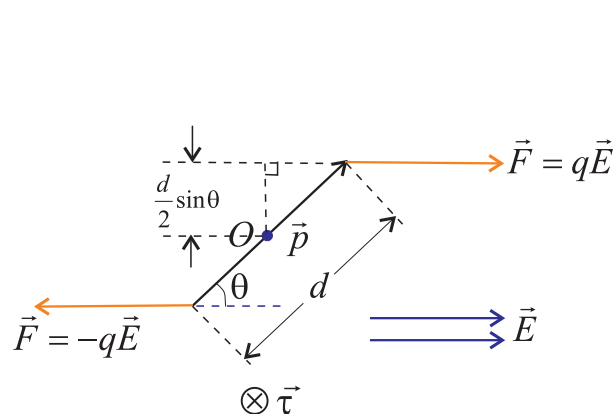
Reminder:



Force $\vec{F}$ exerts at point P.
The force exerts a **torque**
 $\vec{\tau} = \vec{r} \times \vec{F}$ on point P with respect to point O.

Direction of the **torque vector** $\vec{\tau}$ is determined from the **right-hand rule**.

Reference: Halliday Vol.1 Chap 9.1 (Pg.175) *torque*
 Chap 11.7 (Pg.243) *work done*



Net torque $\vec{\tau}$

- direction: clockwise torque
- magnitude:

$$\begin{aligned}
 \tau &= \tau_{+ve} + \tau_{-ve} \\
 &= F \cdot \frac{d}{2} \sin \theta + F \cdot \frac{d}{2} \sin \theta \\
 &= qE \cdot d \sin \theta \\
 &= pE \sin \theta
 \end{aligned}$$

$$\vec{\tau} = \vec{p} \times \vec{E}$$

Energy Consideration:

When the dipole $\vec{p}$ rotates $d\theta$, the E-field does work.

Work done by external E-field on the dipole:

$$dW = -\tau d\theta$$

Negative sign here because torque by E-field acts to *decrease* θ .

BUT: Because E-field is a **conservative force field**^{1 2}, we can define a **potential energy** (U) for the system, so that

$$dU = -dW$$

$\therefore$ For the dipole in external E-field:

$$dU = -dW = pE \sin \theta d\theta$$

$$\begin{aligned}
 \therefore U(\theta) &= \int dU = \int pE \sin \theta d\theta \\
 &= -pE \cos \theta + U_0
 \end{aligned}$$

¹more to come in Chap.4 of notes

²ref. Halliday Vol.1 Pg.257, Chap 12.1

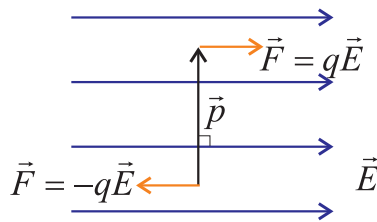
set $U(\theta = 90^\circ) = 0$,

$$\therefore 0 = -pE \cos 90^\circ + U_0$$

$$\therefore U_0 = 0$$

$\therefore$ Potential energy:

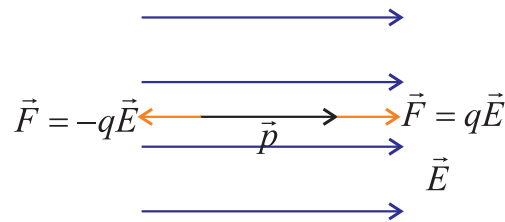
$$U = -pE \cos \theta = -\vec{p} \cdot \vec{E}$$



$$\theta = 90^\circ$$

Torque $|\vec{\tau}| = pE$

$$U = 0 \text{ (define)}$$



$$\theta = 0^\circ$$

Torque $|\vec{\tau}| = 0$

$$U = -pE$$

(based on definition)

**Minimum energy
configuration**

Chapter 3

Electric Flux and Gauss' Law

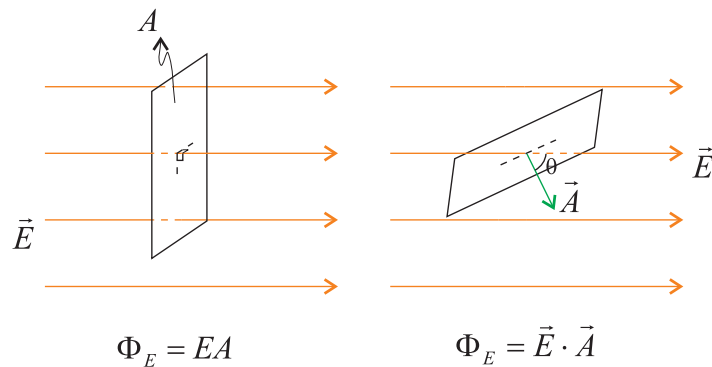
3.1 Electric Flux

Latin: flux = "to flow"

Graphically:

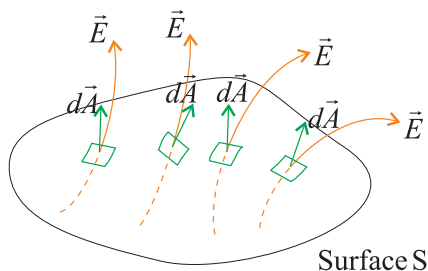
Electric flux Φ_E represents the number of E-field lines crossing a surface.

Mathematically:



Reminder: Vector of the area $\vec{A}$ is perpendicular to the area A .

For non-uniform E-field & surface, direction of the area vector $\vec{A}$ is not uniform.



$d\vec{A}$ = Area vector for small area element dA

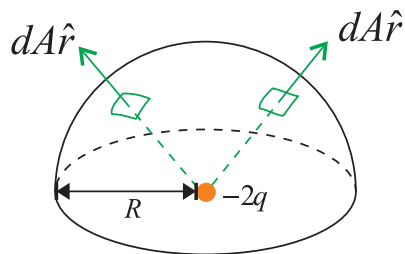
$\therefore$ Electric flux $d\Phi_E = \vec{E} \cdot d\vec{A}$

Electric flux of $\vec{E}$ through surface S: $\Phi_E = \int_S \vec{E} \cdot d\vec{A}$

$\int_S$ = Surface integral over surface S
 = Integration of integral over all area elements on surface S

Example:

S = hemisphere radius R



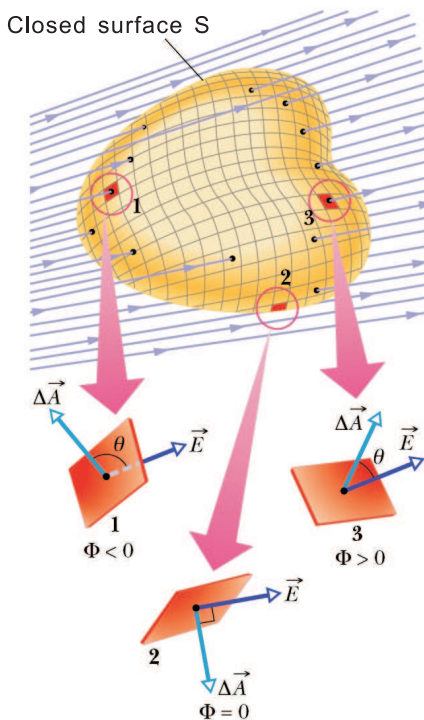
$$\int_S dA = \text{Surface area of } S$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{-2q}{r^2} \hat{r} = \frac{-q}{2\pi\epsilon_0 R^2} \hat{r}$$

For a hemisphere, $d\vec{A} = dA \hat{r}$

$$\begin{aligned} \Phi_E &= \int_S \frac{-q}{2\pi\epsilon_0 R^2} \hat{r} \cdot (dA \hat{r}) \quad (\because \hat{r} \cdot \hat{r} = 1) \\ &= -\frac{q}{2\pi\epsilon_0 R^2} \underbrace{\int_S dA}_{2\pi R^2} \\ &= \frac{-q}{\epsilon_0} \end{aligned}$$

For a closed surface:

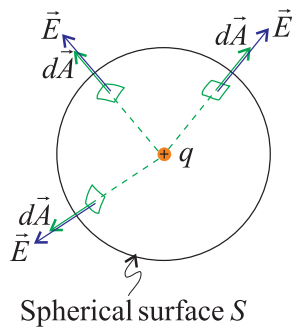


Recall: Direction of area vector $d\vec{A}$ goes from *inside to outside* of closed surface S.

Electric flux over closed surface S: $\Phi_E = \oint_S \vec{E} \cdot d\vec{A}$

$\oint_S$ = Surface integral over closed surface S

Example:



Electric flux of charge q over closed spherical surface of radius R .

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} \hat{r} = \frac{q}{4\pi\epsilon_0 R^2} \hat{r} \quad \text{at the surface}$$

F/A

Again, $d\vec{A} = dA \cdot \hat{r}$

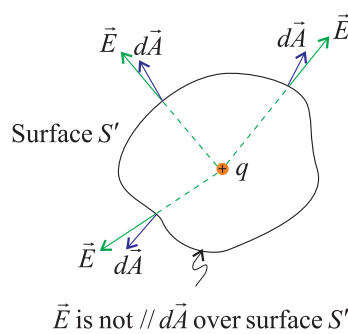
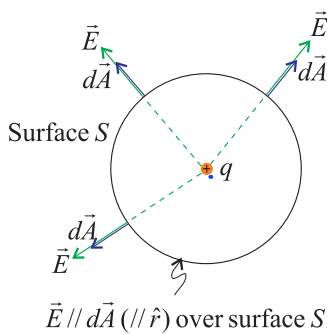
$$\begin{aligned} \therefore \Phi_E &= \oint_S \underbrace{\frac{q}{4\pi\epsilon_0 R^2}}_{\vec{E}} \cdot \underbrace{d\vec{A}}_{\hat{r}} \\ &= \frac{q}{4\pi\epsilon_0 R^2} \underbrace{\oint_S dA}_{\text{Total surface area of } S = 4\pi R^2} \end{aligned}$$

$$\Phi_E = \frac{q}{\epsilon_0}$$

IMPORTANT POINT:

If we remove the spherical symmetry of closed surface S, the total number of E -field lines crossing the surface remains the same.

$\therefore$ The electric flux Φ_E



$$\Phi_E = \oint_S \vec{E} \cdot d\vec{A} = \oint_{S'} \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$$

3.2 Gauss' Law

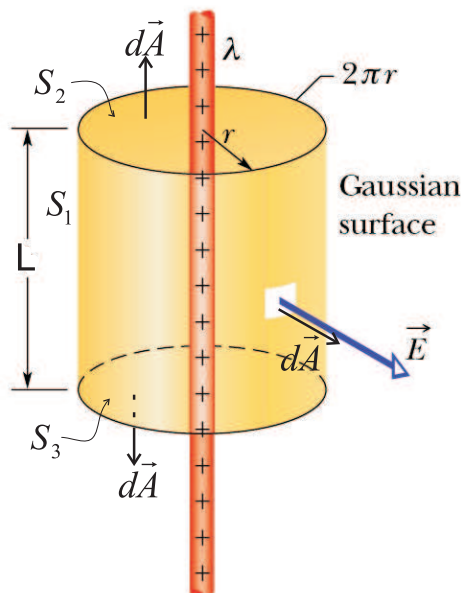
$$\Phi_E = \oint_S \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0} \quad \text{for any closed surface } S$$

And q is the net electric charge enclosed in closed surface S .

- Gauss' Law is valid for *all charge distributions and all closed surfaces*. (*Gaussian surfaces*)
- Coulomb's Law can be derived from Gauss' Law.
- For system with high order of *symmetry*, E-field can be easily determined if we construct *Gaussian surfaces with the same symmetry* and applies Gauss' Law

3.3 E-field Calculation with Gauss' Law

(A) Infinite line of charge



Linear charge density: λ

Cylindrical symmetry.

E-field directs radially outward from the rod.

Construct a Gaussian surface S in the shape of a **cylinder**, making up of a curved surface S_1 , and the top and bottom circles S_2, S_3 .

Gauss' Law:
$$\oint_S \vec{E} \cdot d\vec{A} = \frac{\text{Total charge}}{\epsilon_0} = \frac{\lambda L}{\epsilon_0}$$

$$\lambda = \frac{q}{L}$$

$$\sigma = \frac{q}{A}$$

$$\oint_S \vec{E} \cdot d\vec{A} = \underbrace{\int_{S_1} \vec{E} \cdot d\vec{A}}_{\vec{E} \parallel d\vec{A}} + \underbrace{\int_{S_2} \vec{E} \cdot d\vec{A} + \int_{S_3} \vec{E} \cdot d\vec{A}}_{=0 \because \vec{E} \perp d\vec{A}}$$

$$\therefore E \underbrace{\int_{S_1} dA}_{\text{Total area of surface } S_1} = \frac{\lambda L}{\epsilon_0}$$

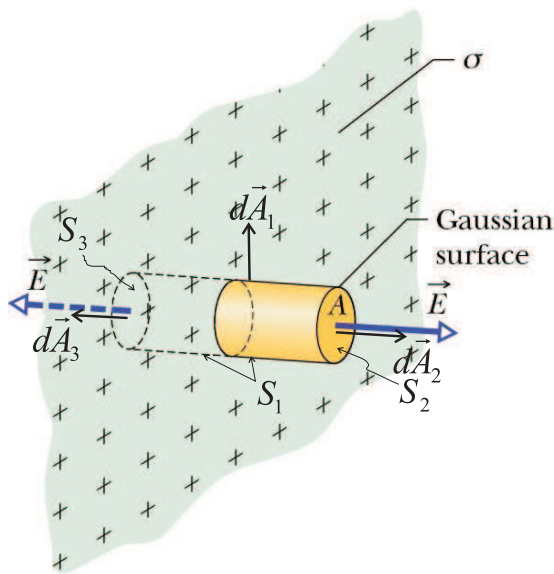
Total area of surface S_1

$$E(2\pi r L) = \frac{\lambda L}{\epsilon_0}$$

$$\therefore E = \frac{\lambda}{2\pi\epsilon_0 r} \quad (\text{Compare with Chapter 2 note})$$

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 r} \hat{r}$$

✓ (B) Infinite sheet of charge



Uniform surface charge density:

σ

Planar symmetry.

E-field directs perpendicular to the sheet of charge.

Construct Gaussian surface S in the shape of a **cylinder (pill box)** of cross-sectional area A .

Gauss' Law: $\oint_S \vec{E} \cdot d\vec{A} = \frac{A\sigma}{\epsilon_0}$

$$\int_{S_1} \vec{E} \cdot d\vec{A} = 0 \quad \because \vec{E} \perp d\vec{A} \text{ over whole surface } S_1$$

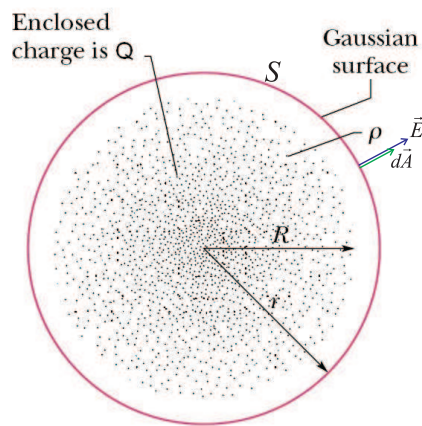
$$\int_{S_2} \vec{E} \cdot d\vec{A} + \int_{S_3} \vec{E} \cdot d\vec{A} = 2EA \quad (\vec{E} \parallel d\vec{A}_2, \vec{E} \parallel d\vec{A}_3)$$

Note: For S_2 , both $\vec{E}$ and $d\vec{A}_2$ point up
 For S_3 , both $\vec{E}$ and $d\vec{A}_3$ point down

$$\therefore 2EA = \frac{A\sigma}{\epsilon_0} \Rightarrow \boxed{E = \frac{\sigma}{2\epsilon_0}} \quad (\text{Compare with Chapter 2 note})$$

✓ (C) Uniformly charged sphere
Total charge = Q
Spherical symmetry.

(a) For $r > R$:



Consider a spherical Gaussian surface S of radius r :

$$\vec{E} \parallel d\vec{A} \parallel \hat{r}$$

$$\text{Gauss' Law: } \oint_S \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

$$\oint_S E \cdot dA = \frac{Q}{\epsilon_0}$$

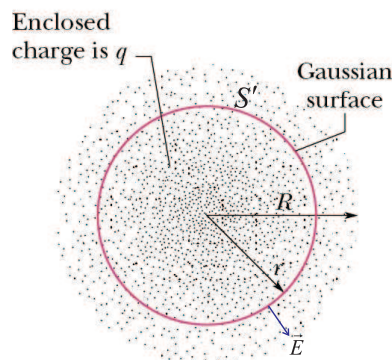
$$E \underbrace{\oint_S dA}_{\text{surface area of } S} = \frac{Q}{\epsilon_0}$$

$$\text{surface area of } S = 4\pi r^2$$

$$\therefore \vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r}; \quad \text{for } r > R$$

$$E = \frac{kQ}{r^2}$$

(b) For $r < R$:



Consider a spherical Gaussian surface S' of radius $r < R$, then total charge included q is *proportional to the volume included by S'*

$$\therefore \frac{q}{Q} = \frac{\text{Volume enclosed by } S'}{\text{Total volume of sphere}}$$

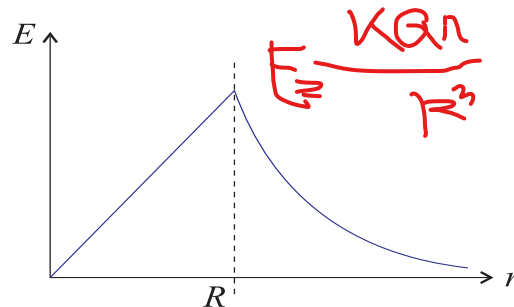
$$\frac{q}{Q} = \frac{4/3 \pi r^3}{4/3 \pi R^3} \Rightarrow q = \frac{r^3}{R^3} Q$$

Gauss' Law: $\oint_{S'} \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$

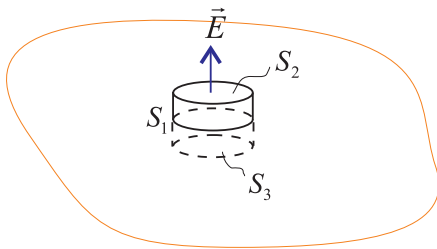
$$E \underbrace{\oint_{S'} dA}_{\text{surface area of } S'} = \frac{r^3}{R^3} \frac{1}{\epsilon_0} \cdot Q$$

surface area of $S' = 4\pi r^2$

$$\therefore \vec{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{R^3} r \hat{r}; \quad \text{for } r \leq R$$



3.4 Gauss' Law and Conductors



For *isolated* conductors, charges are free to move until *all charges lie outside the surface of the conductor*. Also, the E -field at the surface of a conductor is *perpendicular to its surface*. (Why?)

Cross-sectional area A

Consider Gaussian surface S of shape of cylinder:

$$\oint_S \vec{E} \cdot d\vec{A} = \frac{\sigma A}{\epsilon_0}$$

BUT $\int_{S_1} \vec{E} \cdot d\vec{A} = 0 \quad (\because \vec{E} \perp d\vec{A})$
 $\int_{S_3} \vec{E} \cdot d\vec{A} = 0 \quad (\because \vec{E} = 0 \text{ inside conductor})$

$$\begin{aligned} \int_{S_2} \vec{E} \cdot d\vec{A} &= E \underbrace{\int_{S_2} dA}_{\text{Area of } S_2} \quad (\because \vec{E} \parallel d\vec{A}) \\ &= EA \end{aligned}$$

$$\therefore \text{Gauss' Law} \Rightarrow EA = \frac{\sigma A}{\epsilon_0}$$

$$\therefore \boxed{\text{On conductor's surface} \quad E = \frac{\sigma}{\epsilon_0}}$$

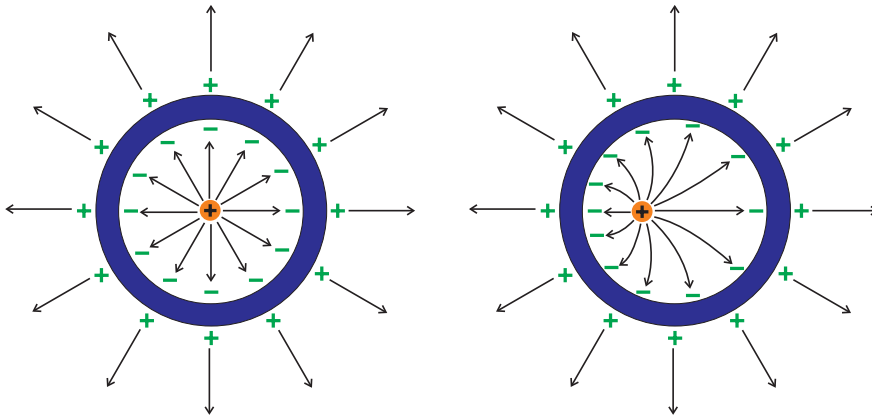
BUT, there's no charge inside conductors.

$$\therefore \boxed{\text{Inside conductors} \quad E = 0} \quad \text{Always!}$$

Notice: Surface charge density on a conductor's surface is *not uniform*.

Example: Conductor with a charge inside

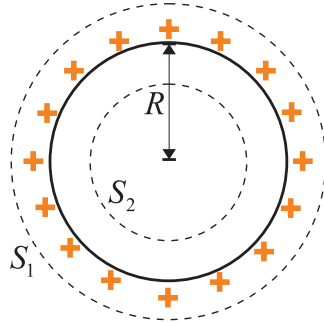
Note: This is not an isolated system (because of the charge inside).



Note: In BOTH cases, $\vec{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} \hat{r}$ outside

Example:

I. Charge sprayed on a conductor sphere:

Total charge = Q

First, we know that charges all move to the *surface* of conductors.

(i) For $r < R$:Consider Gaussian surface S_2

$$\oint_{S_2} \vec{E} \cdot d\vec{A} = 0 \quad (\because \text{no charge inside})$$

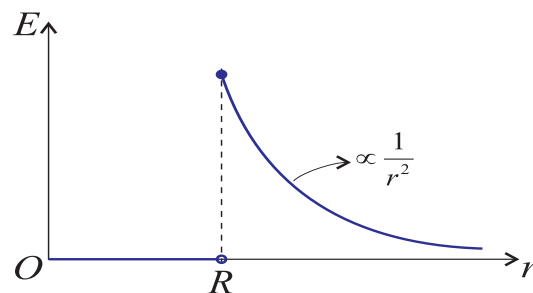
$$\Rightarrow E = 0 \quad \text{everywhere.}$$

(ii) For $r \geq R$:Consider Gaussian surface S_1 :

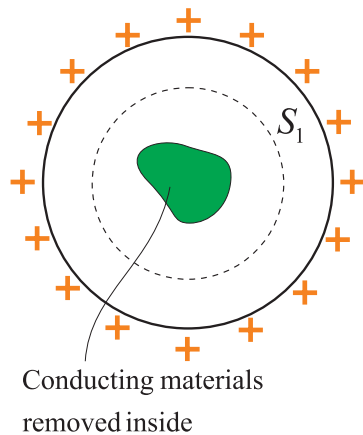
$$\oint_{S_1} \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

$$\underbrace{E \oint_{S_1} d\vec{A}}_{4\pi r^2} = \frac{Q}{\epsilon_0} \quad \begin{array}{l} \text{For a conductor} \\ (\vec{E} \parallel d\vec{A} \parallel \hat{r}) \\ \text{Spherically symmetric} \end{array}$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$



II. Conductor sphere with hole inside:

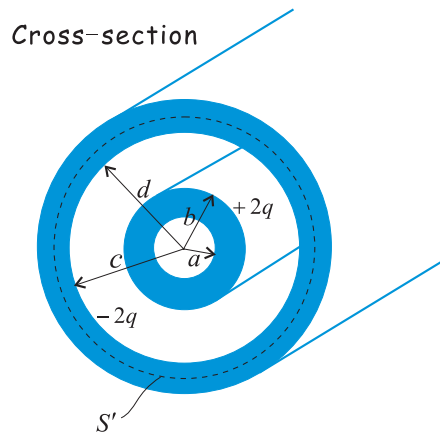


Consider Gaussian surface S_1 : Total charge included = 0

$\therefore$ E-field = 0 inside

The E-field is identical to the case of a solid conductor!!

III. A long hollow cylindrical conductor:



Example:

Inside hollow cylinder ($+2q$)

$$\begin{cases} \text{Inner radius} & a \\ \text{Outer radius} & b \end{cases}$$

Outside hollow cylinder ($-3q$)

$$\begin{cases} \text{Inner radius} & c \\ \text{Outer radius} & d \end{cases}$$

Question: Find the charge on each surface of the conductor.

For the inside hollow cylinder, charges distribute only on the surface.

$\therefore$ Inner radius a surface, charge = 0
and Outer radius b surface, charge = $+2q$

For the outside hollow cylinder, charges do not distribute only on outside.

$\therefore$ It's not an isolated system. (There are charges inside!)

Consider Gaussian surface S' inside the conductor:

E-field always = 0

$\therefore$ Need charge $-2q$ on radius c surface to balance the charge of inner cylinder.

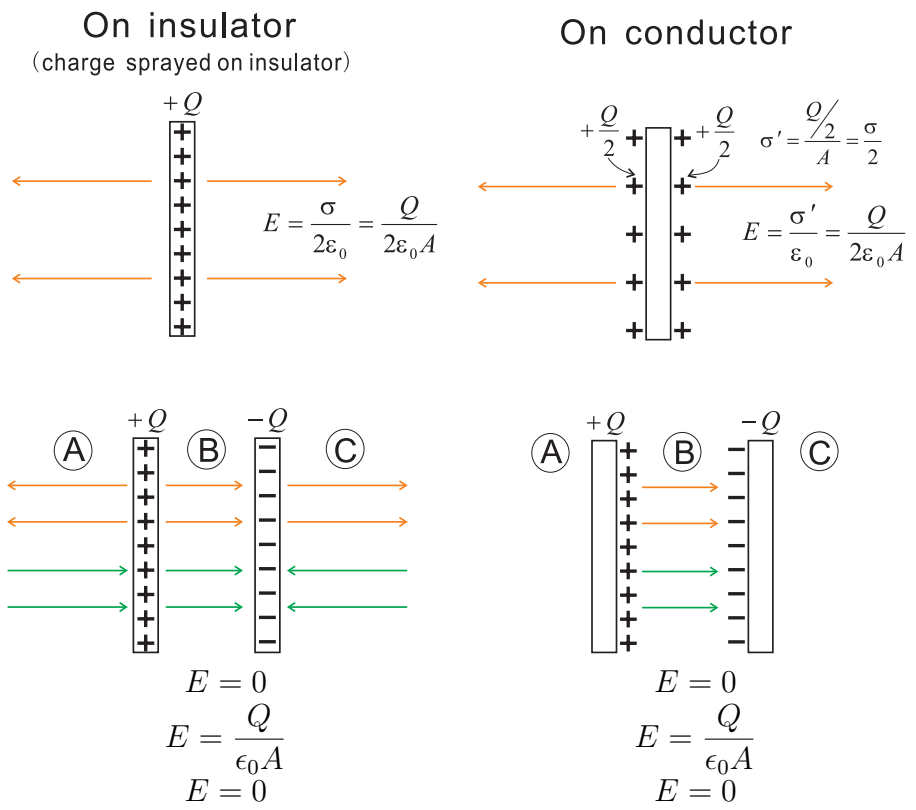
So charge on radius d surface = $-q$. (Why?)

IV. Large sheets of charge:

Total charge Q on sheet of area A ,

$$\therefore \text{Surface charge density } \sigma = \frac{Q}{A}$$

By principle of superposition



$$E = \frac{Q}{\epsilon_0} \quad \left[\sigma = \frac{Q}{A} \right]$$

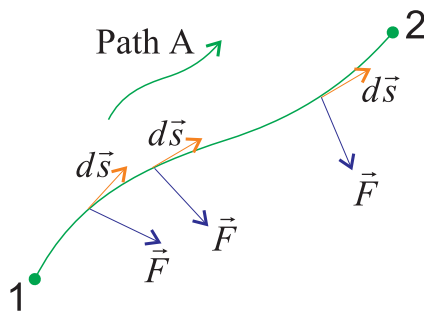
Chapter 4

Electric Potential

4.1 Potential Energy and Conservative Forces

(Read Halliday Vol.1 Chap.12)

Electric force is a **conservative force**



Work done by the electric force $\vec{F}$ as a charge moves an infinitesimal distance $d\vec{s}$ along *Path A* = dW

Note: $d\vec{s}$ is in the *tangent* direction of the curve of *Path A*.

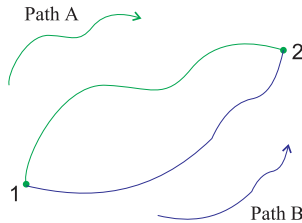
$$dW = \vec{F} \cdot d\vec{s}$$

$\therefore$ Total work done W by force $\vec{F}$ in moving the particle from Point 1 to Point 2

$$W = \int_1^2 \vec{F} \cdot d\vec{s} \text{ Path A}$$

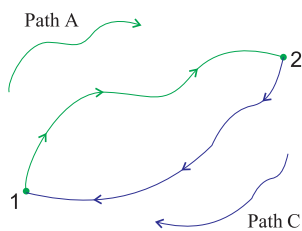
$$\begin{aligned} \int_1^2 \text{ Path A} &= \text{Path Integral} \\ &= \text{Integration over Path A from Point 1 to Point 2.} \end{aligned}$$

DEFINITION: A force is **conservative** if the work done on a particle by the force is *independent of the path taken*.



$\therefore$ For conservative forces,

$$\int_{\text{Path A}}^2 \vec{F} \cdot d\vec{s} = \int_{\text{Path B}}^2 \vec{F} \cdot d\vec{s}$$



Let's consider a path starting at point 1 to 2 through *Path A* and from 2 to 1 through *Path C*

$$\begin{aligned} \text{Work done} &= \int_{\text{Path A}}^2 \vec{F} \cdot d\vec{s} + \int_{\text{Path C}}^1 \vec{F} \cdot d\vec{s} \\ &= \int_{\text{Path A}}^2 \vec{F} \cdot d\vec{s} - \int_{\text{Path B}}^2 \vec{F} \cdot d\vec{s} \end{aligned}$$

DEFINITION: The work done by a **conservative force** on a particle when it moves around a closed path returning to its initial position is zero.

MATHEMATICALLY, $\vec{\nabla} \times \vec{F} = 0$ everywhere for conservative force $\vec{F}$

Conclusion: Since the work done by a conservative force $\vec{F}$ is *path-independent*, we can define a quantity, **potential energy**, that depends only on the *position* of the particle.

Convention: We define **potential energy** U such that

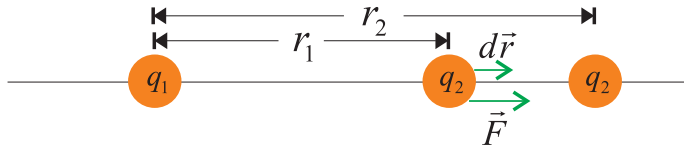
$$dU = -W = - \int \vec{F} \cdot d\vec{s}$$

$\therefore$ For particle moving from 1 to 2

$$\int_1^2 dU = U_2 - U_1 = - \int_1^2 \vec{F} \cdot d\vec{s}$$

where U_1, U_2 are **potential energy** at position 1, 2.

Example:



Suppose charge q_2 moves from point 1 to 2.

$$\begin{aligned}
 \text{From definition: } U_2 - U_1 &= - \int_1^2 \vec{F} \cdot d\vec{r} \\
 &= - \int_{r_1}^{r_2} F dr \quad (\because \vec{F} \parallel d\vec{r}) \\
 &= - \int_{r_1}^{r_2} \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} dr \\
 (\because \int \frac{dr}{r^2} &= -\frac{1}{r} + C) &= \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r} \Big|_{r_1}^{r_2} \\
 &= \frac{1}{4\pi\epsilon_0} q_1 q_2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right)
 \end{aligned}$$

Note:

- (1) This result is generally true for 2-Dimension or 3-D motion.
- (2) If q_2 moves away from q_1 ,
then $r_2 > r_1$, we have
 - If q_1, q_2 are of *same* sign,
then $\Delta U < 0$, $\Delta W > 0$
(ΔW = Work done by electric *repulsive* force)
 - If q_1, q_2 are of *different* sign,
then $\Delta U > 0$, $\Delta W < 0$
(ΔW = Work done by electric *attractive* force)
- (3) If q_2 moves towards q_1 ,
then $r_2 < r_1$, we have
 - If q_1, q_2 are of *same* sign,
then $\Delta U > 0$, $\Delta W < 0$
 - If q_1, q_2 are of *different* sign,
then $\Delta U < 0$, $\Delta W > 0$

(4) Note: It is the *difference* in potential energy that is important.

REFERENCE POINT: $U(r = \infty) = 0$

$$\therefore U_{\infty} - U_1 = \frac{1}{4\pi\epsilon_0} q_1 q_2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right)$$

↓
 ∞

$$\boxed{U(r) = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r}}$$

If q_1, q_2 same sign, then $U(r) > 0$ for all r
 If q_1, q_2 opposite sign, then $U(r) < 0$ for all r

(5) Conservation of Mechanical Energy:

For a system of charges with no external force,

$$E = K + U = \text{Constant}$$

↙ ↘
(Kinetic Energy) (Potential Energy)

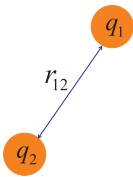
or $\boxed{\Delta E = \Delta K + \Delta U = 0}$

Potential Energy of A System of Charges

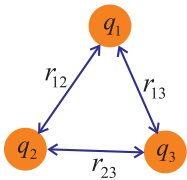
Example: P.E. of 3 charges q_1, q_2, q_3

Start: q_1, q_2, q_3 all at $r = \infty, U = 0$

Step1:  Move q_1 from ∞ to its position $\Rightarrow U = 0$

Step2:  Move q_2 from ∞ to new position $\Rightarrow$

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$$

Step3:  Move q_3 from ∞ to new position $\Rightarrow$ Total P.E.

$$U = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right]$$

Step4: What if there are 4 charges?

4.2 Electric Potential

Consider a charge q at center, we consider its effect on test charge q_0

DEFINITION: We define electric potential V so that

$$\Delta V = \frac{\Delta U}{q_0} = \frac{-\Delta W}{q_0}$$

($\therefore V$ is the P.E. per unit charge)

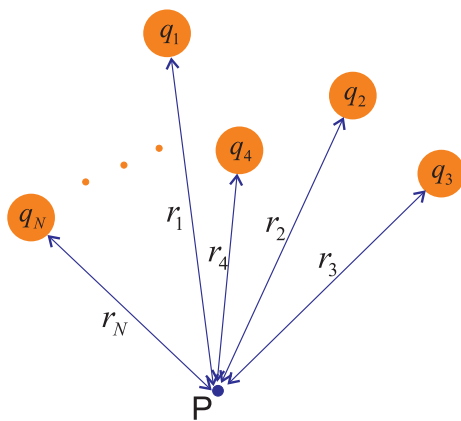
- Similarly, we take $V(r = \infty) = 0$.
- Electric Potential is a **scalar**.
- Unit: $\text{Volt}(V) = \text{Joules/Coulomb}$
- For a single point charge:

$$V(r) = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r}$$

- Energy Unit: $\Delta U = q\Delta V$

$$\text{electron - Volt}(eV) = \underbrace{1.6 \times 10^{-19}}_{\text{charge of electron}} J$$

Potential For A System of Charges



For a total of N point charges, the potential V at any point P can be derived from the **principle of superposition**.

Recall that potential due to q_1 at point P: $V_1 = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1}{r_1}$

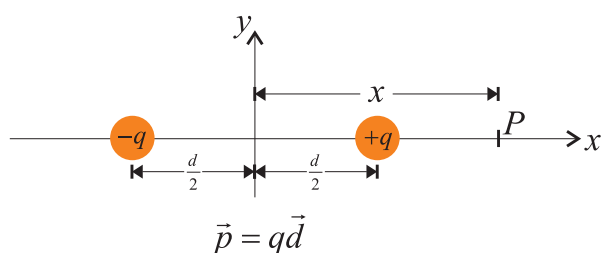
$\therefore$ Total potential at point P due to N charges:

$$\begin{aligned} V &= V_1 + V_2 + \dots + V_N \quad (\text{principle of superposition}) \\ &= \frac{1}{4\pi\epsilon_0} \left[\frac{q_1}{r_1} + \frac{q_2}{r_2} + \dots + \frac{q_N}{r_N} \right] \end{aligned}$$

$$V = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \frac{q_i}{r_i}$$

Note: For $\vec{E}, \vec{F}$, we have a sum of vectors
 For V, U , we have a sum of scalars

Example: Potential of an electric dipole



Consider the potential of point P at distance $x > \frac{d}{2}$ from dipole.

$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{+q}{x - \frac{d}{2}} + \frac{-q}{x + \frac{d}{2}} \right]$$

Special Limiting Case: $x \gg d$

$$\frac{1}{x \mp \frac{d}{2}} = \frac{1}{x} \cdot \frac{1}{1 \mp \frac{d}{2x}} \simeq \frac{1}{x} \left[1 \pm \frac{d}{2x} \right]$$

$$\therefore V = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{x} \left[1 + \frac{d}{2x} - \left(1 - \frac{d}{2x} \right) \right]$$

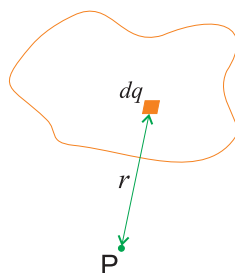
$$V = \frac{p}{4\pi\epsilon_0 x^2} \quad (\text{Recall } p = qd)$$

For a point charge $E \propto \frac{1}{r^2} \quad V \propto \frac{1}{r}$

For a dipole $E \propto \frac{1}{r^3} \quad V \propto \frac{1}{r^2}$

For a quadrupole $E \propto \frac{1}{r^4} \quad V \propto \frac{1}{r^3}$

Electric Potential of Continuous Charge Distribution



For any charge distribution, we write the electrical potential dV due to infinitesimal charge dq :

$$dV = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r}$$

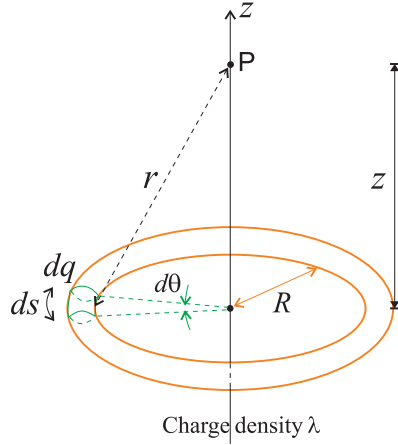
$$\therefore \boxed{V = \int \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r}}$$

charge
distribution

Similar to the previous examples on E-field, for the case of *uniform* charge distribution:

$$\begin{array}{lll} \text{1-D} & \Rightarrow & \text{long rod} & \Rightarrow & dq = \lambda dx \\ \text{2-D} & \Rightarrow & \text{charge sheet} & \Rightarrow & dq = \sigma dA \\ \text{3-D} & \Rightarrow & \text{uniformly charged body} & \Rightarrow & dq = \rho dV \end{array}$$

Example (1): Uniformly-charged ring



Length of the infinitesimal ring element
 $= ds = R d\theta$

$$\begin{aligned} \therefore \text{charge } dq &= \lambda ds \\ &= \lambda R d\theta \end{aligned}$$

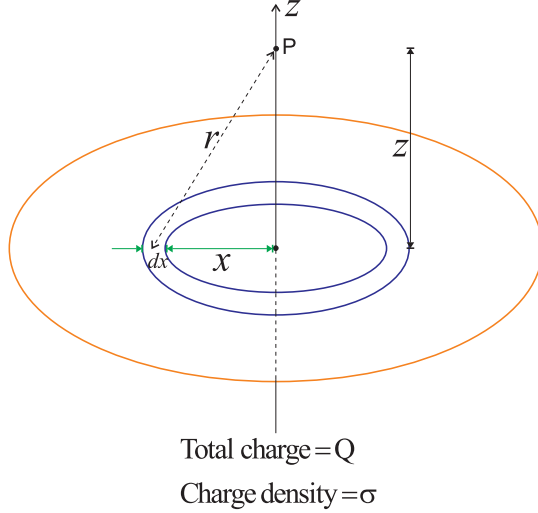
$$dV = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r} = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda R d\theta}{\sqrt{R^2 + z^2}}$$

The integration is around the entire ring.

$$\begin{aligned} \therefore V &= \int_{\text{ring}} dV \\ &= \int_0^{2\pi} \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda R d\theta}{\sqrt{R^2 + z^2}} \\ &= \frac{\lambda R}{4\pi\epsilon_0 \sqrt{R^2 + z^2}} \underbrace{\int_0^{2\pi} d\theta}_{2\pi} \end{aligned}$$

$$\begin{array}{l} \text{Total charge on the} \\ \text{ring} = \lambda \cdot (2\pi R) \end{array} \quad V = \frac{Q}{4\pi\epsilon_0 \sqrt{R^2 + z^2}}$$

$$\text{LIMITING CASE: } z \gg R \Rightarrow V = \frac{Q}{4\pi\epsilon_0 \sqrt{z^2}} = \frac{Q}{4\pi\epsilon_0 |z|}$$

Example (2): Uniformly-charged disk

Using the **principle of superposition**, we will find the potential of a disk of uniform charge density by integrating the potential of *concentric rings*.

$$\therefore dV = \frac{1}{4\pi\epsilon_0} \int_{\text{disk}} \frac{dq}{r}$$

Ring of radius x : $dq = \sigma dA = \sigma (2\pi x dx)$

$$\begin{aligned} \therefore V &= \int_0^R \frac{1}{4\pi\epsilon_0} \cdot \frac{\sigma 2\pi x dx}{\sqrt{x^2 + z^2}} \\ &= \frac{\sigma}{4\epsilon_0} \int_0^R \frac{d(x^2 + z^2)}{(x^2 + z^2)^{1/2}} \\ V &= \frac{\sigma}{2\epsilon_0} (\sqrt{z^2 + R^2} - \sqrt{z^2}) \\ &= \frac{\sigma}{2\epsilon_0} (\sqrt{z^2 + R^2} - |z|) \end{aligned}$$

Recall:
 $|x| = \begin{cases} +x; & x \geq 0 \\ -x; & x < 0 \end{cases}$

Limiting Case:

(1) If $|z| \gg R$

$$\begin{aligned} \sqrt{z^2 + R^2} &= \sqrt{z^2 \left(1 + \frac{R^2}{z^2}\right)} \\ &= |z| \cdot \left(1 + \frac{R^2}{z^2}\right)^{\frac{1}{2}} \quad \left((1+x)^n \approx 1 + nx \text{ if } x \ll 1 \right) \\ &\simeq |z| \cdot \left(1 + \frac{R^2}{2z^2}\right) \quad \left(\frac{|z|}{z^2} = \frac{1}{|z|} \right) \end{aligned}$$

$$\therefore \text{At large } z, V \simeq \frac{\sigma}{2\epsilon_0} \cdot \frac{R^2}{2|z|} = \frac{Q}{4\pi\epsilon_0|z|} \quad (\text{like a point charge})$$

where $Q = \text{total charge on disk} = \sigma \cdot \pi R^2$

(2) If $|z| \ll R$

$$\begin{aligned}\sqrt{z^2 + R^2} &= R \cdot \left(1 + \frac{z^2}{R^2}\right)^{\frac{1}{2}} \\ &\simeq R \left(1 + \frac{z^2}{2R^2}\right)\end{aligned}$$

$$\therefore V \simeq \frac{\sigma}{2\epsilon_0} \left[R - |z| + \frac{z^2}{2R} \right]$$

At $z = 0$, $V = \frac{\sigma R}{2\epsilon_0}$; Let's call this V_0

$$\therefore V(z) = \frac{\sigma R}{2\epsilon_0} \left[1 - \frac{|z|}{R} + \frac{z^2}{2R^2} \right]$$

$$V(z) = V_0 \left[1 - \frac{|z|}{R} + \frac{z^2}{2R^2} \right]$$

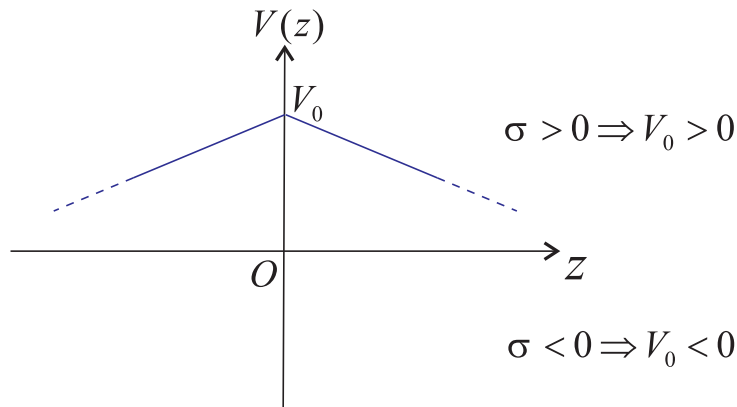
The *key* here is that it is the difference between potentials of two points that is important.

$\Rightarrow$ A convenience reference point to compare in this example is the potential of the charged disk.

$\therefore$ The important quantity here is

$$V(z) - V_0 = -\frac{|z|}{R} V_0 + \underbrace{\frac{z^2}{2R^2} V_0}_{\text{neglected as } z \ll R}$$

$$V(z) - V_0 = -\frac{V_0}{R} |z|$$



4.3 Relation Between Electric Field E and Electric Potential V

(A) To get V from $\vec{E}$:

Recall our definition of the potential V :

$$\Delta V = \frac{\Delta U}{q_0} = -\frac{W_{12}}{q_0}$$

where ΔU is the change in P.E.; W_{12} is the work done in bringing charge q_0 from point 1 to 2.

$$\therefore \Delta V = V_2 - V_1 = \frac{-\int_1^2 \vec{F} \cdot d\vec{s}}{q_0}$$

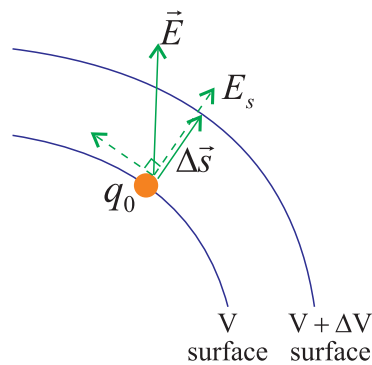
However, the definition of E-field: $\vec{F} = q_0 \vec{E}$

$$\therefore \Delta V = V_2 - V_1 = -\int_1^2 \vec{E} \cdot d\vec{s}$$

Note: The integral on the right hand side of the above can be calculated *along any path from point 1 to 2. (Path-Independent)*

Convention: $V_\infty = 0 \Rightarrow V_P = -\int_\infty^P \vec{E} \cdot d\vec{s}$

(B) To get $\vec{E}$ from V :



(i.e. Potential = V on the surface)

Again, use the definition of V :

$$\Delta U = q_0 \Delta V = \underbrace{-W}_{\text{Work done}}$$

However,

$$\begin{aligned} W &= \underbrace{q_0 \vec{E}}_{\text{Electric force}} \cdot \Delta \vec{s} \\ &= q_0 E_s \Delta s \end{aligned}$$

where E_s is the E-field component along the path $\Delta \vec{s}$.

$$\therefore q_0 \Delta V = -q_0 E_s \Delta s$$

$$\therefore E_s = -\frac{\Delta V}{\Delta s}$$

For infinitesimal Δs ,

$$\therefore \boxed{E_s = -\frac{dV}{ds}}$$

Note: (1) Therefore the E-field component along *any direction* is the negative derivative of the potential *along the same direction*.

(2) If $d\vec{s} \perp \vec{E}$, then $\Delta V = 0$

(3) ΔV is biggest/smallest if $d\vec{s} \parallel \vec{E}$

Generally, for a potential $V(x, y, z)$, the relation between $\vec{E}(x, y, z)$ and V is

$$E_x = -\frac{\partial V}{\partial x} \quad E_y = -\frac{\partial V}{\partial y} \quad E_z = -\frac{\partial V}{\partial z}$$

$\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}$ are **partial derivatives**

For $\frac{\partial}{\partial x}V(x, y, z)$, everything y, z are treated like a *constant* and we only take derivative with respect to x .

Example: If $V(x, y, z) = x^2y - z$

$$\frac{\partial V}{\partial x} =$$

$$\frac{\partial V}{\partial y} =$$

$$\frac{\partial V}{\partial z} =$$

For other co-ordinate systems

(1) Cylindrical:

$$V(r, \theta, z) \quad \left\{ \begin{array}{l} E_r = -\frac{\partial V}{\partial r} \\ E_\theta = -\frac{1}{r} \cdot \frac{\partial V}{\partial \theta} \\ E_z = -\frac{\partial V}{\partial z} \end{array} \right.$$

(2) Spherical:

$$V(r, \theta, \phi) \quad \left\{ \begin{array}{l} E_r = -\frac{\partial V}{\partial r} \\ E_\theta = -\frac{1}{r} \cdot \frac{\partial V}{\partial \theta} \\ E_\phi = -\frac{1}{r \sin \theta} \cdot \frac{\partial V}{\partial \phi} \end{array} \right.$$

Note: Calculating V involves summation of *scalars*, which is easier than adding *vectors* for calculating E-field.

$\therefore$ To find the E-field of a general charge system, we first calculate V , and then derive $\vec{E}$ from the partial derivative.

Example: Uniformly charged disk

From potential calculations:

$$V = \frac{\sigma}{2\epsilon_0}(\sqrt{R^2 + z^2} - |z|) \quad \text{for a point along the z-axis}$$

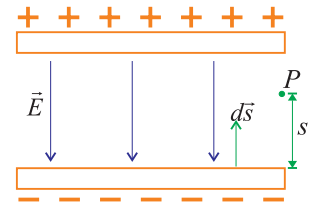
For $z > 0$, $|z| = z$

$$\therefore E_z = -\frac{\partial V}{\partial z} = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{R^2 + z^2}} \right] \quad (\text{Compare with Chap.2 notes})$$

Example: Uniform electric field

(e.g. Uniformly charged +ve and -ve plates)

Consider a path going from the -ve plate to the +ve plate
Potential at point P, V_P can be deduced from definition.



$$\begin{aligned} \text{i.e.} \quad V_P - V_- &= - \int_0^s \vec{E} \cdot d\vec{s} & (V_- = \text{Potential of } -ve \text{ plate}) \\ &= - \int_0^s (-E \, ds) & \because \vec{E}, d\vec{s} \text{ pointing opposite directions} \\ &= E \int_0^s ds = Es \end{aligned}$$

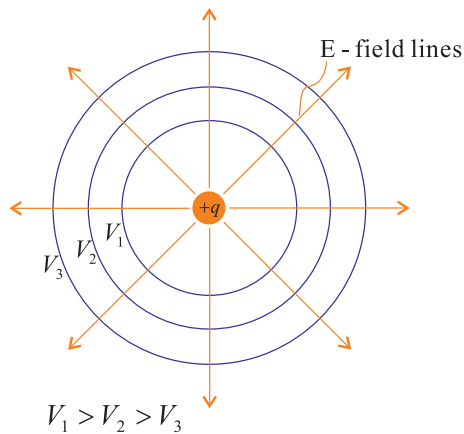
Convenient reference: $V_- = 0$

$$\therefore \boxed{V_P = E \cdot s}$$

4.4 Equipotential Surfaces

Equipotential surface is a surface on which the *potential is constant*.

$$\Rightarrow (\Delta V = 0)$$



$$V(r) = \frac{1}{4\pi\epsilon_0} \cdot \frac{+q}{r} = \text{const}$$

$$\Rightarrow r = \text{const}$$

$\Rightarrow$ Equipotential surfaces are *circles/spherical surfaces*

Note: (1) A charge can move freely on an equipotential surface without any work done.

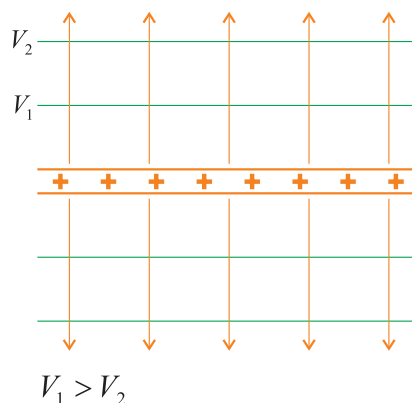
(2) The **electric field lines** must be *perpendicular* to the **equipotential surfaces**. (Why?)

On an equipotential surface, $V = \text{constant}$

$$\Rightarrow \Delta V = 0 \Rightarrow \vec{E} \cdot d\vec{l} = 0, \text{ where } d\vec{l} \text{ is } \textit{tangent} \text{ to equipotential surface}$$

$\therefore \vec{E}$ must be *perpendicular* to equipotential surfaces.

Example: Uniformly charged surface (infinite)



$$\text{Recall} \quad V = V_0 - \frac{\sigma}{2\epsilon_0}|z|$$

$\uparrow$

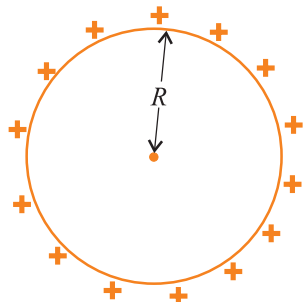
Potential at $z = 0$

Equipotential surface means

$$V = \text{const} \Rightarrow V_0 - \frac{\sigma}{2\epsilon_0}|z| = C$$

$$\Rightarrow |z| = \text{constant}$$

Example: Isolated spherical charged conductors



Recall:

- (1) E-field inside = 0
- (2) charge distributed on the *outside* of conductors.

(i) Inside conductor:

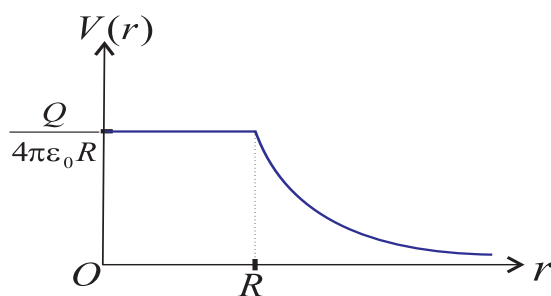
$$\begin{aligned}
 E = 0 &\Rightarrow \Delta V = 0 \text{ everywhere in conductor} \\
 &\Rightarrow V = \text{constant everywhere in conductor} \\
 &\Rightarrow \text{The entire conductor is at the same potential}
 \end{aligned}$$

(ii) Outside conductor:

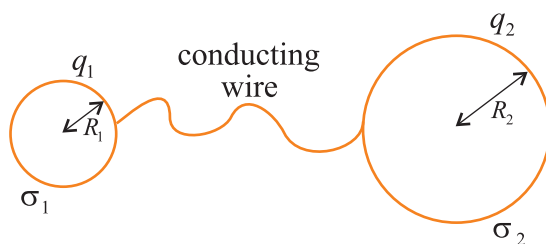
$$V = \frac{Q}{4\pi\epsilon_0 r}$$

$\therefore$ Spherically symmetric (Just like a point charge.)

BUT not true for conductors of arbitrary shape.



Example: Connected conducting spheres



Two conductors connected can be seen as a *single conductor*

$\therefore$ Potential everywhere is identical.

$$\begin{aligned} \text{Potential of radius } R_1 \text{ sphere } V_1 &= \frac{q_1}{4\pi\epsilon_0 R_1} \\ \text{Potential of radius } R_2 \text{ sphere } V_2 &= \frac{q_2}{4\pi\epsilon_0 R_2} \end{aligned}$$

$$\begin{aligned} \mathbf{V}_1 &= \mathbf{V}_2 \\ \Rightarrow \frac{q_1}{R_1} &= \frac{q_2}{R_2} \quad \Rightarrow \quad \frac{q_1}{q_2} = \frac{R_1}{R_2} \end{aligned}$$

Surface charge density

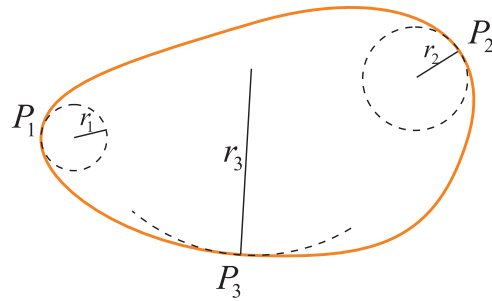
$$\sigma_1 = \frac{q_1}{\underbrace{4\pi R_1^2}_{\text{Surface area of radius } R_1 \text{ sphere}}}$$

$$\therefore \frac{\sigma_1}{\sigma_2} = \frac{q_1}{q_2} \cdot \frac{R_2^2}{R_1^2} = \frac{R_2}{R_1}$$

$\therefore$ If $R_1 < R_2$, then $\sigma_1 > \sigma_2$

And the surface electric field $E_1 > E_2$

For arbitrary shape conductor:



At every point on the conductor, we fit a *circle*. The radius of this circle is the *radius of curvature*.

$$E_3 < E_2 < E_1$$

Note: Charge distribution on a conductor does **not** have to be uniform.

Chapter 5

Capacitance and DC Circuits

5.1 Capacitors

A **capacitor** is a system of *two conductors* that carries *equal and opposite charges*. A capacitor *stores charge and energy* in the form of electro-static field.

We define **capacitance** as

$$C = \frac{Q}{V} \quad \text{Unit: Farad(F)}$$

where

Q = Charge on one plate

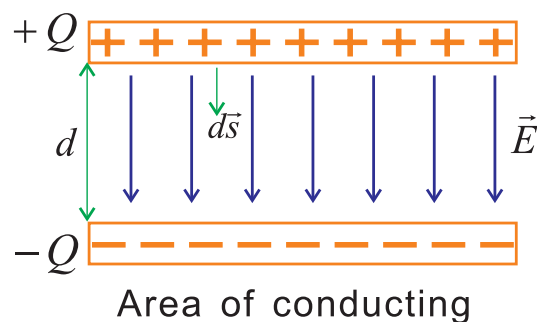
V = Potential difference between the plates

Note: The C of a capacitor is a *constant* that depends only on its shape and material.

i.e. If we increase V for a capacitor, we can increase Q stored.

5.2 Calculating Capacitance

5.2.1 Parallel-Plate Capacitor



(1) Recall from Chapter 3 note,

$$|\vec{E}| = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}$$

(2) Recall from Chapter 4 note,

$$\Delta V = V_+ - V_- = - \int_-^+ \vec{E} \cdot d\vec{s}$$

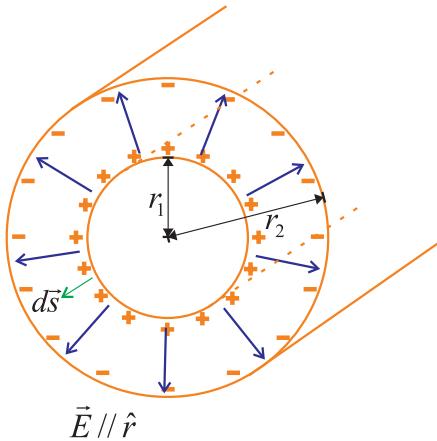
Again, notice that this integral is independent of the path taken.

$\therefore$ We can take the path that is parallel to the $\vec{E}$ -field.

$$\begin{aligned} \therefore \Delta V &= \int_+^- \vec{E} \cdot d\vec{s} \\ &= \int_+^- E \cdot ds \\ &= \frac{Q}{\epsilon_0 A} \underbrace{\int_+^- ds}_{\text{Length of path taken}} \\ &= \frac{Q}{\epsilon_0 A} \cdot d \end{aligned}$$

$$(3) \therefore \boxed{C = \frac{Q}{\Delta V} = \frac{\epsilon_0 A}{d}}$$

5.2.2 Cylindrical Capacitor



Consider two concentric cylindrical wire of inner and outer radii r_1 and r_2 respectively. The length of the capacitor is L where $r_1 < r_2 \ll L$.

- (1) Using Gauss' Law, we determine that the E-field between the conductors is (cf. Chap3 note)

$$\vec{E} = \frac{1}{2\pi\epsilon_0} \cdot \frac{\lambda}{r} \hat{r} = \frac{1}{2\pi\epsilon_0} \cdot \frac{Q}{Lr} \hat{r}$$

where λ is charge per unit length

- (2)

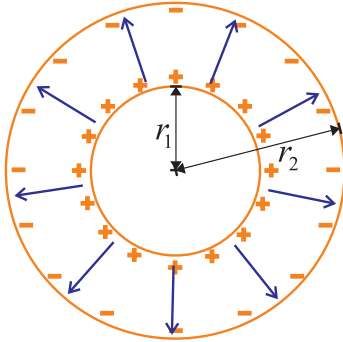
$$\Delta V = \int_+^- \vec{E} \cdot d\vec{s}$$

Again, we choose the path of integration so that $d\vec{s} \parallel \hat{r} \parallel \vec{E}$

$$\therefore \Delta V = \int_{r_1}^{r_2} E dr = \frac{Q}{2\pi\epsilon_0 L} \underbrace{\int_{r_1}^{r_2} \frac{dr}{r}}_{\ln(\frac{r_2}{r_1})}$$

$$\therefore \boxed{C = \frac{Q}{\Delta V} = 2\pi\epsilon_0 \frac{L}{\ln(r_2/r_1)}}$$

5.2.3 Spherical Capacitor



$\vec{E} \parallel \hat{r}$

Choose $d\vec{s} \parallel \hat{r}$

For the space between the two conductors,

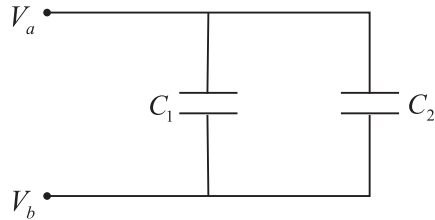
$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{r^2}; \quad r_1 < r < r_2$$

$$\begin{aligned} \Delta V &= \int_+^- \vec{E} \cdot d\vec{s} \\ \text{Choose } d\vec{s} \parallel \hat{r} &= \int_{r_1}^{r_2} \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{r^2} dr \\ &= \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{r_1} - \frac{1}{r_2} \right] \end{aligned}$$

$$\boxed{C = 4\pi\epsilon_0 \left[\frac{r_1 r_2}{r_2 - r_1} \right]}$$

5.3 Capacitors in Combination

(a) Capacitors in Parallel



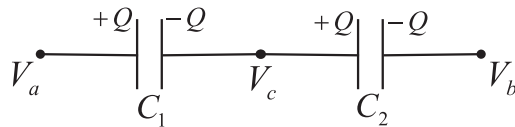
In this case, it's the *potential difference* $V = V_a - V_b$ that is the same across the capacitor.

BUT: Charge on each capacitor different

$$\begin{aligned} \text{Total charge } Q &= Q_1 + Q_2 \\ &= C_1 V + C_2 V \\ Q &= \underbrace{(C_1 + C_2)}_{\text{Equivalent capacitance}} V \end{aligned}$$

$\therefore$ For capacitors in parallel: $C = C_1 + C_2$

(b) Capacitors in Series



The *charge across capacitors* are the same.

BUT: Potential difference (P.D.) across capacitors different

$$\begin{aligned} \Delta V_1 &= V_a - V_c = \frac{Q}{C_1} && \text{P.D. across } C_1 \\ \Delta V_2 &= V_c - V_b = \frac{Q}{C_2} && \text{P.D. across } C_2 \end{aligned}$$

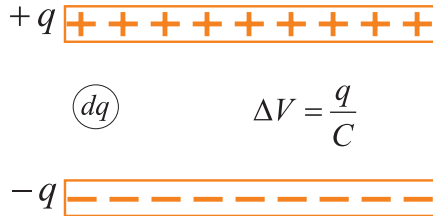
$\therefore$ Potential difference

$$\begin{aligned} \Delta V &= V_a - V_b \\ &= \Delta V_1 + \Delta V_2 \\ \Delta V &= Q \left(\frac{1}{C_1} + \frac{1}{C_2} \right) = \frac{Q}{C} \end{aligned}$$

where C is the **Equivalent Capacitance**

$$\therefore \boxed{\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}}$$

5.4 Energy Storage in Capacitor



In charging a capacitor, *positive charge* is being moved from the *negative plate* to the *positive plate*.
 $\Rightarrow$ NEEDS WORK DONE!

Suppose we move charge dq from $-ve$ to $+ve$ plate, *change in potential energy*

$$dU = \Delta V \cdot dq = \frac{q}{C} dq$$

Suppose we keep putting in a total charge Q to the capacitor, the *total potential energy*

$$U = \int dU = \int_0^Q \frac{q}{C} dq$$

$$\therefore \boxed{U = \frac{Q^2}{2C} = \frac{1}{2} C \Delta V^2} \quad (\because Q = C \Delta V)$$

The energy stored in the capacitor is stored in the **electric field** between the plates.

Note : In a parallel-plate capacitor, the *E-field is constant between the plates*.

$\therefore$ We can consider the E-field energy

$$\text{density } u = \frac{\text{Total energy stored}}{\text{Total volume with E-field}}$$

$$\therefore u = \frac{U}{\underbrace{Ad}_{\text{Rectangular volume}}}$$

Recall

$$\begin{cases} C = \frac{\epsilon_0 A}{d} \\ E = \frac{\Delta V}{d} \end{cases} \Rightarrow \Delta V = Ed$$

$$\therefore u = \frac{1}{2} \left(\overbrace{\frac{\epsilon_0 A}{d}}^C \right) \cdot \left(\overbrace{Ed}^{(\Delta V)^2} \right)^2 \cdot \overbrace{\frac{1}{Ad}}^{\frac{1}{\text{Volume}}}$$

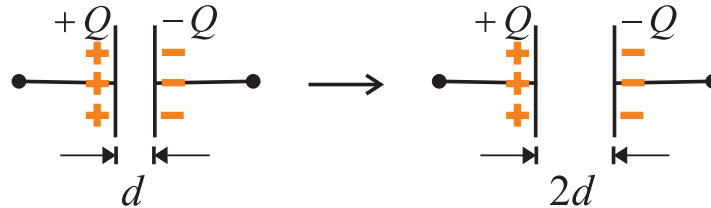
$$\boxed{u = \frac{1}{2} \epsilon_0 E^2}$$

↑

Energy per unit volume
of the electrostatic field

can be generally applied

Example : Changing capacitance



(1) Isolated Capacitor:

Charge on the capacitor plates remains constant.

BUT: $C_{new} = \frac{\epsilon_0 A}{2d} = \frac{1}{2} C_{old}$

$$\therefore U_{new} = \frac{Q^2}{2C_{new}} = \frac{Q^2}{2C_{old}/2} = 2U_{old}$$

$\therefore$ In pulling the plates apart, work done $W > 0$

Summary :

$(V = \frac{Q}{C}) \Rightarrow$	$\frac{Q}{2} \rightarrow Q$	$V \rightarrow 2V$	$C \rightarrow C/2$	
	$\frac{1}{2} \epsilon_0 E^2 =$	$u \rightarrow u$	$E \rightarrow E$	$(E = \frac{V}{d})$
			$U \rightarrow 2U$	$(U = u \cdot \text{volume})$

(2) Capacitor connected to a battery:

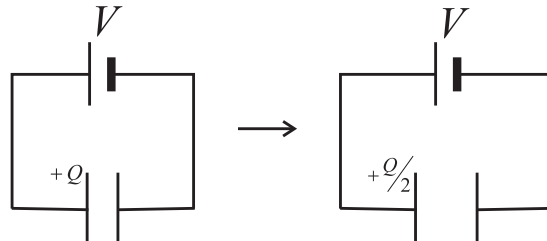
Potential difference between capacitor plates remains constant.

$$U_{new} = \frac{1}{2} C_{new} \Delta V^2 = \frac{1}{2} \cdot \frac{1}{2} C_{old} \Delta V^2 = \frac{1}{2} U_{old}$$

$\therefore$ In pulling the plates apart, work done by battery < 0

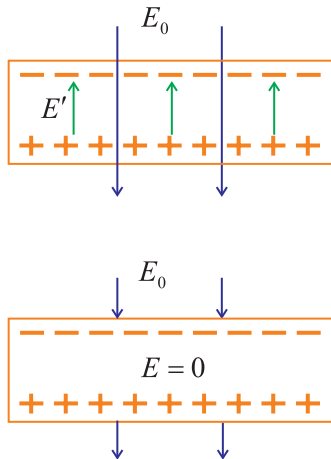
Summary :

$Q \rightarrow Q/2$	$C \rightarrow C/2$
$V \rightarrow V$	$E \rightarrow E/2$
$u \rightarrow u/4$	$U \rightarrow U/2$



5.5 Dielectric Constant

We first recall the case for a *conductor* being placed in an *external E-field* E_0 .



In a conductor, charges are free to move inside so that the *internal E-field* E' set up by these charges

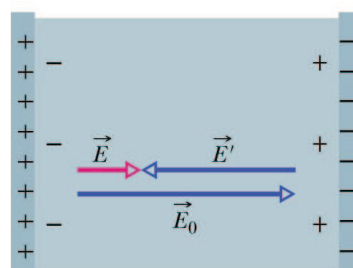
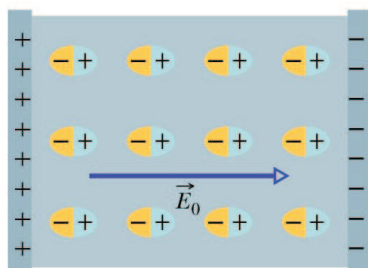
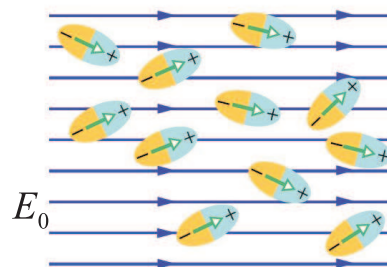
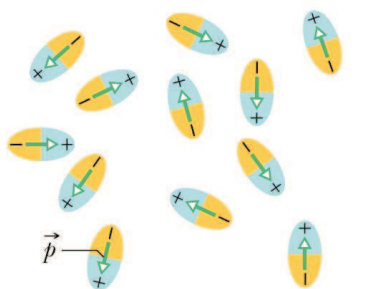
$$E' = -E_0$$

so that E-field inside conductor = 0.

Generally, for **dielectric**, the atoms and molecules behave like a **dipole** in an E-field.



Or, we can envision this so that in the absence of E-field, the *direction of dipole in the dielectric* are randomly distributed.



The aligned dipoles will generate an *induced E-field* E' , where $|E'| < |E_0|$.
We can observe the aligned dipoles in the form of *induced surface charge*.

Dielectric Constant : When a dielectric is placed in an external E-field E_0 , the E-field inside a dielectric is *induced*.
E-field in dielectric

$$E = \frac{1}{K_e} E_0$$

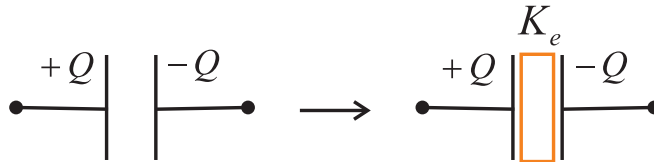
$$K_e = \text{dielectric constant} \geq 1$$

Example :

Vacuum	$K_e = 1$
Porcelain	$K_e = 6.5$
Water	$K_e \sim 80$
Perfect conductor	$K_e = \infty$
Air	$K_e = 1.00059$

5.6 Capacitor with Dielectric

Case I :



Again, the *charge remains constant* after dielectric is inserted.

BUT: $E_{new} = \frac{1}{K_e} E_{old}$

$$\therefore \Delta V = Ed \Rightarrow \Delta V_{new} = \frac{1}{K_e} \Delta V_{old}$$

$$\therefore C = \frac{Q}{\Delta V} \Rightarrow C_{new} = K_e C_{old}$$

For a parallel-plate capacitor with dielectric:

$$C = \frac{K_e \epsilon_0 A}{d}$$

We can also write $C = \frac{\epsilon A}{d}$ in general with

$$\epsilon = K_e \epsilon_0 \quad (\text{called } \mathbf{\textit{permittivity of dielectric}})$$

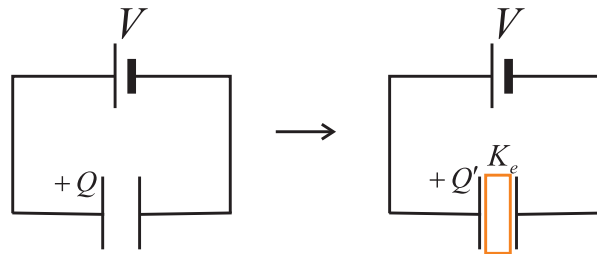
(Recall $\epsilon_0 = \mathbf{\textit{Permittivity of free space}}$)

$$\text{Energy stored } U = \frac{Q^2}{2C};$$

$$\therefore U_{new} = \frac{1}{K_e} U_{old} < U_{old}$$

$\therefore \text{Work done in inserting dielectric} < 0$

Case II : Capacitor connected to a battery



Voltage across capacitor plates *remains constant* after insertion of dielectric.

In both scenarios, the E-field inside capacitor remains constant
($\because E = V/d$)

BUT: How can E-field remain constant?

ANSWER: By having extra charge on capacitor plates.

Recall: For conductors,

$$E = \frac{\sigma}{\epsilon_0} \quad (\text{Chapter 3 note})$$

$$\Rightarrow E = \frac{Q}{\epsilon_0 A} \quad (\sigma = \text{charge per unit area} = Q/A)$$

After insertion of dielectric:

$$E' = \frac{E}{K_e} = \frac{Q'}{K_e \epsilon_0 A}$$

But E-field remains constant!

$$\therefore E' = E \Rightarrow \frac{Q'}{K_e \epsilon_0 A} = \frac{Q}{\epsilon_0 A}$$

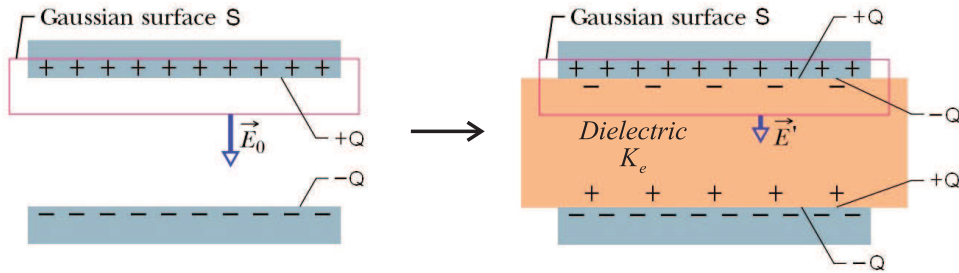
$$\Rightarrow Q' = K_e Q > Q$$

$$\begin{array}{lll}
\therefore \text{ Capacitor} & C = Q/V & \Rightarrow C' \rightarrow K_e C \\
\text{Energy stored} & U = \frac{1}{2} CV^2 & \Rightarrow U' \rightarrow K_e U \\
(\text{i.e. } U_{\text{new}} > U_{\text{old}}) & &
\end{array}$$

$$\therefore \text{ Work done to insert dielectric } > 0$$

5.7 Gauss' Law in Dielectric

The Gauss' Law we've learned is applicable in *vacuum only*. Let's use the capacitor as an example to examine Gauss' Law in dielectric.



Free charge on plates	$\pm Q$	$\pm Q$
Induced charge on dielectric	0	$\mp Q'$

$$\begin{array}{ll}
\text{Gauss' Law} & \text{Gauss' Law:} \\
\oint_S \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0} & \oint_S \vec{E}' \cdot d\vec{A} = \frac{Q - Q'}{\epsilon_0} \\
\Rightarrow E_0 = \frac{Q}{\epsilon_0 A} & \therefore E' = \frac{Q - Q'}{\epsilon_0 A} \quad (2)
\end{array} \quad (1)$$

However, we define $E' = \frac{E_0}{K_e}$ (3)

From (1), (2), (3) $\therefore \frac{Q}{K_e \epsilon_0 A} = \frac{Q}{\epsilon_0 A} - \frac{Q'}{\epsilon_0 A}$

$$\therefore \text{ Induced charge density } \sigma' = \frac{Q'}{A} = \sigma \left(1 - \frac{1}{K_e} \right) < \sigma$$

where σ is free charge density.

Recall Gauss' Law in Dielectric:

$$\begin{array}{ccccc}
\epsilon_0 \oint_S \vec{E}' \cdot d\vec{A} & = & Q & - & Q' \\
\uparrow & & \uparrow & & \uparrow \\
\text{E-field in dielectric} & & \text{free charge} & & \text{induced charge}
\end{array}$$

$$\begin{aligned}\Rightarrow \epsilon_0 \oint_S \vec{E}' \cdot d\vec{A} &= Q - Q\left[1 - \frac{1}{K_e}\right] \\ \Rightarrow \epsilon_0 \oint_S \vec{E}' \cdot d\vec{A} &= \frac{Q}{K_e}\end{aligned}$$

$$\boxed{\oint_S K_e \vec{E}' \cdot d\vec{A} = \frac{Q}{\epsilon_0}} \quad \begin{array}{l} \text{Gauss' Law} \\ \text{in dielectric} \end{array}$$

Note :

- (1) This goes back to the Gauss' Law in vacuum with $E = \frac{E_0}{K_e}$ for dielectric
- (2) Only *free charges* need to be considered, even for dielectric where there are *induced charges*.
- (3) Another way to write:

$$\oint_S \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon}$$

where $\vec{E}$ is E-field in dielectric, $\epsilon = K_e \epsilon_0$ is Permittivity

Energy stored with dielectric:

Total energy stored: $U = \frac{1}{2} CV^2$

With dielectric, recall $C = \frac{K_e \epsilon_0 A}{d}$

$$V = Ed$$

$\therefore$ Energy stored per unit volume:

$$\boxed{u_e = \frac{U}{Ad} = \frac{1}{2} K_e \epsilon_0 E^2}$$

$$\text{and } u_{\text{dielectric}} = K_e u_{\text{vacuum}}$$

$\therefore$ More energy is stored per unit volume in dielectric than in vacuum.

5.8 Ohm's Law and Resistance

ELECTRIC CURRENT is defined as the flow of electric charge through a cross-sectional area.

$$\boxed{i = \frac{dQ}{dt}} \quad \begin{array}{l} \text{Unit: Ampere (A)} \\ = \text{C/second} \end{array}$$

Convention :

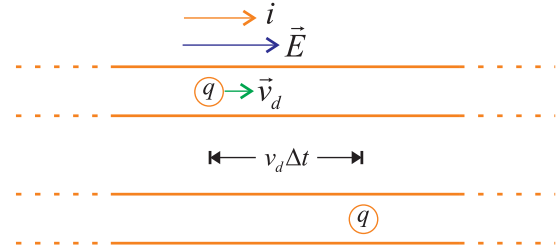
- (1) Direction of current is the direction of *flow of positive charge*.
- (2) Current is NOT a vector, but the **current density** is a **vector**.

$\vec{j}$ = charge flow per unit time per unit area

$$\boxed{i = \int \vec{j} \cdot d\vec{A}}$$

Drift Velocity :

Consider a current i flowing through a cross-sectional area A :



$\therefore$ In time Δt , total charges passing through segment:

$$\Delta Q = q \underbrace{A(V_d \Delta t)}_{\text{Volume of charge passing through}} n$$

where q is charge of the current carrier, n is density of charge carrier per unit volume

$$\therefore \text{Current: } \boxed{i = \frac{\Delta Q}{\Delta t} = nqAv_d}$$

$$\text{Current Density: } \boxed{\vec{j} = nq\vec{v}_d}$$

Note : For metal, the charge carriers are the free electrons inside.

$\therefore \vec{j} = -ne\vec{v}_d$ for metals

$\therefore$ Inside metals, $\vec{j}$ and $\vec{v}_d$ are in *opposite direction*.

We define a general property, **conductivity** (σ), of a material as:

$$\boxed{\vec{j} = \sigma \vec{E}}$$

Note : In general, σ is NOT a constant number, but rather a *function of position and applied E-field*.

A more commonly used property, **resistivity** (ρ), is defined as $\rho = \frac{1}{\sigma}$

$$\therefore \vec{E} = \rho \vec{j}$$

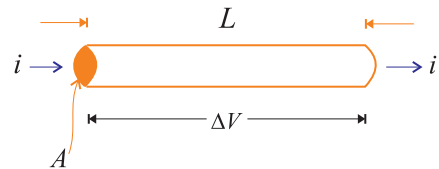
Unit of ρ : Ohm-meter (Ωm)
where Ohm (Ω) = Volt/Ampere

OHM'S LAW:

Ohmic materials have resistivity that are *independent of the applied electric field*.
i.e. metals (in not too high E-field)

Example :

Consider a **resistor** (ohmic material) of length L and cross-sectional area A .



$\therefore$ Electric field inside conductor:

$$\Delta V = \int \vec{E} \cdot d\vec{s} = E \cdot L \Rightarrow E = \frac{\Delta V}{L}$$

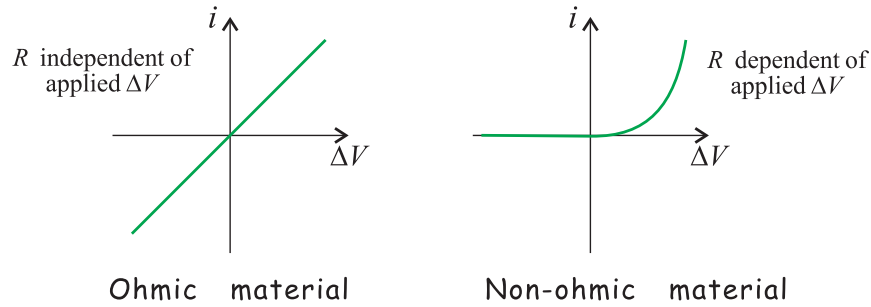
Current density: $j = \frac{i}{A}$

$$\begin{aligned} \therefore \rho &= \frac{E}{j} \\ \rho &= \frac{\Delta V}{L} \cdot \frac{1}{i/A} \end{aligned}$$

$$\boxed{\frac{\Delta V}{i} = R = \rho \frac{L}{A}}$$

where R is the **resistance** of the conductor.

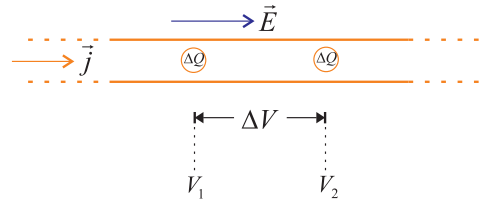
Note: $\Delta V = iR$ is NOT a statement of Ohm's Law. It's just a definition for resistance.



(Read Chap. 29-4 of Halliday Vol 2)

ENERGY IN CURRENT:

Assuming a charge ΔQ enters with potential V_1 and leaves with potential V_2 :



$\therefore$ Potential energy lost in the wire:

$$\begin{aligned}\Delta U &= \Delta Q V_2 - \Delta Q V_1 \\ \Delta U &= \Delta Q (V_2 - V_1)\end{aligned}$$

$\therefore$ Rate of energy lost per unit time

$$\frac{\Delta U}{\Delta t} = \frac{\Delta Q}{\Delta t} (V_2 - V_1)$$

Joule's heating

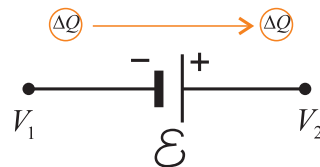
$$P = i \cdot \Delta V = \text{Power dissipated in conductor}$$

For a resistor R , $P = i^2 R = \frac{\Delta V^2}{R}$

5.9 DC Circuits

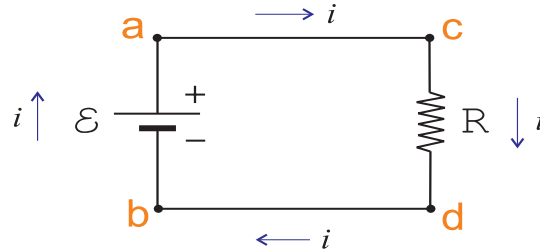
A **battery** is a device that *supplies electrical energy* to maintain a current in a circuit.

In moving from point 1 to 2, electric potential energy increase by $\Delta U = \Delta Q (V_2 - V_1) = \text{Work done by } \mathcal{E}$



Define $\mathcal{E} = \text{Work done/charge} = V_2 - V_1$

Example :



$$\left. \begin{array}{l} V_a = V_c \\ V_b = V_d \end{array} \right\} \text{ assuming }^{(1)} \text{ perfect conducting wires.}$$

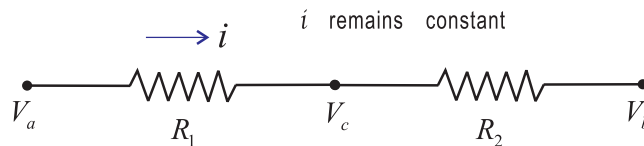
$$\text{By Definition: } V_c - V_d = iR$$

$$V_a - V_b = \mathcal{E}$$

$$\therefore \mathcal{E} = iR \Rightarrow i = \frac{\mathcal{E}}{R}$$

Also, we have assumed⁽²⁾ zero resistance inside battery.

Resistance in combination :

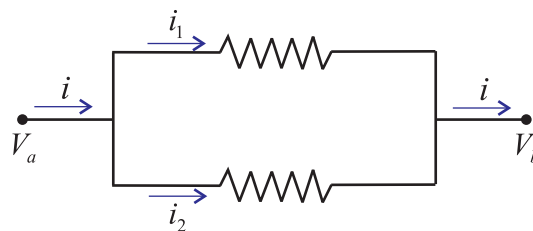


Potential difference (P.D.)

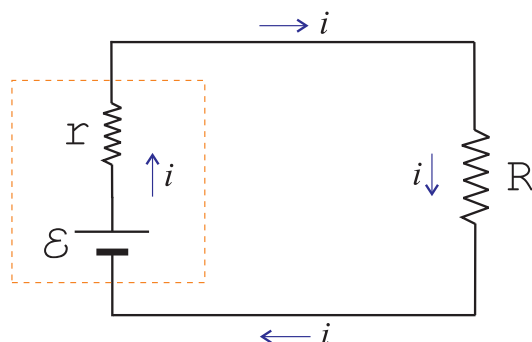
$$\begin{aligned} V_a - V_b &= (V_a - V_c) + (V_c - V_b) \\ &= iR_1 + iR_2 \end{aligned}$$

$\therefore$ Equivalent Resistance

$$\begin{aligned} R &= R_1 + R_2 && \text{for resistors in series} \\ \frac{1}{R} &= \frac{1}{R_1} + \frac{1}{R_2} && \text{for resistors in parallel} \end{aligned}$$



Example :



For real battery, there is an **internal resistance** that we cannot ignore.

$$\begin{aligned}\therefore \mathcal{E} &= i(R + r) \\ i &= \frac{\mathcal{E}}{R + r}\end{aligned}$$

Joule's heating in resistor R :

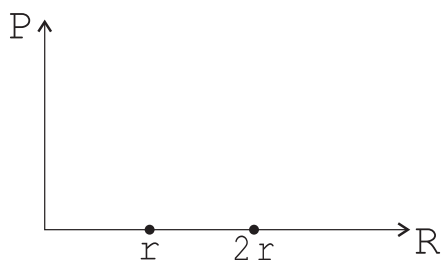
$$\begin{aligned}P &= i \cdot (\text{P.D. across resistor } R) \\ &= i^2 R \\ P &= \frac{\mathcal{E}^2 R}{(R + r)^2}\end{aligned}$$

Question: What is the value of R to obtain *maximum* Joule's heating?

Answer: We want to find R to *maximize* P .

$$\frac{dP}{dR} = \frac{\mathcal{E}^2}{(R + r)^2} - \frac{\mathcal{E}^2 2R}{(R + r)^3}$$

$$\begin{aligned}\text{Setting } \frac{dP}{dR} = 0 &\Rightarrow \frac{\mathcal{E}^2}{(R + r)^3} [(R + r) - 2R] = 0 \\ &\Rightarrow r - R = 0 \\ &\Rightarrow R = r\end{aligned}$$

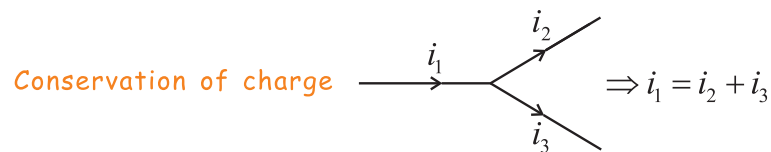


ANALYSIS OF COMPLEX CIRCUITS:

KIRCHHOFF'S LAWS:

(1) First Law (Junction Rule):

Total current entering a junction equal to the total current leaving the junction.

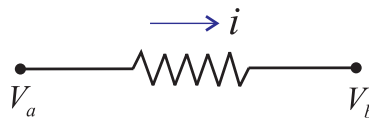


(2) Second Law (Loop Rule):

The sum of potential differences around a complete circuit loop is zero.

Convention :

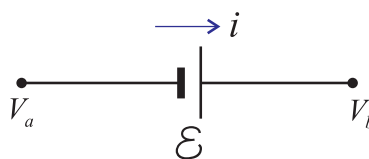
(i)



$$V_a > V_b \Rightarrow \text{Potential difference} = -iR$$

i.e. Potential *drops* across resistors

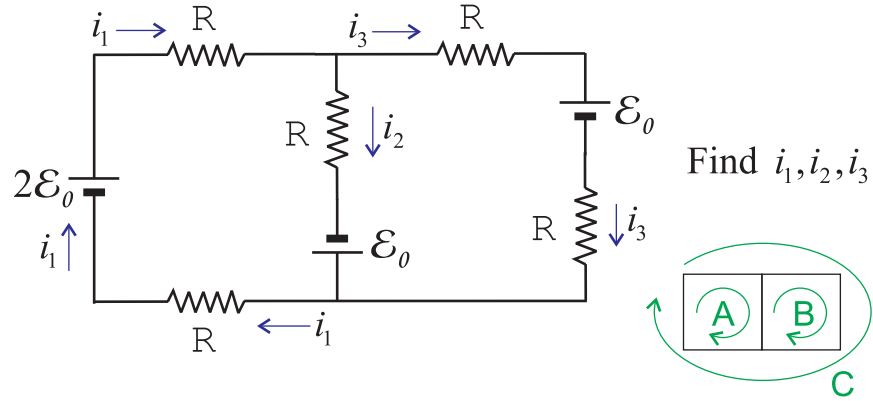
(ii)



$$V_b > V_a \Rightarrow \text{Potential difference} = +\mathcal{E}$$

i.e. Potential *rises* across the negative plate of the battery.

Example :



By junction rule:

$$i_1 = i_2 + i_3 \quad (5.1)$$

By loop rule:

$$\text{Loop A} \Rightarrow 2\mathcal{E}_0 - i_1 R - i_2 R + \mathcal{E}_0 - i_1 R = 0 \quad (5.2)$$

$$\text{Loop B} \Rightarrow -i_3 R - \mathcal{E}_0 - i_3 R - \mathcal{E}_0 + i_2 R = 0 \quad (5.3)$$

$$\text{Loop C} \Rightarrow 2\mathcal{E}_0 - i_1 R - i_3 R - \mathcal{E}_0 - i_3 R - i_1 R = 0 \quad (5.4)$$

BUT: $(5.4) = (5.2) + (5.3)$

General rule: Need only 3 equations for 3 current

$$i_1 = i_2 + i_3 \quad (5.1)$$

$$3\mathcal{E}_0 - 2i_1 R - i_2 R = 0 \quad (5.2)$$

$$-2\mathcal{E}_0 + i_2 R - 2i_3 R = 0 \quad (5.3)$$

Substitute (5.1) into (5.2) :

$$\begin{aligned} 3\mathcal{E}_0 - 2(i_2 + i_3)R - i_2 R &= 0 \\ \Rightarrow 3\mathcal{E}_0 - 3i_2 R - 2i_3 R &= 0 \end{aligned} \quad (5.4)$$

Subtract (5.3) from (5.4), i.e. $(5.4) - (5.3)$

$$3\mathcal{E}_0 - (-2\mathcal{E}_0) - 3i_2 R - i_2 R = 0$$

$$\Rightarrow \boxed{i_2 = \frac{5}{4} \cdot \frac{\mathcal{E}_0}{R}}$$

Substitute i_2 into (5.3) :

$$-2\mathcal{E}_0 + \left(\frac{5}{4} \cdot \frac{\mathcal{E}_0}{R}\right)R - 2i_3 R = 0$$

$$\Rightarrow \boxed{i_3 = -\frac{3}{8} \cdot \frac{\mathcal{E}_0}{R}}$$

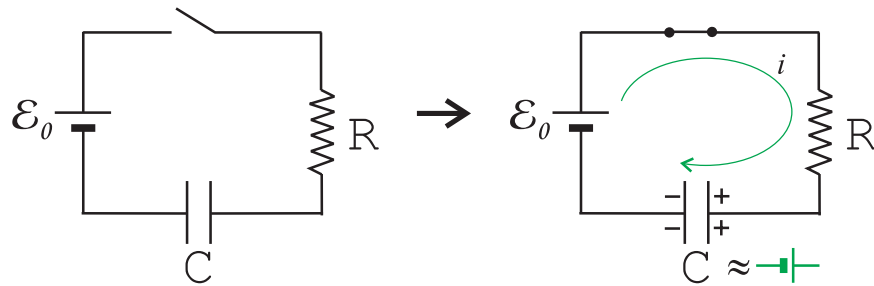
Substitute i_2, i_3 into (5.1) :

$$\boxed{i_1 = \left(\frac{5}{4} - \frac{3}{8}\right) \frac{\mathcal{E}_0}{R} = \frac{7}{8} \cdot \frac{\mathcal{E}_0}{R}}$$

Note: A *negative* current means that it is flowing in *opposite direction* from the one assumed.

5.10 RC Circuits

(A) *Charging* a capacitor with battery:



Using the loop rule:

$$+\mathcal{E}_0 - \underbrace{iR}_{\substack{\text{P.D.} \\ \text{across } R}} - \underbrace{\frac{Q}{C}}_{\substack{\text{P.D.} \\ \text{across } C}} = 0$$

Note: Direction of i is chosen so that the current represents the rate at which the charge on the capacitor is *increasing*.

$$\begin{aligned} \therefore \mathcal{E} &= R \overbrace{\frac{dQ}{dt}}^i + \frac{Q}{C} && \text{1st order differential eqn.} \\ \Rightarrow \frac{dQ}{\mathcal{E}C - Q} &= \frac{dt}{RC} \end{aligned}$$

Integrate both sides and use the initial condition:

$t = 0, \quad Q \text{ on capacitor} = 0$

$$\int_0^Q \frac{dQ}{\mathcal{E}C - Q} = \int_0^t \frac{dt}{RC}$$

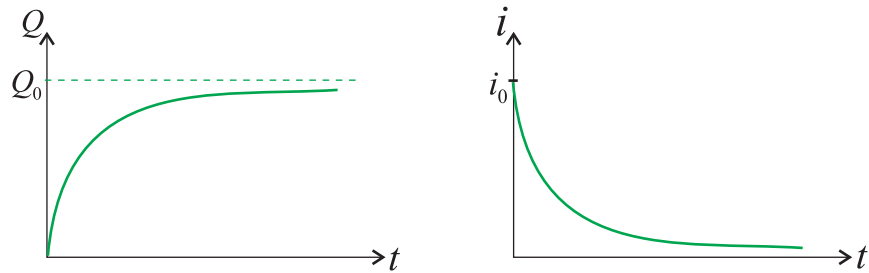
$$\begin{aligned}
& -\ln(\mathcal{E}C - Q)\Big|_0^Q = \frac{t}{RC}\Big|_0^t \\
\Rightarrow & -\ln(\mathcal{E}C - Q) + \ln(\mathcal{E}C) = \frac{t}{RC} \\
\Rightarrow & \ln\left(\frac{1}{1 - \frac{Q}{\mathcal{E}C}}\right) = \frac{t}{RC} \\
\Rightarrow & \frac{1}{1 - \frac{Q}{\mathcal{E}C}} = e^{t/RC} \\
\Rightarrow & \frac{Q}{\mathcal{E}C} = 1 - e^{-t/RC} \\
\Rightarrow & \boxed{Q(t) = \mathcal{E}C(1 - e^{-t/RC})}
\end{aligned}$$

Note: (1) At $t = 0$, $Q(t = 0) = \mathcal{E}C(1 - 1) = 0$

(2) As $t \rightarrow \infty$, $Q(t \rightarrow \infty) = \mathcal{E}C(1 - 0) = \mathcal{E}C$
= Final charge on capacitor (Q_0)

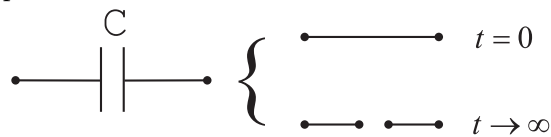
(3) Current:

$$\begin{aligned}
i &= \frac{dQ}{dt} \\
&= \mathcal{E}C\left(\frac{1}{RC}\right)e^{-t/RC} \\
i(t) &= \frac{\mathcal{E}}{R}e^{-t/RC} \\
\begin{cases} i(t = 0) &= \frac{\mathcal{E}}{R} = \text{Initial current} = i_0 \\ i(t \rightarrow \infty) &= 0 \end{cases}
\end{aligned}$$



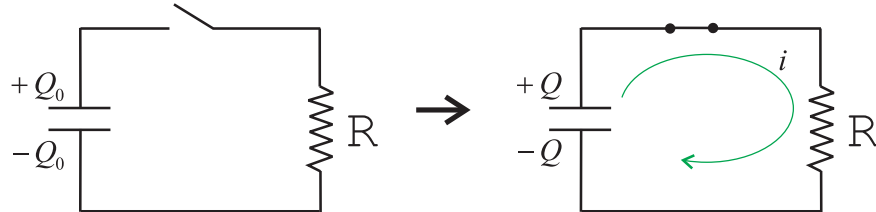
(4) At time = 0, the capacitor acts like *short circuit* when there is *zero charge on the capacitor*.

(5) As time $\rightarrow \infty$, the capacitor is *fully charged* and current = 0, it acts like a *open circuit*.



- (6) $\tau_c = RC$ is called the **time constant**. It's the time it takes for the charge to reach $(1 - \frac{1}{e}) Q_0 \simeq 0.63Q_0$

(B) *Discharging* a charged capacitor:



Note: Direction of i is chosen so that the current represents the rate at which the charge on the capacitor is *decreasing*.

$$\therefore i = -\frac{dQ}{dt}$$

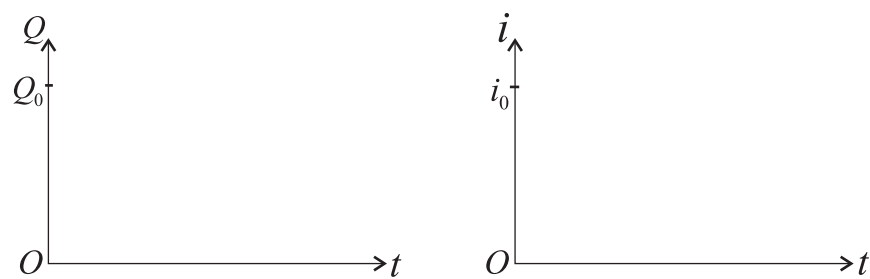
Loop Rule:

$$\begin{aligned} V_c - iR &= 0 \\ \Rightarrow \frac{Q}{C} + \frac{dQ}{dt}R &= 0 \\ \Rightarrow \frac{dQ}{dt} &= -\frac{1}{RC}Q \end{aligned}$$

Integrate both sides and use the initial condition:

$t = 0$, Q on capacitor $= Q_0$

$$\begin{aligned} \int_{Q_0}^Q \frac{dQ}{Q} &= -\frac{1}{RC} \int_0^t dt \\ \Rightarrow \ln Q - \ln Q_0 &= -\frac{t}{RC} \\ \Rightarrow \ln\left(\frac{Q}{Q_0}\right) &= -\frac{t}{RC} \\ \Rightarrow \frac{Q}{Q_0} &= e^{-t/RC} \\ \Rightarrow Q(t) &= Q_0 e^{-t/RC} \\ (i = -\frac{dQ}{dt}) \Rightarrow i(t) &= \frac{Q_0}{RC} e^{-t/RC} \\ (\text{At } t = 0) \Rightarrow i(t = 0) &= \frac{1}{R} \cdot \underbrace{\frac{Q_0}{C}}_{\text{Initial P.D. across capacitor}} \\ i_0 &= \frac{V_0}{R} \end{aligned}$$



$$\text{At } t = RC = \tau \quad Q(t = RC) = \frac{1}{e} Q_0 \simeq 0.37 Q_0$$

Chapter 6

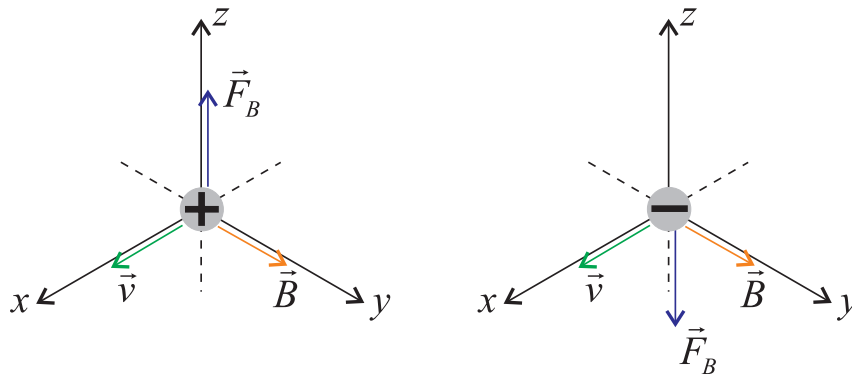
Magnetic Force

6.1 Magnetic Field

For stationary charges, they experienced an **electric force** in an **electric field**.
For moving charges, they experienced a **magnetic force** in a **magnetic field**.

$$\begin{aligned}\text{Mathematically, } \vec{F}_E &= q\vec{E} \quad (\text{electric force}) \\ \vec{F}_B &= q\vec{v} \times \vec{B} \quad (\text{magnetic force})\end{aligned}$$

Direction of the magnetic force determined from *right hand rule*.



Magnetic field $\vec{B}$: Unit = Tesla (T)
1T = 1C moving at 1m/s experiencing 1N

Common Unit: 1 Gauss (G) = 10^{-4}T $\approx$ magnetic field on earth's surface

Example: What's the force on a 0.1C charge moving at velocity $\vec{v} = (10\hat{j} - 20\hat{k})\text{ms}^{-1}$ in a magnetic field $\vec{B} = (-3\hat{i} + 4\hat{k}) \times 10^{-4}\text{T}$

$$\vec{F} = q\vec{v} \times \vec{B}$$

$$\begin{aligned}
 &= +0.1 (10\hat{j} - 20\hat{k}) \times (-3\hat{i} + 4\hat{k}) \times 10^{-4} N \\
 &= 10^{-5} (-30 \cdot -\hat{k} + 40\hat{i} + 60\hat{j} + 0) N
 \end{aligned}$$

Effects of magnetic field is usually quite small.

$$\begin{aligned}
 \vec{F} &= q\vec{v} \times \vec{B} \\
 |\vec{F}| &= qvB \sin \theta, \quad \text{where } \theta \text{ is the angle between } \vec{v} \text{ and } \vec{B}
 \end{aligned}$$

$\therefore$ Magnetic force is *maximum* when $\theta = 90^\circ$ (i.e. $\vec{v} \perp \vec{B}$)

Magnetic force is *minimum* (0) when $\theta = 0^\circ, 180^\circ$ (i.e. $\vec{v} \parallel \vec{B}$)

Graphical representation of B-field: **Magnetic field lines**

Compared with **Electric field lines**:

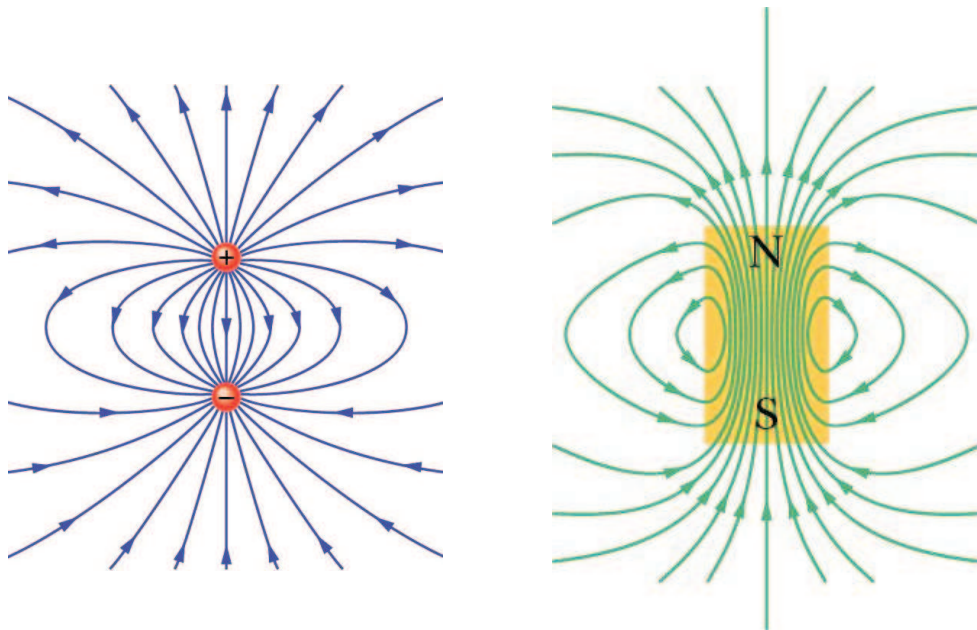
Similar characteristics :

- (1) Direction of E-field/B-field indicated by *tangent* of the field lines.
- (2) Magnitude of E-field/B-field indicated by *density* of the field lines.

Differences :

- (1) $\vec{F}_E \parallel$ E-field lines; $\vec{F}_B \perp$ B-field lines
- (2) E-field line begins at positive charge and ends at negative charge; B-field line forms a closed loop.

Example : Chap35, Pg803 Halliday

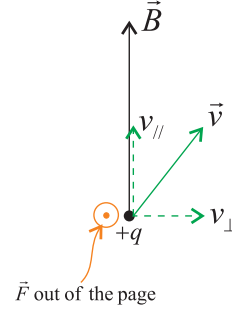


<u>Note:</u> Isolated magnetic monopoles do not exist.

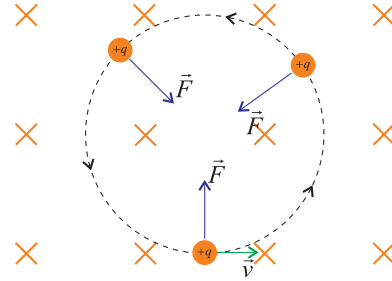
6.2 Motion of A Point Charge in Magnetic Field

Since $\vec{F}_B \perp \vec{v}$, therefore B-field only changes the *direction* of the velocity but not its *magnitude*.

Generally, $\vec{F}_B = q\vec{v} \times \vec{B} = qv_\perp B$,
 $\therefore$ We only need to consider the motion component $\perp$ to B-field.



We have *circular motion*. Magnetic force provides the *centripetal force* on the moving charge particles.



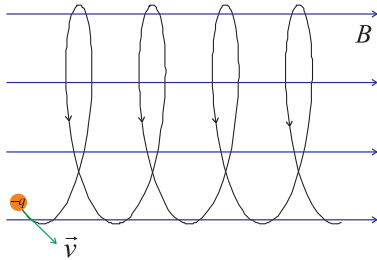
$$\begin{aligned}\therefore F_B &= m \frac{v^2}{r} \\ |q|vB &= m \frac{v^2}{r} \\ \therefore r &= \frac{mv}{|q|B}\end{aligned}$$

where r is radius of circular motion.

Time for moving around one orbit:

$$T = \frac{2\pi r}{v} = \frac{2\pi m}{qB} \quad \text{Cyclotron Period}$$

- (1) Independent of v (non-relativistic)
- (2) Use it to measure m/q



Generally, charged particles with constant velocity moves in **helix** in the presence of constant B-field.

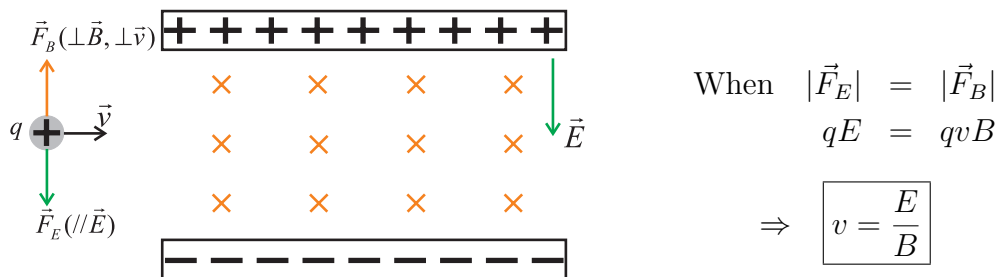
Note :

- (1) B-field does NO work on particles.
- (2) B-field does NOT change K.E. of particles.

Particle Motion in Presence of E-field & B-field:

$$\boxed{\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}} \quad \text{Lorentz Force}$$

Special Case : $\vec{E} \perp \vec{B}$



$\therefore$ For charged particles moving at $v = E/B$, they will pass through the crossed E and B fields without vertical displacement.

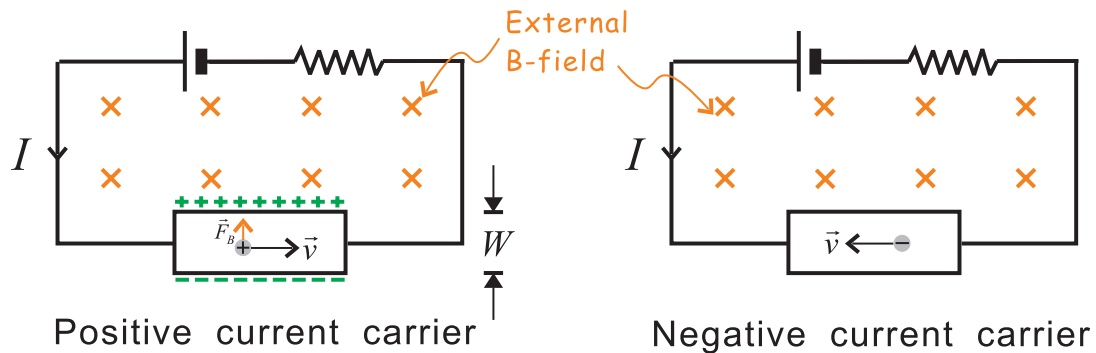
$\Rightarrow$ **velocity selector**

Applications :

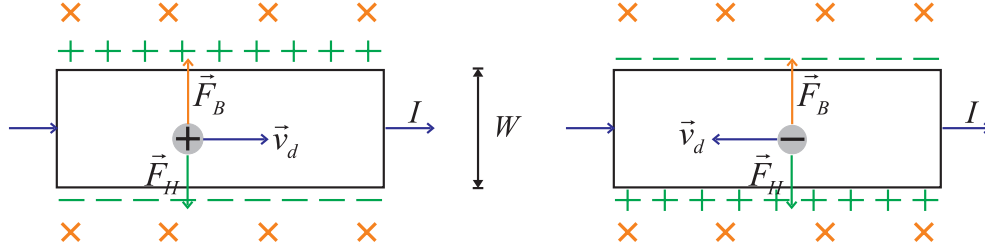
- Cyclotron (Lawrence & Livingston 1934)
- Measuring e/m for electrons (Thomson 1897)
- Mass Spectrometer (Aston 1919)

6.3 Hall Effect

Charges travelling in a conducting wire will be *pushed to one side of the wire by the external magnetic field*. This separation of charge in the wire is called the **Hall Effect**.



The separation will stop when F_B experienced by the current carrier is *balanced* by the force $\vec{F}_H$ caused by the E-field set up by the separated charges.



Define :

$$\begin{aligned}\Delta V_H &= \text{Hall Voltage} \\ &= \text{Potential difference across the conducting strip}\end{aligned}$$

$$\therefore \text{E-field from separated charges: } E_H = \frac{\Delta V_H}{W}$$

where $W = \text{width of conducting strip}$

In equilibrium: $q\vec{E}_H + q\vec{v}_d \times \vec{B} = 0$, where $\vec{v}_d$ is drift velocity

$$\therefore \frac{\Delta V_H}{W} = v_d B$$

Recall from Chapter 5,

$$i = nqAv_d$$

where n is density of charge carrier,

A is cross-sectional area = width $\times$ thickness = $W \cdot t$

$$\therefore \frac{\Delta V_H}{W} = \frac{i}{nqWt} B$$

$$\Rightarrow \boxed{n = \frac{iB}{qt\Delta V_H}} \quad \text{To determine density of charge carriers}$$

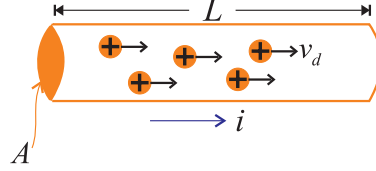
Suppose we determine n for a particular metal ($\therefore q = e$), then we can *measure B-field strength by measuring the Hall voltage*:

$$\boxed{B = \frac{net}{i} \Delta V_H}$$

6.4 Magnetic Force on Currents

Current = many charges moving together

Consider a wire segment, length L , carrying current i in a magnetic field.



$$\text{Total magnetic force} = \underbrace{(q\vec{v}_d \times \vec{B})}_{\text{force on one charge carrier}} \cdot \underbrace{nAL}_{\text{Total number of charge carrier}}$$

Recall $i = nqv_dA$

$$\therefore \boxed{\text{Magnetic force on current } \vec{F} = i\vec{L} \times \vec{B}}$$

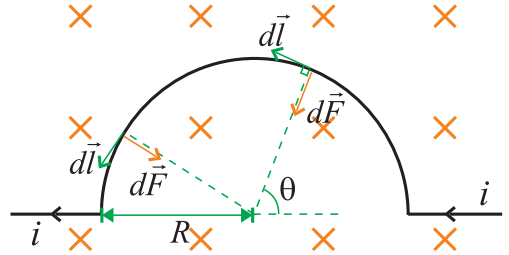
where $\vec{L}$ = Vector of which: $|\vec{L}|$ = length of current segment; direction = direction of current

For an infinitesimal wire segment $d\vec{l}$

$$\boxed{d\vec{F} = i d\vec{l} \times \vec{B}}$$

Example 1: Force on a semicircle current loop

$$\begin{aligned} d\vec{l} &= \text{Infinitesimal arc length element} \perp \vec{B} \\ \therefore dl &= R d\theta \\ \therefore dF &= iRB d\theta \end{aligned}$$



By symmetry argument, we only need to consider vertical forces, $dF \cdot \sin \theta$

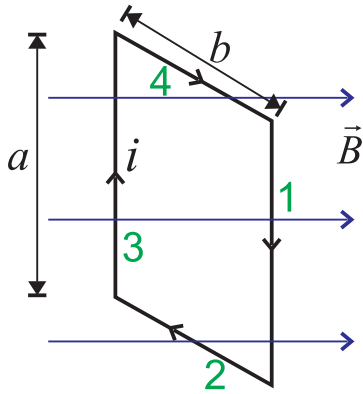
$$\begin{aligned} \therefore \text{Net force } F &= \int_0^\pi dF \sin \theta \\ &= iRB \int_0^\pi \sin \theta d\theta \\ F &= 2iRB \text{ (downward)} \end{aligned}$$

Method 2: Write $d\vec{l}$ in $\hat{i}, \hat{j}$ components

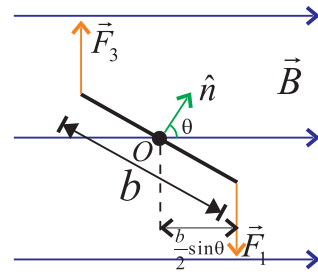
$$\begin{aligned}
 d\vec{l} &= -dl \sin \theta \hat{i} + dl \cos \theta \hat{j} \\
 &= R d\theta (-\sin \theta \hat{i} + \cos \theta \hat{j}) \\
 \vec{B} &= -B \hat{k} \quad (\text{into the page}) \\
 \therefore d\vec{F} &= i d\vec{l} \times \vec{B} \\
 &= -iRB \sin \theta d\theta \hat{j} - iRB \cos \theta \hat{i}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \vec{F} &= \int_0^\pi d\vec{F} \\
 &= -iRB \left[\int_0^\pi \sin \theta d\theta \hat{j} + \int_0^\pi \cos \theta d\theta \hat{i} \right] \\
 &= -2iRB \hat{j}
 \end{aligned}$$

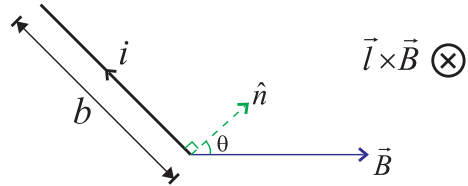
Example 2: Current loop in B-field



View from top

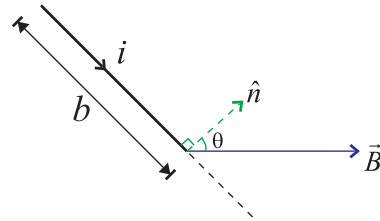


For segment 2:



$$F_2 = ibB \sin(90^\circ + \theta) = ibB \cos \theta \quad (\text{pointing downward})$$

For segment 4:



$$F_2 = ibB \sin(90^\circ - \theta) = ibB \cos \theta \quad (\text{pointing upward})$$

For segment1: $F_1 = iaB$

For segment3: $F_3 = iaB$

$\therefore$ Net force on the current loop = 0

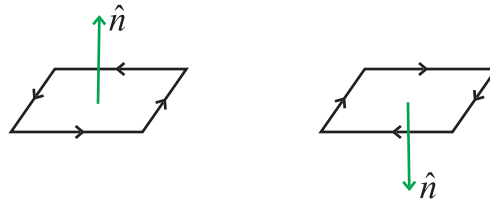
But, net torque on the loop about O

$$\begin{aligned}
 &= \tau_1 + \tau_3 \\
 &= iaB \cdot \frac{b}{2} \sin \theta + iaB \cdot \frac{b}{2} \sin \theta \\
 &= i \underbrace{ab}_A B \sin \theta \\
 &\quad A = \text{area of loop}
 \end{aligned}$$

Suppose the loop is a coil with N turns of wires:

$$\text{Total torque } \boxed{\tau = NiAB \sin \theta}$$

Define: Unit vector $\hat{n}$ to represent the area-vector (using right hand rule)



Then we can rewrite the torque equation as

$$\boxed{\vec{\tau} = NiA \hat{n} \times \vec{B}}$$

Define: $NiA \hat{n} = \vec{\mu}$ = Magnetic dipole moment of loop

$$\boxed{\vec{\tau} = \vec{\mu} \times \vec{B}}$$

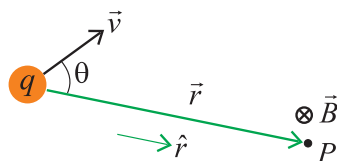
Chapter 7

Magnetic Field

7.1 Magnetic Field

A moving charge $\left\{ \begin{array}{l} \text{experiences magnetic force in B-field.} \\ \text{can generate B-field.} \end{array} \right.$

Magnetic field $\vec{B}$ due to moving point charge:

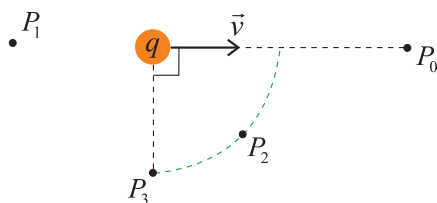


$$\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{q\vec{v} \times \hat{r}}{r^2} = \frac{\mu_0}{4\pi} \cdot \frac{q\vec{v} \times \vec{r}}{r^3}$$

where $\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A (N/A}^2\text{)}$

Permeability of free space (Magnetic constant)

$$|\vec{B}| = \frac{\mu_0}{4\pi} \cdot \frac{qv \sin \theta}{r^2} \quad \left\{ \begin{array}{ll} \text{maximum} & \text{when } \theta = 90^\circ \\ \text{minimum} & \text{when } \theta = 0^\circ/180^\circ \end{array} \right.$$

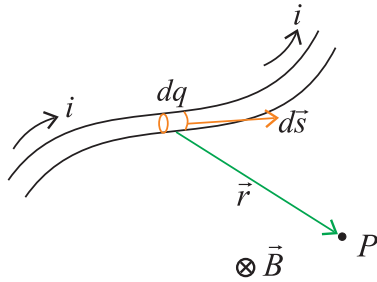


$$\begin{array}{l} \vec{B} \text{ at } P_0 = 0 = \vec{B} \text{ at } P_1 \\ \vec{B} \text{ at } P_2 < \vec{B} \text{ at } P_3 \end{array}$$

However, a single moving charge will NOT generate a steady magnetic field.

stationary charges generate steady E-field.

steady currents generate steady B-field.



Magnetic field at point P can be obtained by *integrating* the contribution from individual current segments.
(**Principle of Superposition**)

$$\therefore d\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{dq \vec{v} \times \hat{r}}{r^2}$$

Notice: $dq \vec{v} = dq \cdot \frac{d\vec{s}}{dt} = i d\vec{s}$

$$\boxed{d\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{i d\vec{s} \times \hat{r}}{r^2}} \quad \text{Biot-Savart Law}$$

For current around a whole circuit:

$$\vec{B} = \int_{\text{entire circuit}} d\vec{B} = \int_{\text{entire circuit}} \frac{\mu_0}{4\pi} \cdot \frac{i d\vec{s} \times \hat{r}}{r^2}$$

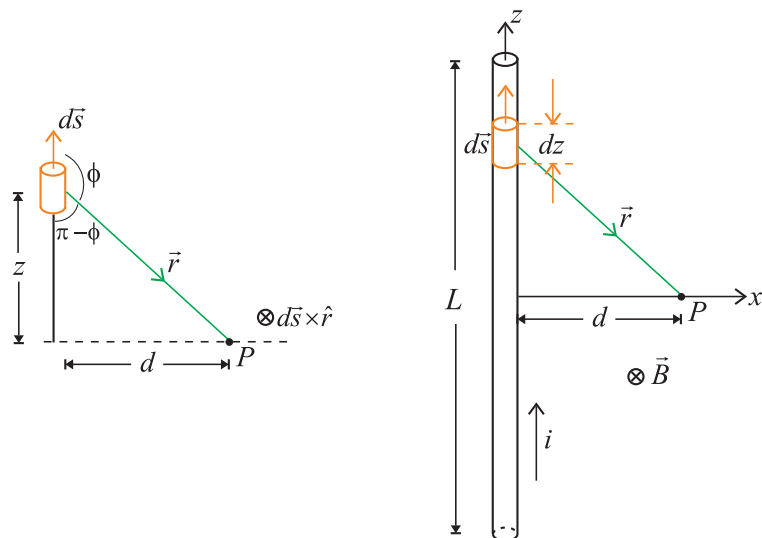
Biot-Savart Law is to *magnetic field* as

Coulomb's Law is to *electric field*.

Basic element of E-field: *Electric charges* dq

Basic element of B-field: *Current element* $i d\vec{s}$

Example 1 : Magnetic field due to straight current segment



$$\begin{aligned}
\therefore |\vec{d\vec{s}} \times \hat{r}| &= dz \sin \phi \\
&= dz \sin(\pi - \phi) \quad (\text{Trigonometry Identity}) \\
&= dz \cdot \frac{d}{r} = \frac{d \cdot dz}{\sqrt{d^2 + z^2}}
\end{aligned}$$

$$dB = \frac{\mu_0}{4\pi} \cdot \frac{i dz}{r^2} \cdot \frac{d}{r} = \frac{\mu_0 i}{4\pi} \cdot \frac{d}{(d^2 + z^2)^{3/2}} dz$$

$$\therefore B = \int_{-L/2}^{+L/2} dB = \frac{\mu_0 i d}{4\pi} \int_{-L/2}^{+L/2} \frac{dz}{(d^2 + z^2)^{3/2}}$$

$$B = \frac{\mu_0 i}{4\pi d} \cdot \frac{z}{(z^2 + d^2)^{1/2}} \Big|_{-L/2}^{+L/2}$$

$$B = \frac{\mu_0 i}{4\pi d} \cdot \frac{L}{(\frac{L^2}{4} + d^2)^{1/2}}$$

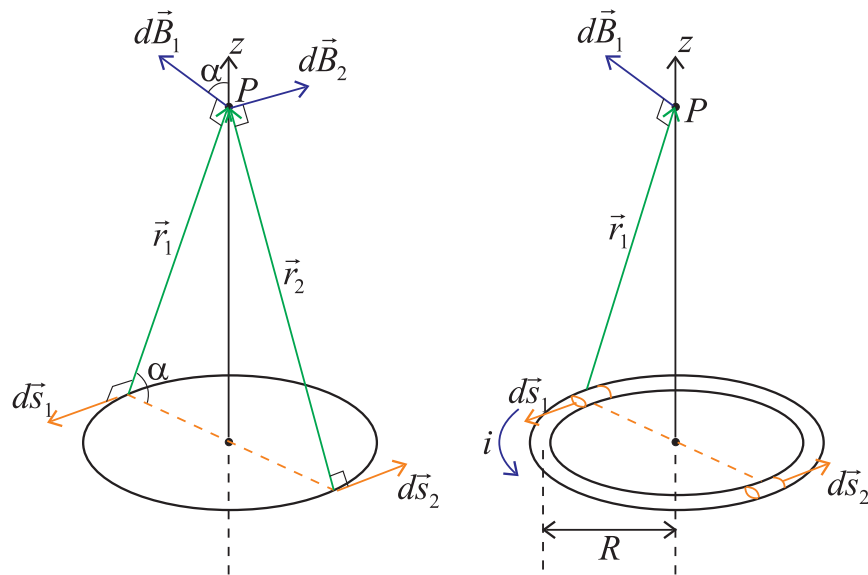
Limiting Cases : When $L \gg d$ (B-field due to long wire)

$$\left(\frac{L^2}{4} + d^2\right)^{-1/2} \approx \left(\frac{L^2}{4}\right)^{-1/2} = \frac{2}{L}$$

$$\therefore B = \frac{\mu_0 i}{2\pi d}; \quad \text{direction of B-field determined from right-hand screw rule}$$

Recall : $E = \frac{\lambda}{2\pi\epsilon_0 d}$ for an infinite long line of charge.

Example 2 : A circular current loop



Notice that for every current element $id\vec{s}_1$, generating a magnetic field $d\vec{B}_1$ at point P , there is an opposite current element $id\vec{s}_2$, generating B-field $d\vec{B}_2$ so that

$$d\vec{B}_1 \sin \alpha = -d\vec{B}_2 \sin \alpha$$

$\therefore$ Only vertical component of B-field needs to be considered at point P .

$$dB = \frac{\mu_0}{4\pi} \cdot \frac{id\vec{s} \sin \overbrace{90^\circ}^{\because d\vec{s} \perp \hat{r}}}{r^2}$$

$\therefore$ B-field at point P :

$$B = \int_{\text{around circuit}} dB \underbrace{\cos \alpha}_{\text{consider vertical component}}$$

$$\begin{aligned} \therefore B &= \int_0^{2\pi} \frac{\mu_0 i \cos \alpha}{4\pi r^2} \cdot \underbrace{ds}_{Rd\theta} \\ &= \frac{\mu_0 i}{4\pi} \cdot \frac{R}{r^3} \underbrace{\int_0^{2\pi} ds}_{\text{Integrate around circumference of circle} = 2\pi R} \\ \therefore B &= \frac{\mu_0 i R^2}{2r^3} \end{aligned}$$

$B = \frac{\mu_0 i R^2}{2(R^2 + z^2)^{3/2}} ; \quad \text{direction of B-field determined from right-hand screw rule}$
--

Limiting Cases :

(1) B-field at center of loop:

$$z = 0 \quad \Rightarrow \quad \boxed{B = \frac{\mu_0 i}{2R}}$$

(2) For $z \gg R$,

$$B = \frac{\mu_0 i R^2}{2z^3 \left(1 + \frac{R^2}{z^2}\right)^{3/2}} \approx \frac{\mu_0 i R^2}{2z^3} \propto \frac{1}{z^3}$$

Recall E-field for an electric dipole: $E = \frac{p}{4\pi\epsilon_0 x^3}$

$\therefore$ A circular current loop is also called a **magnetic dipole**.

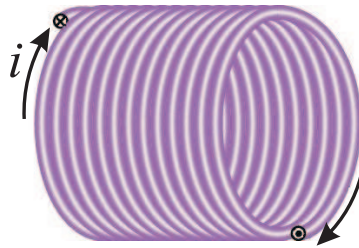
(3) A current arc:



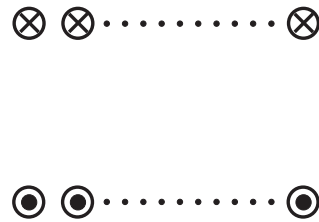
$$\begin{aligned}
 B &= \int_{\text{around circuit}} dB \underbrace{\cos \alpha}_{\substack{z=0 \Rightarrow \\ \alpha=0 \text{ here.}}} \\
 &\quad R\theta = \text{length of arc} \\
 &= \frac{\mu_0 i}{4\pi} \cdot \underbrace{\frac{R}{r^3}}_{R=r} \cdot \int_0^\theta \underbrace{ds}_{R d\theta} \\
 &\quad \text{when } \alpha = 0 \\
 B &= \frac{\mu_0 i \theta}{4\pi R}
 \end{aligned}$$

Example 3 : Magnetic field of a solenoid

Solenoid is used to produce a *strong and uniform* magnetic field inside its coils.



Solenoid



Tightly-packed coils of wire

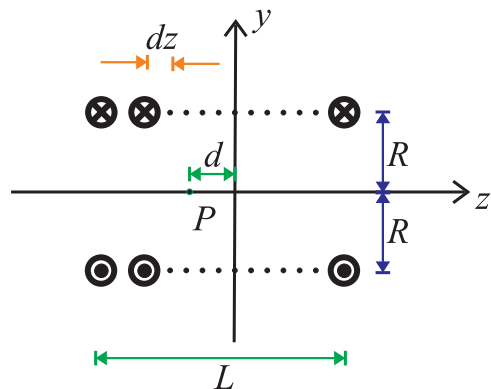
Consider a solenoid of length L consisting of N turns of wire.

Define: n = Number of turns per unit length = $\frac{N}{L}$

Consider B-field at distance d from the center of the solenoid:

For a segment of length dz , number of current turns = ndz

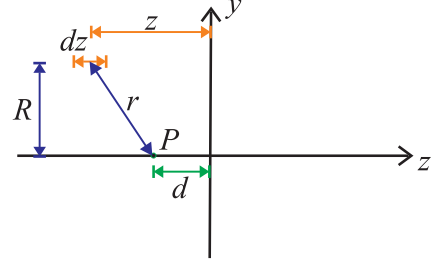
$\therefore$ Total current = $ni dz$



Using the result from one coil in Example 2, we get B-field from coils of length dz at distance z from center:

$$dB = \frac{\mu_0(ni dz)R^2}{2r^3}$$

However $r = \sqrt{R^2 + (z - d)^2}$



$$\begin{aligned} \therefore B &= \int_{-L/2}^{+L/2} dB \quad \text{(Integrating over the entire solenoid)} \\ &= \frac{\mu_0 ni R^2}{2} \int_{-L/2}^{+L/2} \frac{dz}{[R^2 + (z - d)^2]^{3/2}} \\ B &= \frac{\mu_0 ni}{2} \left[\frac{\frac{L}{2} + d}{\sqrt{R^2 + (\frac{L}{2} + d)^2}} + \frac{\frac{L}{2} - d}{\sqrt{R^2 + (\frac{L}{2} - d)^2}} \right] \\ &\quad \text{along negative } z \text{ direction} \end{aligned}$$

Ideal Solenoid :

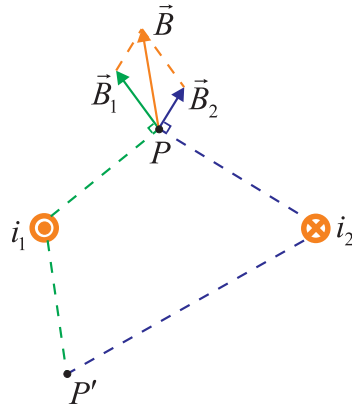
$$\begin{aligned} L &\gg R \\ \text{then } B &= \frac{\mu_0 ni}{2} [1 + 1] \end{aligned}$$

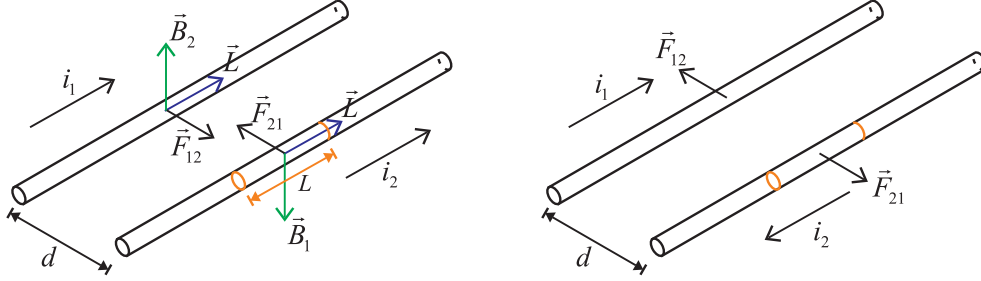
$$\therefore \boxed{B = \mu_0 ni ; \quad \text{direction of B-field determined from right-hand screw rule}}$$

Question : What is the B-field at the end of an ideal solenoid? $B = \frac{\mu_0 ni}{2}$

7.2 Parallel Currents

Magnetic field at point P $\vec{B}$ due to two currents i_1 and i_2 is the *vector sum* of the $\vec{B}$ fields $\vec{B}_1$, $\vec{B}_2$ due to individual currents. (**Principle of Superposition**)



Force Between Parallel Currents :

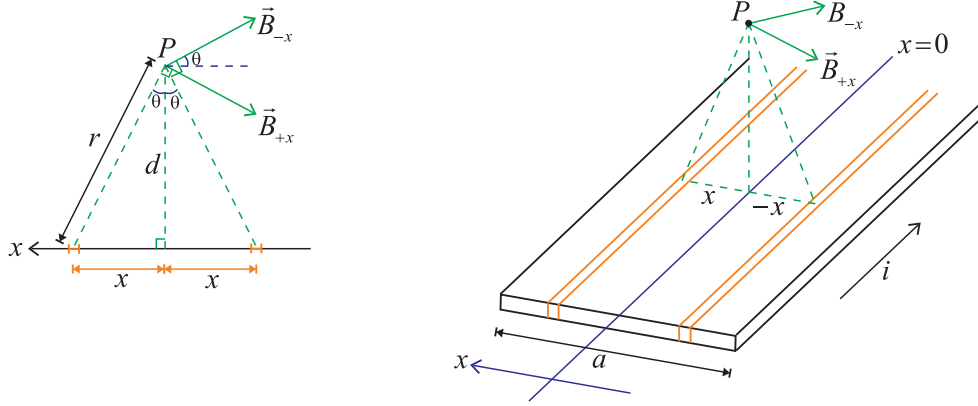
Consider a segment of length L on i_2 :

$$\vec{B}_1 = \frac{\mu_0 i_1}{2\pi d} \quad (\text{pointing down}) \qquad \vec{B}_2 = \frac{\mu_0 i_1}{2\pi d} \quad (\text{pointing up})$$

Force on i_2 coming from i_1 :

$$|\vec{F}_{21}| = i_2 \vec{L} \times \vec{B}_1 = \frac{\mu_0 L i_1 i_2}{2\pi d} = |\vec{F}_{12}| \quad (\text{Def'n of ampere, } A)$$

$\therefore$ Parallel currents attract, anti-parallel currents repel.

Example : Sheet of current

Consider an infinitesimal wire of width dx at position x , there exists another element at $-x$ so that vertical $\vec{B}$ -field components of $\vec{B}_{+x}$ and $\vec{B}_{-x}$ cancel.

$\therefore$ Magnetic field due to dx wire:

$$dB = \frac{\mu_0 \cdot di}{2\pi r} \quad \text{where } di = i\left(\frac{dx}{a}\right)$$

$\therefore$ Total B-field (*pointing along $-x$ axis*) at point P :

$$B = \int_{-a/2}^{+a/2} dB \cos \theta = \int_{-a/2}^{+a/2} \frac{\mu_0 i}{2\pi a} \cdot \frac{dx}{r} \cdot \cos \theta$$

Variable transformation (Goal: change r, x to d, θ , then integrate over θ):

$$\begin{cases} d = r \cos \theta & \Rightarrow r = d \sec \theta \\ x = d \tan \theta & \Rightarrow dx = d \sec^2 \theta d\theta \end{cases}$$

Limits of integration: $-\theta_0$ to θ_0 , where $\tan \theta_0 = \frac{a}{2d}$

$$\begin{aligned} \therefore B &= \frac{\mu_0 i}{2\pi a} \int_{-\theta_0}^{\theta_0} \frac{d \sec^2 \theta d\theta}{d \sec \theta} \cdot \cos \theta \\ &= \frac{\mu_0 i}{2\pi a} \int_{-\theta_0}^{\theta_0} d\theta \\ B &= \frac{\mu_0 i \theta_0}{\pi a} = \frac{\mu_0 i}{\pi a} \tan^{-1} \left(\frac{a}{2d} \right) \end{aligned}$$

Limiting Cases :

(1) $d \gg a$

$$\tan \theta = \frac{a}{2d} \Rightarrow \theta \approx \frac{a}{2d}$$

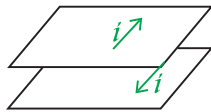
$$\therefore B = \frac{\mu_0 i}{2\pi a} \quad \text{B-field due to infinite long wire}$$

(2) $d \ll a$ (*Infinite sheet of current*)

$$\tan \theta = \frac{a}{2d} \rightarrow \infty \Rightarrow \theta = \frac{\pi}{2}$$

$$\therefore B = \frac{\mu_0 i}{2a} \quad \text{Constant!}$$

Question : Large sheet of opposite flowing currents.



What's the B-field between & outside the sheets?

7.3 Ampère's Law

In our study of electricity, we notice that the **inverse square force law** leads to **Gauss' Law**, which is useful for *finding E-field for systems with high level of symmetry*.

For magnetism, Gauss' Law is simple

$$\oint_S \vec{B} \cdot d\vec{A} = 0 \quad \because \text{There is no magnetic monopole.}$$

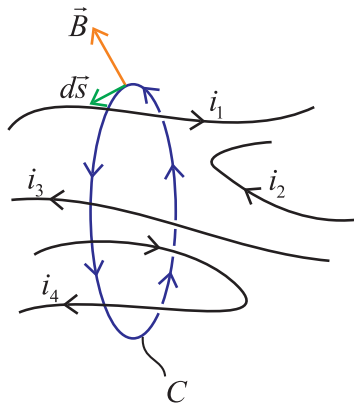
A more useful law for calculating B-field for highly symmetric situations is **the Ampère's Law**:

$$\oint_C \vec{B} \cdot d\vec{s} = \mu_0 i$$

$\oint_C$ = Line integral evaluated around a closed loop C (**Amperian curve**)

i = Net current that penetrates the area bounded by curve C^* (topological property)

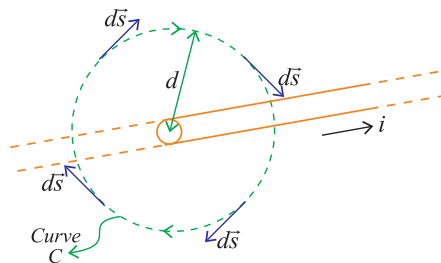
Convention : Use the **right-hand screw rule** to determine the *sign* of current.



$$\begin{aligned} \oint_C \vec{B} \cdot d\vec{s} &= \mu_0(i_1 - i_3 + i_4 - i_2) \\ &= \mu_0(i_1 - i_3) \end{aligned}$$

Applications of the Ampere's Law :

(1) Long-straight wire



Construct an Amperian curve of radius d :

By symmetry argument, we know $\vec{B}$ -field only has *tangential component*

$$\therefore \oint_C \vec{B} \cdot d\vec{s} = \mu_0 i$$

Take $d\vec{s}$ to be the tangential vector around the circular path:

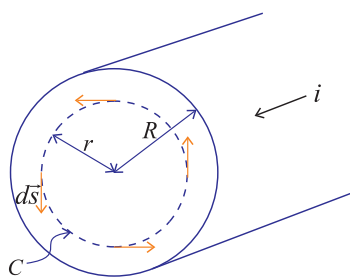
$$\begin{aligned}\therefore \vec{B} \cdot d\vec{s} &= B ds \\ B \oint_C ds &= \mu_0 i \\ \text{Circumference} \\ \text{of circle} &= 2\pi d \\ \therefore B(2\pi d) &= \mu_0 i\end{aligned}$$

B-field due to long,
straight current

$$B = \frac{\mu_0 i}{2\pi d}$$

(Compare with 7.1 Example 1)

(2) Inside a current-carrying wire

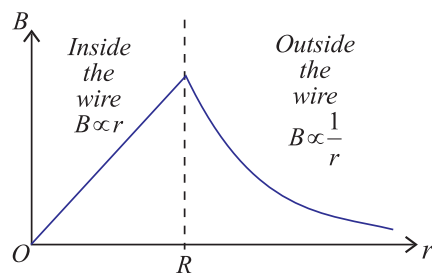


Again, symmetry argument implies that $\vec{B}$ is *tangential* to the Amperian curve and $\vec{B} \rightarrow B(r)\hat{\theta}$

Consider an Amperian curve of radius $r(< R)$

$$\oint_C \vec{B} \cdot d\vec{s} = B \oint_C ds = B(2\pi r) = \mu_0 i_{\text{included}}$$

But $i_{\text{included}} \propto$ cross-sectional area of C



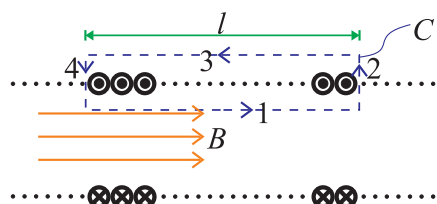
$$\begin{aligned}\therefore \frac{i_{\text{included}}}{i} &= \frac{\pi r^2}{\pi R^2} \\ \therefore i_{\text{included}} &= \frac{r^2}{R^2} i\end{aligned}$$

$$\therefore B = \frac{\mu_0 i}{2\pi R^2} \cdot r \propto r$$

Recall: Uniformly charged infinite long rod

(3) Solenoid (Ideal)

Consider the rectangular Amperian curve 1234.



$$\begin{aligned}
 \oint_C \vec{B} \cdot d\vec{s} &= \int_1 \vec{B} \cdot d\vec{s} + \int_2 \cancel{\vec{B} \cdot d\vec{s}} + \int_3 \cancel{\vec{B} \cdot d\vec{s}} + \int_4 \cancel{\vec{B} \cdot d\vec{s}} \\
 \int_2 &= \int_4 = 0 \quad \because \quad \begin{cases} \vec{B} \cdot d\vec{s} = 0 & \text{inside solenoid} \\ \vec{B} = 0 & \text{outside solenoid} \end{cases} \\
 \int_3 &= 0 \quad \because \quad \vec{B} = 0 \quad \text{outside solenoid} \\
 \therefore \quad \oint_C \vec{B} \cdot d\vec{s} &= \int_1 \vec{B} \cdot d\vec{s} = Bl = \mu_0 i_{tot}
 \end{aligned}$$

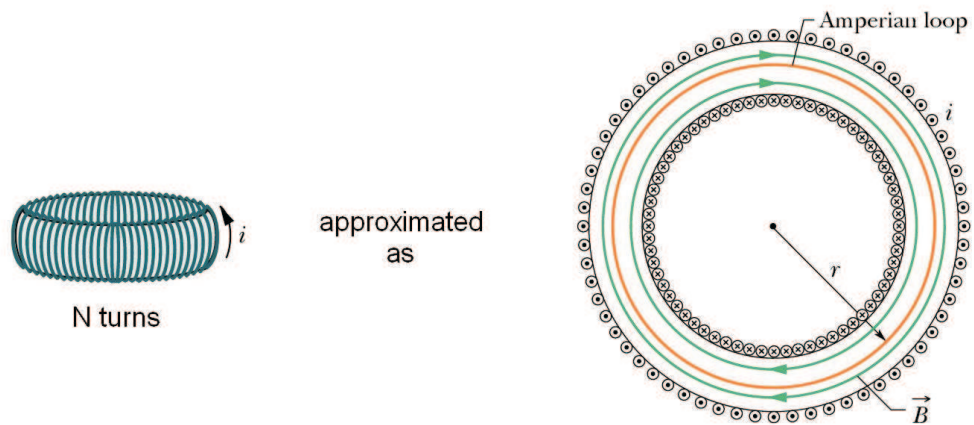
But $i_{tot} = \underbrace{nl}_{\text{Number of coils included}} \cdot i$

$$\therefore \boxed{B = \mu_0 ni}$$

Note :

- (i) The assumption that $\vec{B} = 0$ outside the ideal solenoid is only *approximate*. (Halliday, Pg.763)
- (ii) B-field everywhere inside the solenoid is a *constant*. (for ideal solenoid)

(4) Toroid (A *circular solenoid*)



By symmetry argument, the B-field lines form *concentric circles inside the toroid*.

Take Amperian curve C to be a circle of radius r inside the toroid.

$$\begin{aligned}
 \oint_C \vec{B} \cdot d\vec{s} &= B \oint_C ds = B \cdot 2\pi r = \mu_0 (Ni) \\
 \therefore \quad B &= \frac{\mu_0 Ni}{2\pi r} \quad \text{inside toroid}
 \end{aligned}$$

Note :

(i) $B \neq$ constant inside toroid

(ii) Outside toroid:

Take Amperian curve to be circle of radius $r > R$.

$$\oint_C \vec{B} \cdot d\vec{s} = B \oint_C ds = B \cdot 2\pi r = \mu_0 \cdot i_{incl} = 0$$

$$\therefore B = 0$$

Similarly, in the central cavity $B = 0$

7.4 Magnetic Dipole

Recall from §6.4, we define the **magnetic dipole moment** of a rectangular current loop

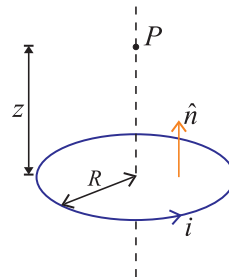
$$\boxed{\vec{\mu} = NiA\hat{n}}$$

where $\hat{n}$ = area unit vector with direction
determined by the right-hand rule
 N = Number of turns in current loop
 A = Area of current loop

This is actually a *general definition* of a magnetic dipole, i.e. we use it for current loops of all shapes.

A common and symmetric example: circular current.

Recall from §7.1 Example 2, magnetic field at point P (height z above the ring)

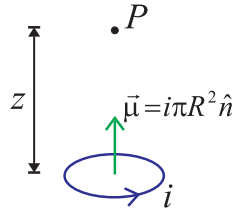


$$\vec{B} = \frac{\mu_0 i R^2 \hat{n}}{2(R^2 + z^2)^{3/2}} = \frac{\mu_0 \vec{\mu}}{2\pi(R^2 + z^2)^{3/2}}$$

At distance $z \gg R$,

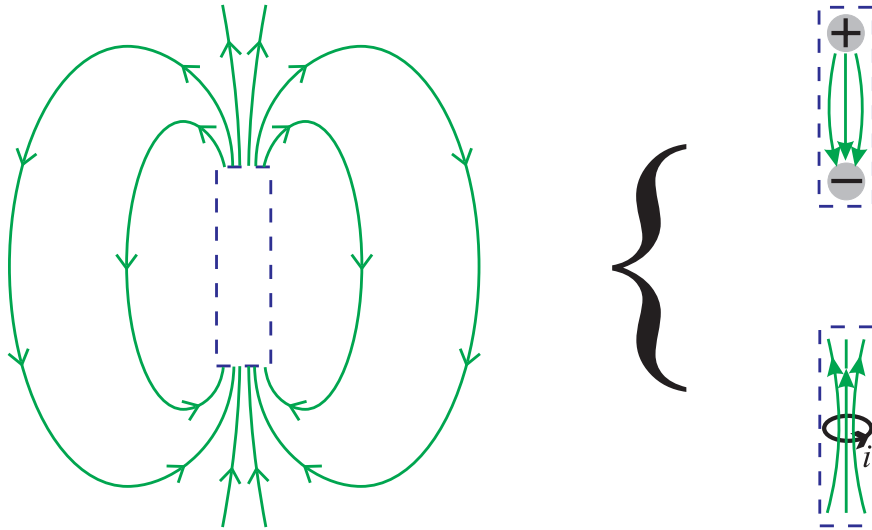
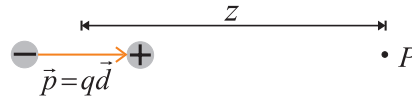
$$\vec{B} = \frac{\mu_0 \vec{\mu}}{2\pi z^3}$$

due to *magnetic dipole*
(for $z \gg R$)



$$\vec{E} = \frac{\vec{p}}{4\pi\epsilon_0 z^3}$$

due to *electric dipole*
(for $z \gg d$)



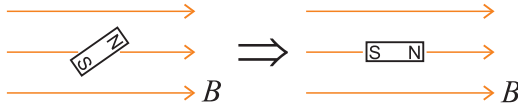
Also, notice $\vec{\mu}$ = magnetic dipole moment $\left[\begin{array}{l} \text{Unit: } \text{Am}^2 \\ J/T \end{array} \right]$

μ_0 = Permeability of free space
 $= 4\pi \times 10^{-7} \text{Tm/A}$

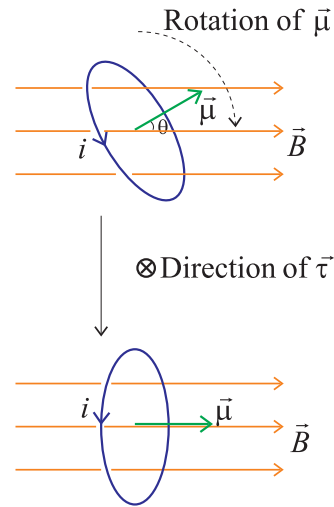
7.5 Magnetic Dipole in A Constant B-field

In the presence of a constant magnetic field, we have shown for a *rectangular current loop*, it experiences a **torque** $\boxed{\vec{\tau} = \vec{\mu} \times \vec{B}}$. It applies to any magnetic dipole in general.

∴ External magnetic field aligns the magnetic dipoles.



Similar to electric dipole in a E-field, we can consider the work done in rotating the magnetic dipole. (refer to Chapter 2)



$$dW = -dU, \quad \text{where } U \text{ is potential energy of dipole}$$

$$U = -\vec{\mu} \cdot \vec{B}$$

Note :

- (1) We cannot define the potential energy of a magnetic field in general. However, we can define the potential energy of a magnetic dipole in a constant magnetic field.
- (2) In a *non-uniform external B-field*, the magnetic dipole will *experience a net force (not only net torque)*

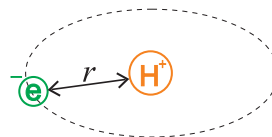
7.6 Magnetic Properties of Materials

Recall intrinsic electric dipole in molecules:



Intrinsic dipole (magnetic) in atoms:

In our classical model of atoms, electrons revolve around a positive nuclear.



$$\therefore \text{ "Current" } i = \frac{e}{P}, \quad \text{where } P \text{ is period of one orbit around nucleus}$$

$$P = \frac{2\pi r}{v}, \quad \text{where } v \text{ is velocity of electron}$$

$\therefore$ **Orbit magnetic dipole** of atom:

$$\mu = iA = \left(\frac{ev}{2\pi r}\right)(\pi r^2) = \frac{erv}{2}$$

Recall: angular momentum of rotation $l = mrv$

$$\therefore \mu = \frac{e}{2m} \cdot l$$

In *quantum mechanics*, we know that

$$l \text{ is quantized, i.e. } l = N \cdot \frac{h}{2\pi}$$

where N = Any positive integer (1,2,3, ...)

h = Planck's constant ($6.63 \times 10^{-34} \text{ J} \cdot \text{s}$)

$\therefore$ **Orbital magnetic dipole moment**

$$\mu_l = \underbrace{\frac{eh}{4m\pi}} \cdot N$$

Bohr's magneton $\mu_B = 9.27 \times 10^{-24} \text{ J/T}$

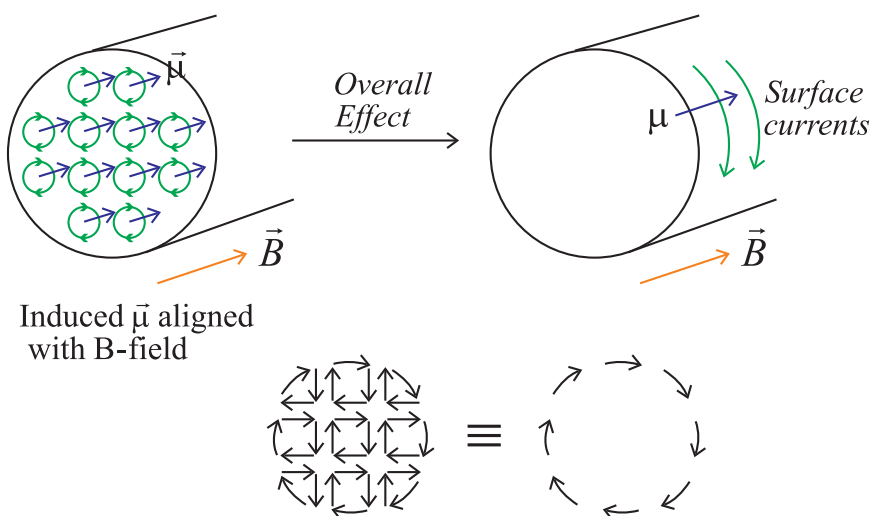
There is another source of intrinsic magnetic dipole moment inside an atom:

Spin dipole moment: coming from the intrinsic "spin" of electrons.

Quantum mechanics suggests that e^- are *always* spinning and it's either an "up" spin or a "down" spin

$$\mu_e = 9.65 \times 10^{-27} \approx \mu_B$$

So can there be induced magnetic dipole?



Recall: for electric field

$$E_{dielectric} = K_e E_{vacuum} ; \quad K_e \geq 1$$

For magnetic field in a material:

$$\vec{B}_{net} = \underbrace{\vec{B}_0}_{\substack{\uparrow \\ \text{applied} \\ \text{B-field}}} + \underbrace{\vec{B}_M}_{\substack{\uparrow \\ \text{B-field produced} \\ \text{by induced dipoles}}}$$

In many materials (except ferromagnets),

$$\vec{B}_M \propto \vec{B}_0$$

Define :

$$\vec{B}_M = \chi_m \vec{B}_0$$

χ_m is a *number* called **magnetic susceptibility**.

$$\begin{aligned} \therefore \vec{B}_{net} &= \vec{B}_0 + \chi_m \vec{B}_0 \\ &= (1 + \chi_m) \vec{B}_0 \end{aligned}$$

$$\vec{B}_{net} = \kappa_m \vec{B}_0 ; \quad \kappa_m = 1 + \chi_m$$

Define : κ_m is a *number* called **relative permeability**.

One more term

Define : the **Magnetization** of a material:

$$\vec{M} = \frac{d\vec{\mu}}{dV} \quad \text{where } \vec{\mu} \text{ is magnetic dipole moment, } V \text{ is volume}$$

(or, the net magnetic dipole moment per unit volume)

In most materials (except ferromagnets),

$$\vec{B}_M = \mu_0 \vec{M}$$

Three types of magnetic materials:

(1) **Paramagnetic:**

$$\begin{aligned} \kappa_m &\geq 1 \\ (\chi_m &\geq 0) \end{aligned} \quad \text{induced magnetic dipoles } \textit{aligned} \\ \text{with the applied B-field.}$$

e.g. Al ($\chi_m \doteq 2.2 \times 10^{-5}$), Mg (1.2×10^{-5}), O_2 (2.0×10^{-6})

(2) **Diamagnetic:**

$$\begin{array}{l} \kappa_m \leq 1 \\ (\chi_m \leq 0) \end{array} , \quad \begin{array}{l} \text{induced magnetic dipoles } \textit{aligned} \\ \textit{opposite with the applied B-field.} \end{array}$$

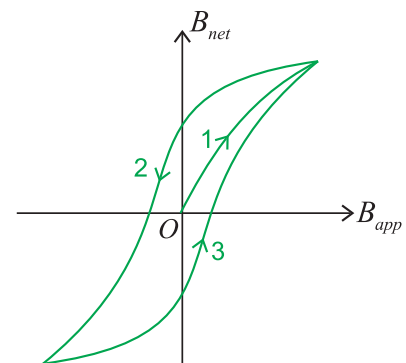
e.g. Cu ($\chi_m \approx -1 \times 10^{-5}$), Ag (-2.6×10^{-5}), N_2 (-5×10^{-9})

(3) **Ferromagnetic:**

e.g. Fe, Co, Ni

Magnetization not linearly proportional to applied field.

$\Rightarrow \frac{B_{net}}{B_{app}}$ not a constant (can be as big as $\sim 5000 - 100,000$)



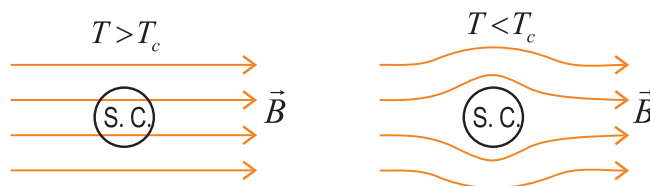
hysteresis curve

(hysteros: [Greek!] later, behind)

Interesting Case : Superconductors

$$\chi_m = -1$$

A perfect diamagnetic.
NO magnetic field inside.



Chapter 8

Faraday's Law of Induction

8.1 Faraday's Law

In the previous chapter, we have shown that *steady electric current* can give *steady magnetic field* because of the symmetry between electricity & magnetism.

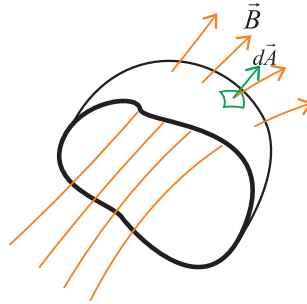
We can ask: *Steady magnetic field* can give *steady electric current*. ×
OR *Changing magnetic field* can give *steady electric current*. ✓

Define :

(1) Magnetic flux through surface S:

$$\Phi_m = \int_S \vec{B} \cdot d\vec{A}$$

Unit of Φ_m : Weber (Wb)
 1Wb = 1Tm²



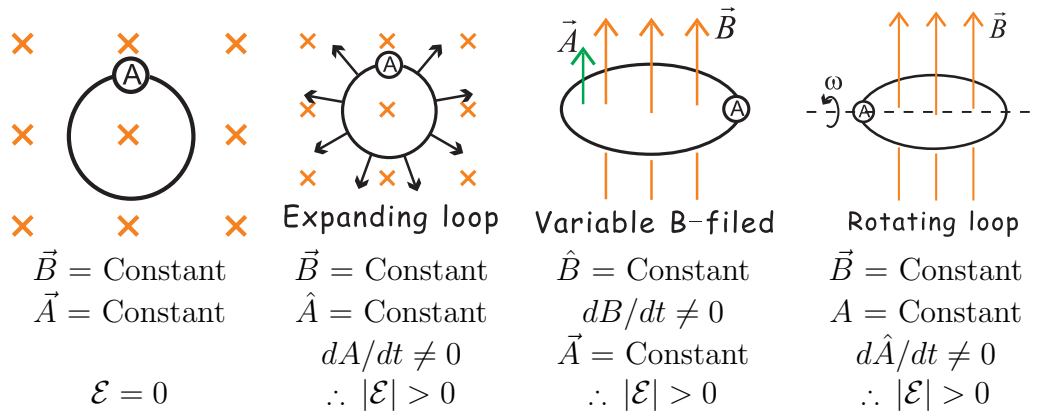
(2) Graphical:

Φ_m = Number of magnetic field lines passing through surface S

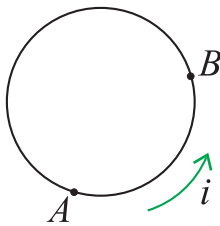
Faraday's law of induction:

$$\text{Induced emf } |\mathcal{E}| = N \left| \frac{d\Phi_m}{dt} \right|$$

where N = Number of coils in the circuit.

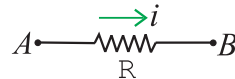


Note : The *induced emf* drives a current throughout the circuit, similar to the function of a *battery*. However, the difference here is that the induced emf is *distributed throughout the circuit*. The consequence is that *we cannot define a potential difference between any two points in the circuit*.



Suppose there is an *induced current* in the loop, can we define ΔV_{AB} ?

Recall:



$$\Delta V_{AB} = V_A - V_B = iR > 0$$

$$\Rightarrow V_A > V_B$$

Going *anti-clockwise* (same as i),

If we start from A, going to B, then we get $V_A > V_B$.

If we start from B, going to A, then we get $V_B > V_A$.

$\therefore$ We cannot define ΔV_{AB} !!

This situation is like when we study *the interior of a battery*.

A battery	}	provides the energy needed to drive the	{	chemical reactions.
The loop	}	charge carriers around the circuit by	{	changing magnetic flux.

sources of emf

non-electric means

8.2 Lenz' Law

- (1) The flux of the magnetic field due to induced current *opposes* the change in flux that causes the induced current.

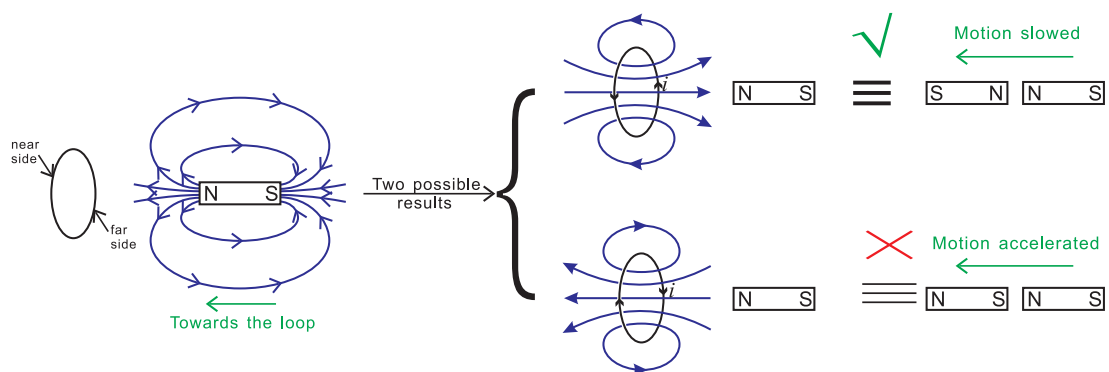
- (2) The induced current is in such a direction as to *oppose* the changes that produces it.
- (3) Incorporating Lenz' Law into Faraday's Law:

$$\mathcal{E} = -N \frac{d\Phi_m}{dt}$$

If $\frac{d\Phi_m}{dt} > 0$, $\Phi_m \uparrow \Rightarrow \mathcal{E}$ appears $\Rightarrow$ Induced current appears.

$\Rightarrow \vec{B}$ -field due to induced current $\Rightarrow$ change in $\Phi_m \xRightarrow{\text{so that}} \Phi_m \downarrow$

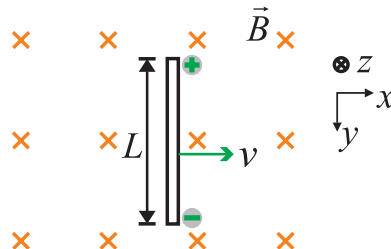
- (4) Lenz' Law is a consequence from *the principle of conservation of energy*.



8.3 Motional EMF

Let's try to look at a special case when the *changing magnetic flux* is carried by *motion in the circuit wires*.

Consider a conductor of length L moving with a velocity v in a magnetic field $\vec{B}$.



Hall Effect for the charge carriers in the rod:

$$\begin{aligned}\vec{F}_E + \vec{F}_B &= 0 \\ \Rightarrow q\vec{E} + q\vec{v} \times \vec{B} &= 0 \quad (\text{where } \vec{E} \text{ is Hall electric field}) \\ \Rightarrow \vec{E} &= -\vec{v} \times \vec{B}\end{aligned}$$

Hall Voltage inside rod:

$$\begin{aligned}\Delta V &= -\int_0^L \vec{E} \cdot d\vec{s} \\ \Delta V &= -EL\end{aligned}$$

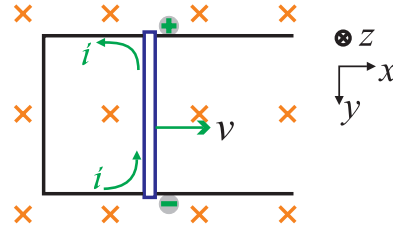
$$\therefore \text{ Hall Voltage : } \boxed{\Delta V = vBL}$$

Now, suppose the moving wire *slides without friction* on a stationary U-shape conductor. The motional emf can drive an electric current i in the U-shape conductor.

$\Rightarrow$ Power is dissipated in the circuit.

$\Rightarrow P_{out} = Vi$ (Joule's heating)

(see Lecture Notes Chapter 4)



What is the source of this power?

Look at the forces acting on the conducting rod:

- Magnetic force:

$$\begin{aligned}\vec{F}_m &= i\vec{L} \times \vec{B} \\ F_m &= iLB \quad (\text{pointing left})\end{aligned}$$

- For the rod to continue to move at constant velocity v , we need to *apply an external force*:

$$\vec{F}_{ext} = -\vec{F}_m = iLB \quad (\text{pointing right})$$

$\therefore$ Power required to keep the rod moving:

$$\begin{aligned}P_{in} &= \vec{F}_{ext} \cdot \vec{v} \\ &= iBLv \\ &= iBL \frac{dx}{dt} \\ &= iB \frac{d(xL)}{dt} \quad (xL = A, \text{ area enclosed by circuit}) \\ &= i \frac{d(BA)}{dt} \quad (BA = \Phi_m, \text{ magnetic flux})\end{aligned}$$

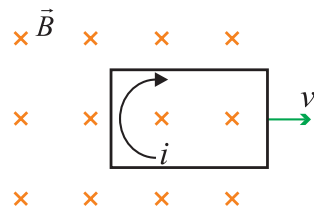
Since energy is not being stored in the system,

$$\begin{aligned}\therefore P_{in} + P_{out} &= 0 \\ iV + i \frac{d\Phi_m}{dt} &= 0\end{aligned}$$

We "prove" Faraday's Law $\Rightarrow \boxed{V = -\frac{d\Phi_m}{dt}}$

Applications :

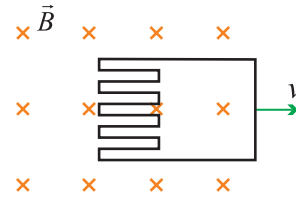
- (1) Eddy current: moving conductors in presence of magnetic field



Induced current

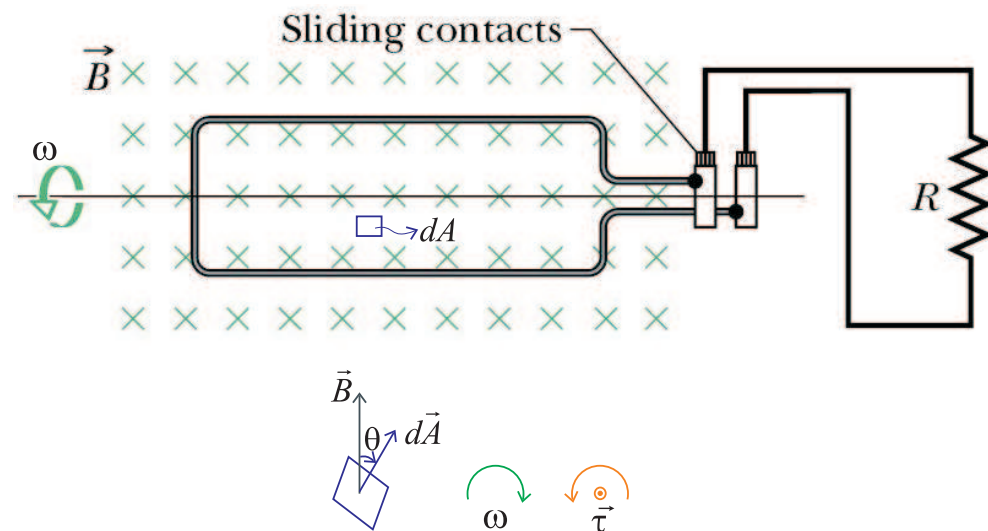
$$\begin{aligned}\Rightarrow \text{Power lost in Joule's heating } \left(\frac{\mathcal{E}^2}{R}\right) \\ \Rightarrow \text{Extra power input to keep moving}\end{aligned}$$

To reduce Eddy currents:



- (2) Generators and Motors:

Assume that the circuit loop is *rotating at a constant angular velocity* ω , (Source of rotation, e.g. steam produced by burner, water falling from a dam)



Magnetic flux through the loop

$$\begin{array}{c} \text{Number of coils} \\ \downarrow \\ \Phi_B = N \int_{\text{loop}} \vec{B} \cdot d\vec{A} = NBA \cos \theta \end{array}$$

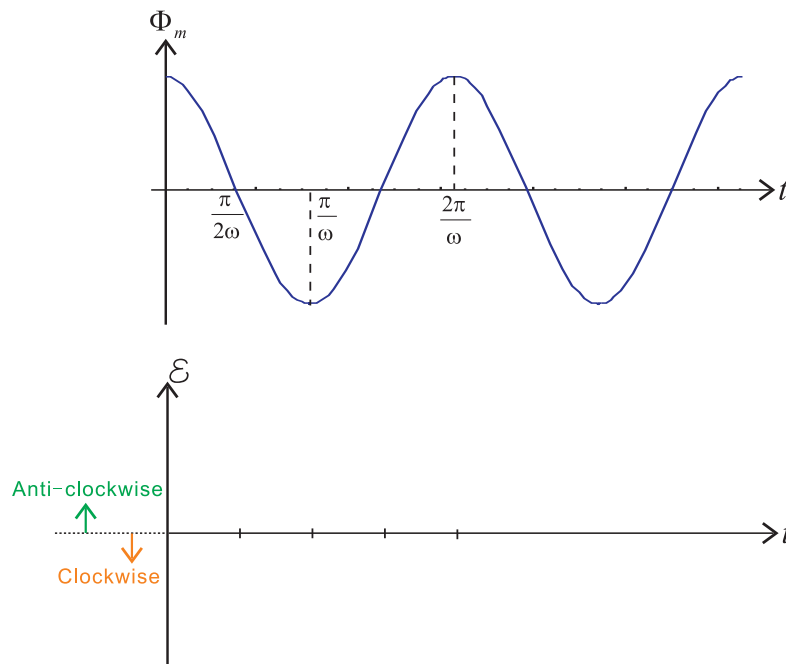
$\downarrow$
changes with time! $\theta = \omega t$

$$\therefore \Phi_B = NBA \cos \omega t$$

$$\begin{aligned} \text{Induced emf: } \mathcal{E} &= -\frac{d\Phi_B}{dt} = -NBA \frac{d}{dt}(\cos \omega t) \\ &= NBA\omega \sin \omega t \end{aligned}$$

$$\text{Induced current: } i = \frac{\mathcal{E}}{R} = \frac{NBA\omega}{R} \sin \omega t$$

Alternating current (AC) voltage generator



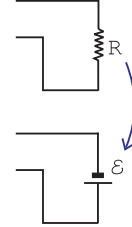
Power has to be provided by the source of rotation to overcome the torque acting on a current loop in a magnetic field.

$$\begin{aligned} \vec{\tau} &= \overbrace{Ni\vec{A}}^{\vec{\mu}} \times \vec{B} \\ \therefore \tau &= NiAB \sin \theta \end{aligned}$$

The net effect of the torque is to *oppose* the rotation of the coil.

An *electric motor* is simply a *generator operating in reverse*.

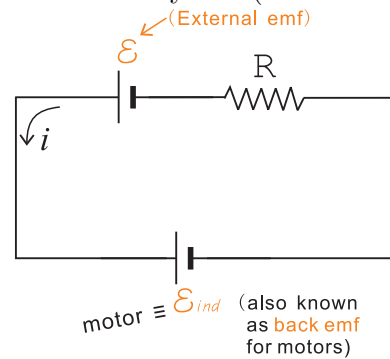
⇒ Replace the load resistance R with a battery of emf $\mathcal{E}$.



With the battery, there is a current in the coil, and it experiences a torque in the B-field.

⇒ Rotation of the coil leads to an induced emf, $\mathcal{E}_{ind}$, in the direction opposite of that of the battery. (Lenz' Law)

$$\therefore i = \frac{\mathcal{E} - \mathcal{E}_{ind}}{R}$$



⇒ As motor speeds up, $\mathcal{E}_{ind} \uparrow$, $\therefore i \downarrow$

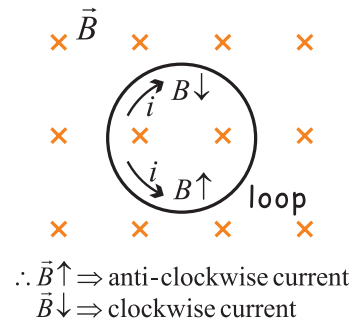
$\therefore$ mechanical power delivered = torque delivered = $NiAB \sin \theta \downarrow$

In conclusion, we can show that

$P_{electric}$	$=$	$i^2 R$	$+$	$P_{mechanical}$
Electric power input				Mechanical power delivered

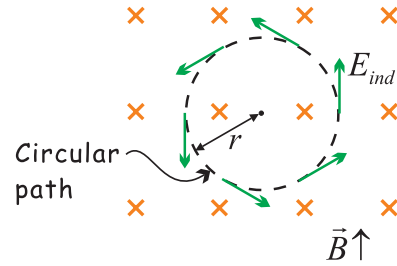
8.4 Induced Electric Field

So far we have discussed that a *change* in magnetic flux will lead in an induced emf distributed in the loop, resulting from an induced E-field.



However, even in the *absence* of the loop (so that there is no induced current), the induced E-field will still accompany a change in magnetic flux.

∴ Consider a circular path in a region with changing magnetic field.



The induced E-field only has tangential components. (i.e. radial E-field = 0)
Why?

Imagine a point charge q_0 travelling around the circular path.

$$\text{Work done by induced E-field} = \underbrace{q_0 E_{ind}}_{\text{force}} \cdot \underbrace{2\pi r}_{\text{distance}}$$

Recall work done also equals to $q_0 \mathcal{E}$, where $\mathcal{E}$ is induced emf

$$\therefore \mathcal{E} = E_{ind} 2\pi r$$

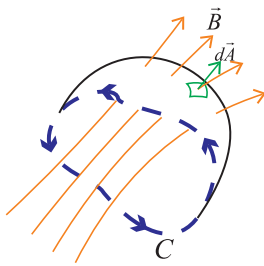
Generally,

$$\mathcal{E} = \oint \vec{E}_{ind} \cdot d\vec{s}$$

where $\oint$ is line integral around a closed loop, $\vec{E}_{ind}$ is induced E-field, $\vec{s}$ is tangential vector of path.

∴ Faraday's Law becomes

$$\oint_C \vec{E}_{ind} \cdot d\vec{s} = -\frac{d}{dt} \int_S \vec{B} \cdot d\vec{A}$$



L.H.S. = Integral around a closed loop C
R.H.S. = Integral over a surface bounded by C

Direction of $d\vec{A}$ determined by direction of line integration C (Right-Hand Rule)

"Regular" E-field

created by charges

E-field lines start from $+ve$ and end on $-ve$ charge



can define electric potential so that we can discuss potential difference between two points

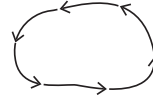


Conservative force field

Induced E-field

created by changing B-field

E-field lines form closed loops



Electric potential cannot be defined (or, potential has no meaning)



Non-conservative force field

The classification of electric and magnetic effects *depend on the frame of reference of the observer*. e.g. For motional emf, observer in the reference frame of the moving loop, will NOT see an induced E-field, just a "regular" E-field.

(Read: Halliday Chap.33-6, 34-7)

Chapter 9

Inductance

9.1 Inductance

An *inductor* stores energy in the *magnetic field* just as a *capacitor* stores energy in the *electric field*.

We have shown earlier that a *changing B-field* will lead to an *induced emf* in a *circuit*.

Question : If a circuit generates a changing magnetic field, does it lead to an induced emf in the same circuit? **YES! Self-Inductance**

The **inductance L** of any current element is

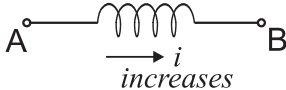
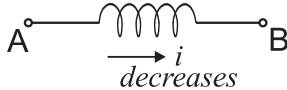
$$\mathcal{E}_L = \Delta V_L = -L \frac{di}{dt}$$

The negative sign comes from Lenz Law.

$$\text{Unit of L: Henry(H)} \quad 1\text{H} = 1 \cdot \frac{\text{Vs}}{\text{A}}$$

- All circuit elements (including resistors) have some inductance.
- Commonly used inductors: solenoids, toroids
- circuit symbol: 

Example : Solenoid

	
	
$\mathcal{E}_L = V_B - V_A = -L \frac{di}{dt} < 0$ $\therefore V_B < V_A$	$\mathcal{E}_L = V_B - V_A = -L \frac{di}{dt} > 0$ $V_B > V_A$

Recall *Faraday's Law*,

$$\mathcal{E}_L = -N \frac{d\Phi_B}{dt} = -\frac{d}{dt} (N\Phi_B)$$

where Φ_B is magnetic flux, $N\Phi_B$ is flux linkage.

$\therefore$ Alternative definition of Inductance:

$$-\frac{d}{dt} (N\Phi_B) = -L \frac{di}{dt} \Rightarrow \boxed{L = \frac{N\Phi_B}{i}}$$

$\therefore$ Inductance is also *flux linkage per unit current*.

Calculating Inductance:

(1) Solenoid:

To first order approximation,



$$B = \mu_0 n i$$

where $n = N/L =$ Number of coils per unit length.

Consider a subsection of length l of the solenoid:

$$\begin{aligned} \text{Flux linkage} &= N \Phi_B \\ &= nl \cdot BA \end{aligned} \quad \begin{array}{l} \text{where } A \text{ is} \\ \text{cross-sectional area} \end{array}$$

$$\therefore \boxed{\begin{aligned} L &= \frac{N\Phi_B}{i} = \mu_0 n^2 l A \\ \frac{L}{l} &= \mu_0 n^2 A = \text{Inductance per unit length} \end{aligned}}$$

Notice :

- (i) $L \propto n^2$
- (ii) The inductance, like the capacitance, depends only on geometric factors, not on i .

(2) Toroid:

Recall: B-field lines are concentric circles.

Inside the toroid:

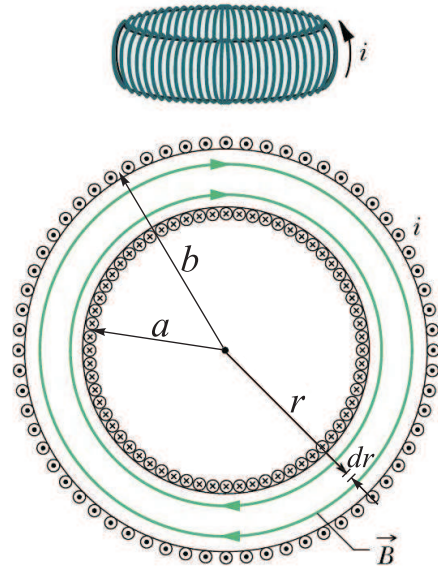
$$B = \frac{\mu_0 i N}{2\pi r}$$

(NOT a constant)

where r is the distance from center.

Outside the toroid:

$$B = 0$$



Flux linkage through the toroid

$$\begin{aligned} N\Phi_B &= N \int \vec{B} \cdot d\vec{a} \quad \left\{ \begin{array}{l} \text{Notice } \vec{B} \parallel d\vec{a} \\ \text{Write } da = h dr \end{array} \right\} \boxed{\text{KEY}} \\ &= \frac{\mu_0 i N^2}{2\pi} \int_a^b \frac{h dr}{r} \\ &= \frac{\mu_0 i N^2 h}{2\pi} \ln\left(\frac{b}{a}\right) \end{aligned}$$

$$\therefore \text{ Inductance } L = \frac{N\Phi_B}{i} = \frac{\mu_0 N^2 h}{2\pi} \ln\left(\frac{b}{a}\right)$$

Again, $L \propto N^2$

Inductance with magnetic materials :

We showed earlier that for capacitors:

$$\left\{ \begin{array}{l} \vec{E} \rightarrow \vec{E}/\kappa_e \\ C \rightarrow \kappa_e C \end{array} \right. \quad \begin{array}{l} \text{(after insertion of} \\ \text{dielectric } \kappa_e > 1) \end{array}$$

For inductors, we first know that

$$\vec{B} \rightarrow \kappa_m \vec{B} \quad \begin{array}{l} \text{(after insertion of} \\ \text{magnetic material)} \end{array}$$

$$\text{Inductance } L = \frac{N\Phi_B}{i}$$

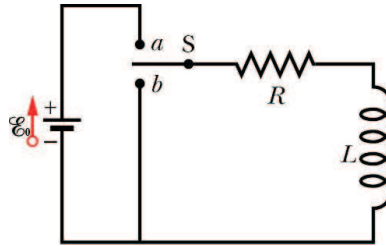
$$\text{However } \Phi_B = \int \vec{B} \cdot d\vec{A} \rightarrow \kappa_m \Phi_B$$

$$\therefore \boxed{L \rightarrow \kappa_m L} \quad (\text{after insertion of magnetic material})$$

$\therefore$ To increase inductance, fill the interior of inductor with *ferromagnetic materials*. ($\times 10^3 - 10^4$)

9.2 LR Circuits

(A) "Charging" an inductor



When the switch is adjusted to position a,

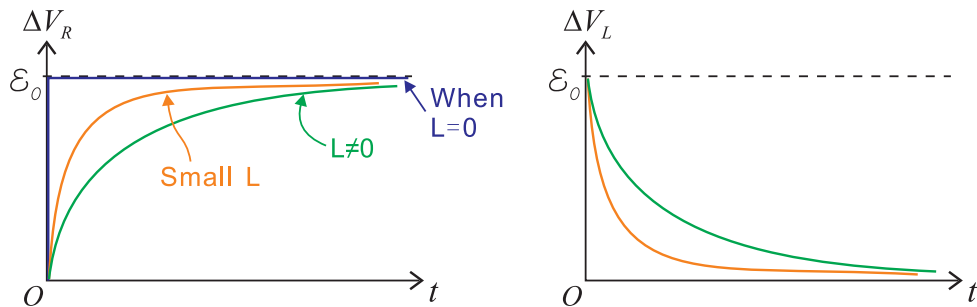
By *loop rule* (clockwise) :

$$\begin{aligned} \mathcal{E}_0 - \Delta V_R + \Delta V_L &= 0 \\ &\downarrow \quad \quad \downarrow \\ \mathcal{E}_0 - iR - L \frac{di}{dt} &= 0 \\ \therefore \frac{di}{dt} + \frac{R}{L} i &= \frac{\mathcal{E}_0}{L} \quad \text{First Order Differential Equation} \end{aligned}$$

Similar to the equation for charging a capacitor! (Chap5)

Solution: $i(t) = \frac{\mathcal{E}_0}{R} (1 - e^{-t/\tau_L})$
 where $\tau_L = \text{Inductive time constant} = L/R$

$$\begin{aligned} \therefore |\Delta V_R| &= iR = \mathcal{E}_0 (1 - e^{-t/\tau_L}) \\ |\Delta V_L| &= L \frac{di}{dt} = L \cdot \frac{\mathcal{E}_0}{R} \cdot \frac{1}{\tau_L} \cdot e^{-t/\tau_L} = \mathcal{E}_0 e^{-t/\tau_L} \end{aligned}$$



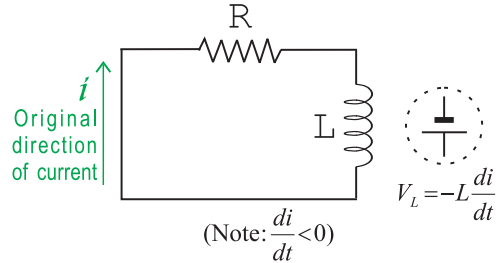
(B) "Discharging" an inductor

When the switch is adjusted at position b after the inductor has been "charged" (i.e. current $i = \mathcal{E}_0/R$ is flowing in the circuit.).

By loop rule:

$$\begin{array}{rcl} \Delta V_L & - & \Delta V_R = 0 \\ \downarrow & & \downarrow \\ -L \frac{di}{dt} & - & iR = 0 \end{array}$$

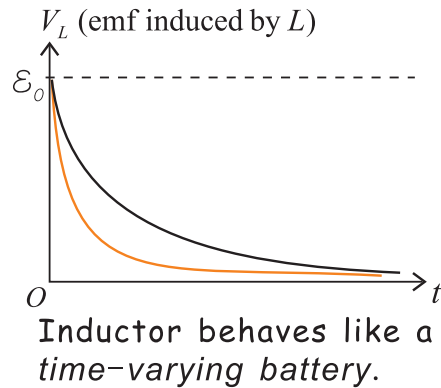
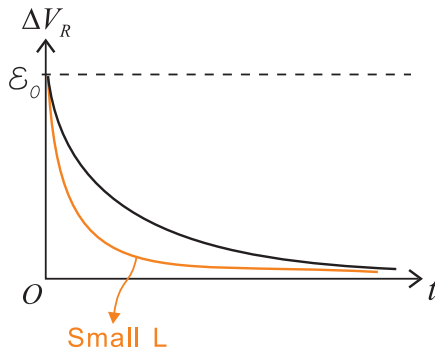
(Treat inductor as source of emf)



$$\therefore \frac{di}{dt} + \frac{R}{L} i = 0 \quad \text{Discharging a capacitor (Chap5)}$$

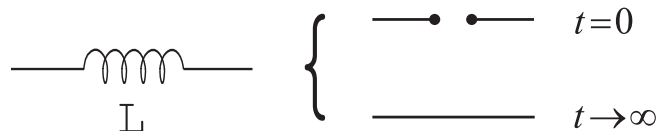
$$i(t) = i_0 e^{-t/\tau_L}$$

where $i_0 = i(t=0)$ = Current when the circuit just switch to position b.



Summary : During charging of inductor,

1. At $t = 0$, inductor acts like *open circuit* when *current flowing is zero*.
2. At $t \rightarrow \infty$, inductor acts like *short circuit* when *current flowing is stabilized at maximum*.



3. Inductors are used everyday in switches for safety concerns.

9.3 Energy Stored in Inductors

Inductors stored *magnetic energy* through the *magnetic field* stored in the circuit. Recall the equation for charging inductors:

$$\mathcal{E}_0 - iR - L \frac{di}{dt} = 0$$

Multiply both sides by i :

$$\underbrace{\mathcal{E}_0 i}_{\text{Power input by emf}} = \underbrace{i^2 R}_{\text{Joule's heating}} + \underbrace{Li \frac{di}{dt}}_{\text{Power stored in inductor}}$$

(Energy supplied to one charge = $q\mathcal{E}_0$) (Power dissipated by resistor)

$$\therefore \text{Power stored in inductor} = \frac{dU_B}{dt} = Li \frac{di}{dt}$$

Integrating both sides and use initial condition

$$\text{At } t = 0, \quad i(t = 0) = U_B(t = 0) = 0$$

$$\therefore \boxed{\text{Energy stored in inductor: } U_B = \frac{1}{2} Li^2}$$

Energy Density Stored in Inductors :

Consider an *infinitely long* solenoid of cross-sectional area A .

For a portion l of the solenoid, we know from §8.1,

$$L = \mu_0 n^2 l A$$

$\therefore$ Energy stored in inductor:

$$U_B = \frac{1}{2} Li^2 = \frac{1}{2} \mu_0 n^2 i^2 \underbrace{l A}_{\text{Volume of solenoid}}$$

$\therefore$ **Energy density** (= Energy stored per unit volume) inside inductor:

$$u_B = \frac{U_B}{lA} = \frac{1}{2} \mu_0 n^2 i^2$$

Recall magnetic field inside solenoid (Chap7)

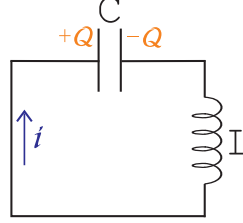
$$B = \mu_0 n i$$

$$\therefore \boxed{u_B = \frac{B^2}{2\mu_0}}$$

This is a *general result* of the *energy stored in a magnetic field*.

9.4 LC Circuit (Electromagnetic Oscillator)

Initial charge on capacitor = Q
 Initial current = 0
 No battery.



Assume current i to be in the direction that *increases* charge on the *positive* capacitor plate.

$$\Rightarrow \quad i = \frac{dQ}{dt} \quad (9.1)$$

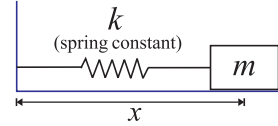
By *Lenz Law*, we also know the "poles" of the inductor.

$$\begin{aligned} \text{Loop rule:} \quad V_C + V_L &= 0 \\ -\frac{Q}{C} - L \frac{di}{dt} &= 0 \end{aligned} \quad (9.2)$$

Combining equations (9.1) and (9.2), we get

$$\boxed{\frac{d^2Q}{dt^2} + \frac{1}{LC} Q = 0}$$

This is similar to the equation of motion of a *simple harmonic oscillator*:



$$\boxed{\frac{d^2x}{dt^2} + \frac{k}{m} x = 0}$$

Another approach (*conservation of energy*)

Total energy stored in circuit:

$$\begin{aligned} U &= U_E + U_B \\ &\quad \downarrow \quad \quad \downarrow \\ U &= \frac{Q^2}{2C} + \frac{1}{2} Li^2 \end{aligned}$$

Since the resistance in the circuit is zero, *no energy is dissipated in the circuit*.

$\therefore$ Energy contained in the circuit is *conserved*.

$$\begin{aligned} \therefore \quad \frac{dU}{dt} &= 0 \\ \Rightarrow \quad \frac{Q}{C} \cdot \frac{dQ}{dt} + L i \frac{di}{dt} &= 0 \quad (\because i = \frac{dQ}{dt}) \end{aligned}$$

$$\Rightarrow L \frac{di}{dt} + \frac{Q}{C} = 0$$

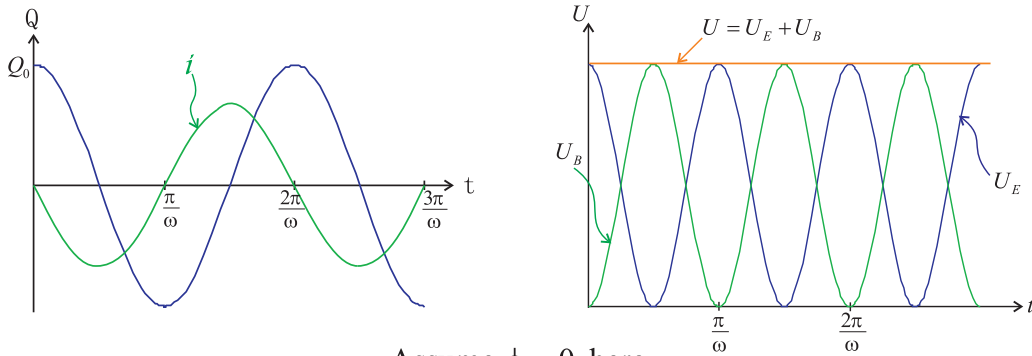
$$\Rightarrow \boxed{\frac{d^2 Q}{dt^2} + \frac{1}{LC} Q = 0}$$

The solution to this differential equation is in the form

$$\begin{aligned} Q(t) &= Q_0 \cos(\omega t + \phi) \\ \therefore \frac{dQ}{dt} &= -\omega Q_0 \sin(\omega t + \phi) \\ \frac{d^2 Q}{dt^2} &= -\omega^2 Q_0 \cos(\omega t + \phi) \\ &= -\omega^2 Q \\ \therefore \frac{d^2 Q}{dt^2} + \omega^2 Q &= 0 \\ \therefore \boxed{\omega^2 = \frac{1}{LC}} &\quad \begin{array}{l} \text{Angular frequency} \\ \text{of the LC oscillator} \end{array} \end{aligned}$$

Also, Q_0 , ϕ are constants derived from the initial conditions. (Two initial conditions, e.g. $Q(t=0)$, and $i(t=0) = \left. \frac{dQ}{dt} \right|_{t=0}$ are required.)

$$\begin{aligned} \text{Energy stored in } C &= \frac{Q^2}{2C} = \frac{Q_0^2}{2C} \cos^2(\omega t + \phi) \\ \text{Energy stored in } L &= \frac{1}{2} L i^2 = \frac{1}{2} L \omega^2 Q_0^2 \sin^2(\omega t + \phi) \\ \boxed{\therefore L \omega^2 = \frac{1}{C}} &= \frac{Q_0^2}{2C} \sin^2(\omega t + \phi) \\ \therefore \text{Total energy stored} &= \frac{Q_0^2}{2C} \\ &= \text{Initial energy stored in capacitor} \end{aligned}$$

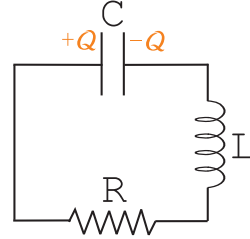


Assume $\phi = 0$ here

9.5 RLC Circuit (Damped Oscillator)

In real life circuit, there's *always* resistance.

In this case, energy stored in the LC oscillator is *NOT* conserved,



and $\frac{dU}{dt} = \text{Power dissipated in the resistor} = -i^2 R$ (*Joule's heating*)

Negative sign shows that energy U is *decreasing*.

$$\therefore Li \frac{di}{dt} + \frac{Q}{C} \cdot \frac{dQ}{dt} = -i^2 R$$

$$\Rightarrow \boxed{\frac{d^2 Q}{dt^2} + \frac{R}{L} \cdot \frac{dQ}{dt} + \frac{1}{LC} Q = 0}$$

This is similar to the equation of motion of a *damped harmonic oscillator* (e.g. if a mass-spring system faces a frictional force $\vec{F} = -b\vec{v}$).

Solution to the equation is in the form $Q(t) = e^{\lambda t}$

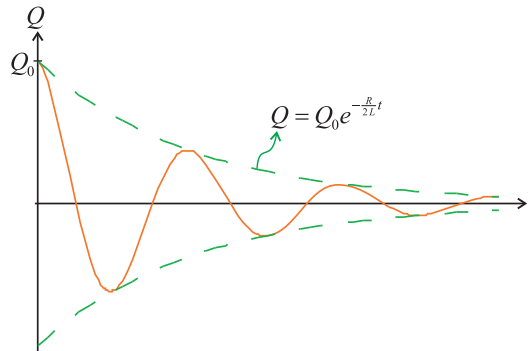
If damping is not too big (i.e. R not too big), solution would become

$$Q(t) = Q_0 \underbrace{e^{-\frac{R}{2L}t}}_{\text{exponential decay term}} \underbrace{\cos(\omega_1 t + \phi)}_{\text{oscillating term}}$$

where $\omega_1^2 = \frac{1}{LC} - \left(\frac{R}{2L}\right)^2$

$$\boxed{\omega_1^2 = \omega^2 - \left(\frac{R}{2L}\right)^2}$$

Damped oscillator always oscillates at a *lower* frequency than the *natural frequency* of the oscillator. (Refer to *Halliday*, Vol1, Chap17 for more details.)



Check this at home: What is $U_E(t) + U_B(t)$ for the case when damping is small? (i.e. $R \ll \omega$)

Chapter 10

AC Circuits

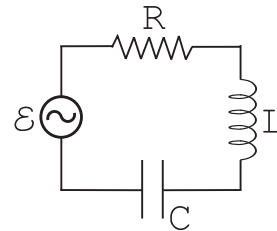
10.1 Alternating Current (AC) Voltage

Recall that an *AC generator* described in Chapter 9 generates a *sinusoidal emf*.

$$\text{i.e. } \mathcal{E} = \mathcal{E}_m \sin(\omega t + \delta)$$

Note :

This circuit is the RLC circuit with one *additional element : the time varying AC power supply*. This is similar to a *driven (damped) oscillator*.



$$L \frac{d^2 Q}{dt^2} + R \frac{dQ}{dt} + \frac{1}{C} Q = \mathcal{E}_m \sin(\omega t + \delta)$$

The general solution consists of two parts:

transient : rapidly dies away in a few cycles (not interesting)

steady state : $Q(t), i(t)$ varies *sinusoidally* with the same frequency as input

Note : Current does NOT vary at frequency $\omega_1^2 = \frac{1}{LC} - \left(\frac{R}{2L}\right)^2$

Since we only concern about the *steady state solution*, therefore we can take any time as starting reference time = 0

For convenience, we can write

$$\boxed{\mathcal{E} = \mathcal{E}_m \sin \omega t}$$

And we can write

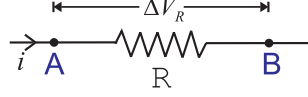
$$\boxed{i = i_m \sin(\omega t - \phi)}$$

where i_m is current amplitude, ϕ is phase constant.

Our goal is to determine i_m and ϕ .

10.2 Phase Relation Between i, V for R,L and C

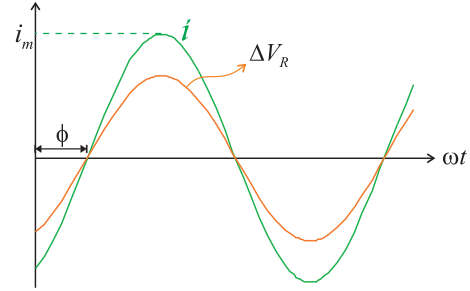
(A) Resistive Element



$$\Delta V_R = V_A - V_B = iR$$

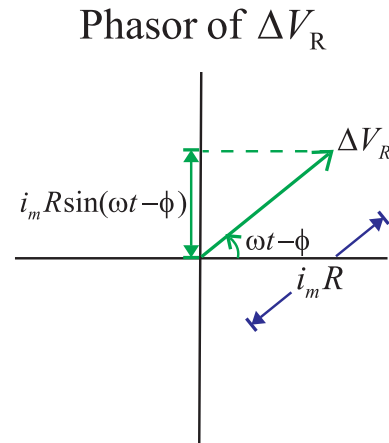
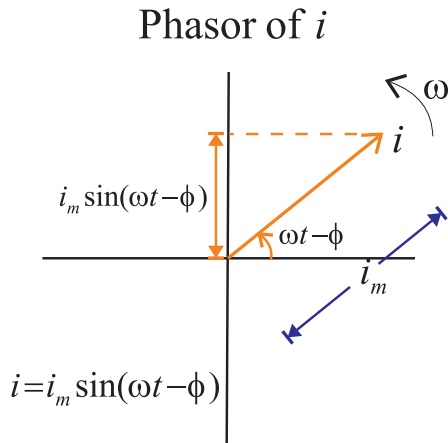
$$\therefore \Delta V_R = i_m R \sin(\omega t - \phi)$$

ΔV_R and i are *in phase*, i.e. what's inside the "sine bracket" (**phase**) is the same for ΔV_R and i .



Graphically, we introduce **phasor diagrams** properties of **phasors**:

- (1) *Length* of a phasor is proportional to the *maximum value*.
- (2) *Projection* of a phasor *onto the vertical axis* gives *instantaneous value*.
- (3) **Convention:** Phasors rotate *anti-clockwise* in a uniform circular motion with *angular velocity*.

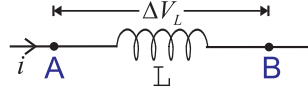


$$\therefore \Delta V_R = (\Delta V_R)_m \sin(\omega t - \phi)$$

$$(\Delta V_R)_m = i_m R$$

"Ohm's Law like" relation for AC resistor

(B) The Inductive Element



Potential drop across inductor

$$\Delta V_L = V_A - V_B = -\mathcal{E}_L = L \frac{di}{dt}$$

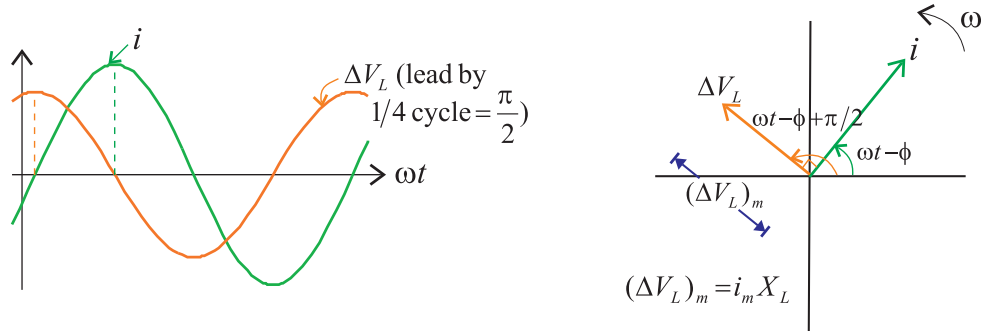
$$\begin{aligned} \therefore \Delta V_L &= Li_m \omega \cos(\omega t - \phi) \\ &= Li_m \omega \sin(\omega t - \phi + \frac{\pi}{2}) \quad [\because \cos \theta = \sin(\theta + \frac{\pi}{2})] \\ &= i_m X_L \sin(\omega t - \phi + \frac{\pi}{2}) \end{aligned}$$

$$(\Delta V_L)_m = i_m X_L$$

"Ohm's Law like" relation for AC inductor

where $X_L = \text{Inductive Reactance}$

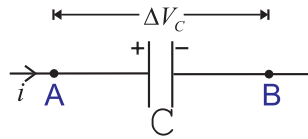
$$X_L = \omega L$$



$$\begin{aligned} \text{As } i \uparrow, V_A > V_B &\therefore \Delta V_L > 0 \\ i \downarrow, V_A < V_B &\therefore \Delta V_L < 0 \end{aligned}$$

$$\begin{array}{llll} \Delta V_L & \text{leads} & i & \text{by } \frac{\pi}{2} \\ i & \text{lags} & \Delta V_L & \text{by } \frac{\pi}{2} \end{array}$$

(C) Capacitive Element



$$\Delta V_C = V_A - V_B = \frac{Q}{C}$$

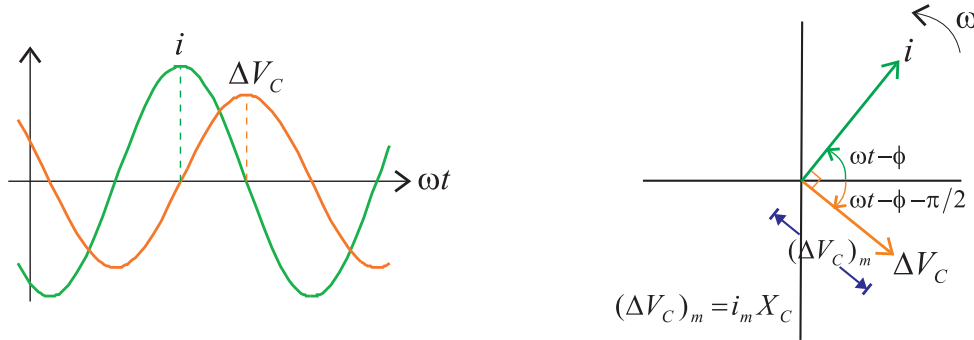
where Q = charge on the positive plate of the capacitor.

$$\begin{aligned}\therefore i = \frac{dQ}{dt} &\Rightarrow Q = \int i dt \\ &= \int i_m \sin(\omega t - \phi) dt \\ &= -\frac{i_m}{\omega} \cos(\omega t - \phi)\end{aligned}$$

$$\begin{aligned}\therefore \Delta V_C &= -\frac{i_m}{\omega C} \cos(\omega t - \phi) \\ &= i_m X_C \sin(\omega t - \phi - \frac{\pi}{2}) \quad [\because -\cos \theta = \sin(\theta - \frac{\pi}{2})]\end{aligned}$$

$$\therefore \boxed{(\Delta V_C)_m = i_m X_C} \quad \text{"Ohm's Law like" relation for AC capacitor}$$

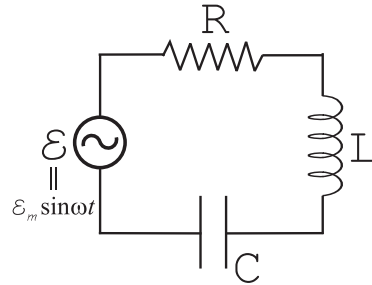
where $\boxed{X_C = \frac{1}{\omega C}} = \text{Capacitive Reactance}$



ΔV_C	lags	i	by	$\frac{\pi}{2}$
i	leads	ΔV_C	by	$\frac{\pi}{2}$

10.3 Single Loop RLC AC Circuit

Given that $\mathcal{E} = \mathcal{E}_m \sin \omega t$, we want to find i_m and ϕ so that we can write $i = i_m \sin(\omega t - \phi)$



$$\begin{aligned}\text{Loop rule:} \quad &\mathcal{E} - \Delta V_R - \Delta V_L - \Delta V_C = 0 \\ \Rightarrow &\mathcal{E} = \Delta V_R + \Delta V_L + \Delta V_C\end{aligned}$$

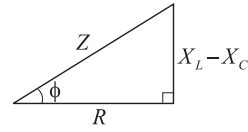
Using results from the previous section, we can write

$$\begin{aligned}\mathcal{E}_m \sin \omega t &= i_m R \sin(\omega t - \phi) \\ &\quad + i_m X_L \cos(\omega t - \phi) - i_m X_C \cos(\omega t - \phi) \\ \mathcal{E}_m \sin \omega t &= i_m [R \sin(\omega t - \phi) + (X_L - X_C) \cos(\omega t - \phi)]\end{aligned}$$

Answer :

1. Take $\tan \phi = \frac{X_L - X_C}{R}$

2. Define $Z = \sqrt{R^2 + (X_L - X_C)^2}$



as the **impedance** of the circuit.

3. Then

$$i_m = \frac{\mathcal{E}_m}{Z} \quad \text{or} \quad \mathcal{E}_m = i_m Z$$

"Ohm's Law like" relation
for AC RLC circuits

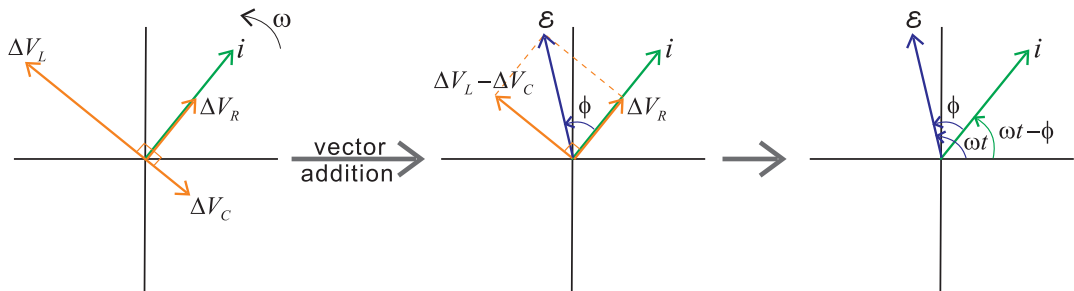
Check :

$$\begin{aligned}R.H.S. &= i_m Z \left[\frac{R}{Z} \sin(\omega t - \phi) + \frac{X_L - X_C}{Z} \cos(\omega t - \phi) \right] \\ &= i_m Z [\cos \phi \sin(\omega t - \phi) + \sin \phi \cos(\omega t - \phi)]\end{aligned}$$

$$\left(\begin{array}{l} \text{Use the relation:} \\ \sin(A + B) = \sin A \cos B + \cos A \sin B \\ \text{Here: } A = \omega t - \phi, \quad B = \phi \end{array} \right)$$

$$\begin{aligned}&= i_m Z \sin(\omega t - \phi + \phi) \\ &= i_m z \sin \omega t \\ &= L.H.S. \quad \text{if} \quad \mathcal{E}_m = i_m Z \quad \text{QED.}\end{aligned}$$

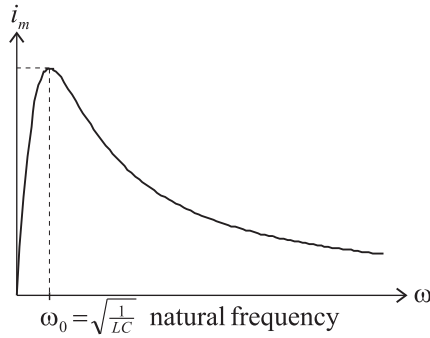
Phasor Approach :



10.4 Resonance

$i_m = \frac{\mathcal{E}_m}{Z}$ is at *maximum* for an AC circuit of *fixed input frequency* ω when Z is at *minimum*.

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$



is at a minimum for a fixed ω when

$$X_L - X_C = \omega L - \frac{1}{\omega C} = 0$$

$$\Rightarrow \omega L = \frac{1}{\omega C}$$

$$\Rightarrow \boxed{\omega^2 = \frac{1}{LC}}$$

same as that for
a RLC circuit

In Hong Kong, the AC power input is 50Hz.
(In US, as mentioned in *Halliday*, is 60Hz.)

$$\therefore \omega = 2\pi f = 314.2s^{-1}$$

10.5 Power in AC Circuits

Consider the *Power dissipated by R* in an AC circuit:

$$P = i^2 R = i_m^2 R \sin^2(\omega t - \phi)$$

The *average* power dissipated in each cycle:

$$P_{ave} = \frac{\int_0^{2\pi/\omega} P dt}{2\pi/\omega} \quad \left(\frac{2\pi}{\omega} \text{ is period of each cycle}\right)$$

$$\begin{aligned} \int_0^{2\pi/\omega} P dt &= i_m^2 R \int_0^{2\pi/\omega} \sin^2(\omega t - \phi) dt \\ &= i_m^2 R \int_0^{2\pi/\omega} \frac{1}{2} [1 - \cos 2(\omega t - \phi)] dt \\ &= i_m^2 R \cdot \left[\frac{t}{2} - \frac{\sin^2(\omega t - \phi)}{4\omega} \right] \Big|_0^{2\pi/\omega} \\ &= i_m^2 R \cdot \frac{1}{2} \cdot \frac{2\pi}{\omega} \end{aligned}$$

$$\therefore \boxed{\begin{array}{l} P_{ave} = \frac{i_m^2}{2} R = i_{rms}^2 R \\ \text{where } i_{rms} = \text{root-mean-square current} \end{array}}$$

$$i_{rms} = \frac{i_m}{\sqrt{2}} \quad \because \text{Current is a sinusoidal func.}$$

Symbol : $\langle P \rangle = P_{ave} = \text{Average of } P \text{ over time}$

For sine and cosine functions of time:

Average : $\langle \sin \omega t \rangle = \langle \cos \omega t \rangle = 0$

Amplitude : Peak value, e.g. $\mathcal{E}_m, i_m, (\Delta V_R)_m, \dots$

Root-Mean-Square(RMS) : It's a measure of the "time-averaged" deviation from zero.

$$x_{rms} = \sqrt{\langle x^2 \rangle}$$

For sines and cosines, for whatever quantity x :

$$x_{rms} = \frac{x_m}{\sqrt{2}} \quad (x_m \text{ is amplitude})$$

For an AC resistor circuit:

$$\boxed{\langle P \rangle = i_{rms}^2 R = \frac{\mathcal{E}_{rms}^2}{R}}$$

Laws for DC circuits can be used to describe AC circuits if we *use rms values* for i and $\mathcal{E}$.

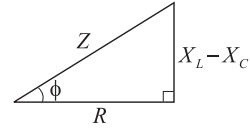
For general AC circuits:

$$\begin{aligned} P &= \mathcal{E}i = \overbrace{\mathcal{E}_m \sin \omega t}^{\mathcal{E}} \cdot \overbrace{i_m \sin(\omega t - \phi)}^i \\ &= \mathcal{E}_m i_m \sin \omega t [\sin \omega t \cos \phi - \cos \omega t \sin \phi] \\ P &= \mathcal{E}_m i_m \left[\underbrace{\sin^2 \omega t}_{\frac{1}{2}} \cos \phi - \underbrace{\sin \omega t \cos \omega t}_{0} \sin \phi \right] \\ &\hspace{15em} \text{(check this!)} \end{aligned}$$

$$\langle P \rangle = \frac{\mathcal{E}_m i_m}{2} \cos \phi$$

$$\boxed{\langle P \rangle = \mathcal{E}_{rms} i_{rms} \underbrace{\cos \phi}_{\text{power factor}}}$$

$$\begin{aligned} \text{Recall} \quad \tan \phi &= \frac{X_L - X_C}{R} \\ \therefore \cos \phi &= \frac{R}{Z} \end{aligned}$$



Maximum power dissipated in circuit when

$$\cos \phi = 1$$

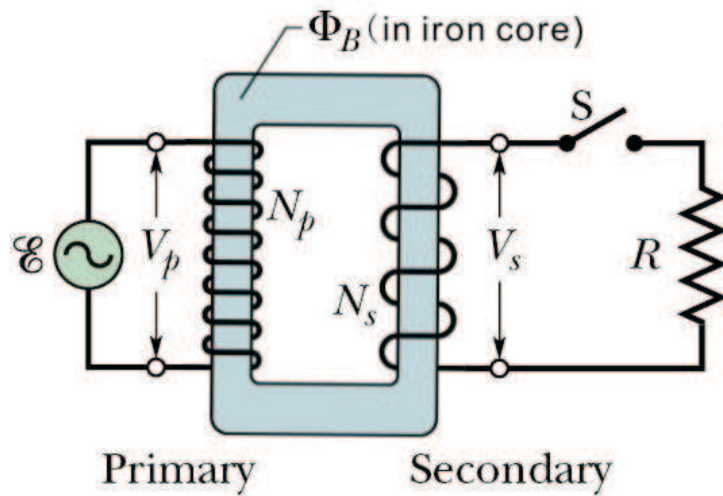
Two possibilities:

$$(1) X_L = X_C = 0$$

$$(2) X_L - X_C = 0 \Rightarrow X_L = X_C \Rightarrow \omega L = \frac{1}{\omega C} \Rightarrow \omega^2 = \frac{1}{LC}$$

(Resonance Condition)

10.6 The Transformer



Power dissipated in resistor

$$\langle P \rangle = i_{rms}^2 R$$

$\therefore$ For power transmission, we'd like to *keep i_{rms} at minimum.*

$\Rightarrow$ HIGH potential difference across transmission wires. (So that total power transmitted $P = i_{rms} \mathcal{E}_{rms}$ is constant.)

However, for home safety, we would like LOW emf supply.

Solution : Transformers

Primary : Number of winding = N_P

Secondary : Number of winding = N_S

In primary circuit, $R_P \approx C_P \approx 0$

$\therefore$ Pure inductive

$$\text{Power factor : } \cos \phi = \frac{R}{Z} \approx 0$$

$\therefore$ No power delivered from emf to transformer.

The *varying* current ($\because$ AC!) in the primary produces an *induced emf* in the secondary coils. *Assuming* perfect magnetic flux linkage:

$$\begin{aligned} & \text{emf per turn in primary} \\ &= \text{emf per turn in secondary} \\ &= -\frac{d\Phi_B}{dt} \end{aligned}$$

$$\begin{aligned} \text{emf per turn in primary} &= \frac{\Delta V_P}{N_P} & (\Delta V_P \text{ is P.D. across primary}) \\ \text{emf per turn in secondary} &= \frac{\Delta V_S}{N_S} \\ \Rightarrow \boxed{\frac{\Delta V_P}{\Delta V_S} = \frac{N_P}{N_S}} \end{aligned}$$

If $N_P > N_S$, then $\Delta V_P > \Delta V_S$ *Step-Down*
 If $N_P < N_S$, then $\Delta V_P < \Delta V_S$ *Step-Up*

Consider power in circuit:

$$i_P \Delta V_P = i_S \Delta V_S$$

In the secondary, we have

$$\Delta V_S = i_S R$$

Combining the 3 equations, we have

$$\boxed{\Delta V_P = \left(\frac{N_P}{N_S}\right)^2 R \cdot i_P}$$

$$\text{"Equivalence Resistor"} = \left(\frac{N_P}{N_S}\right)^2 R$$

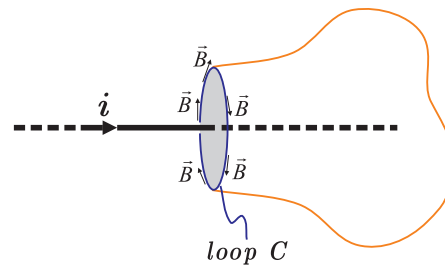
Chapter 11

Displacement Current and Maxwell's Equations

11.1 Displacement Current

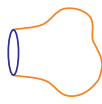
We saw in Chap.7 that we can use **Ampère's law** to calculate magnetic fields due to currents.

We know that the integral $\oint_C \vec{B} \cdot d\vec{s}$ around any close loop C is equal to $\mu_0 i_{incl}$, where i_{incl} = *current passing an area bounded by the closed curve C* .



e.g.

 = Flat surface bounded by loop C

 = Curved surface bounded by loop C

If **Ampère's law** is true all the time, then the i_{incl} *determined should be independent of the surface chosen.*

Let's consider a simple case: *charging a capacitor*.

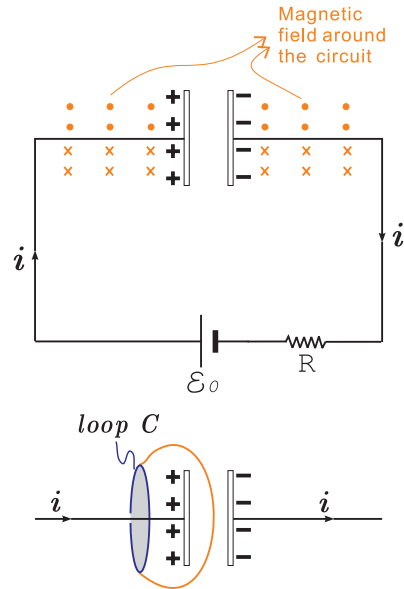
From Chap.5, we know there is a current flowing $i(t) = \frac{\mathcal{E}_0}{R} e^{-t/RC}$, which leads to a magnetic field observed $\vec{B}$. With Ampère's law, $\oint_C \vec{B} \cdot d\vec{s} = \mu_0 i_{incl}$. BUT WHAT IS i_{incl} ?

If we look at , $i_{incl} = i(t)$

If we look at , $i_{incl} = 0$

($\because$ There is no charge flow between the capacitor plates.)

$\therefore$ Ampère's law is either WRONG or INCOMPLETE.



Two observations:

1. While there is no current between the capacitor's plates, there is a *time-varying electric field between the plates of the capacitor*.
2. We know *Ampère's law is mostly correct from measurements of B-field around circuits*.

$\Downarrow$

Can we revise Ampère's law to fix it?

Electric field between capacitor's plates: $E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$, where Q = charge on capacitor's plates, A = Area of capacitor's plates.

$$\therefore Q = \varepsilon_0 \underbrace{E \cdot A}_{\text{Electric flux}} = \varepsilon_0 \Phi_E$$

$\therefore$ We can define

$$\frac{dQ}{dt} = \varepsilon_0 \frac{d\Phi_E}{dt} = i_{disp}$$

where i_{disp} is called **Displacement Current** (first proposed by Maxwell). Maxwell first proposed that this is the missing term for the Ampère's law:

$$\oint_C \vec{B} \cdot d\vec{s} = \mu_0 \left(i_{incl} + \varepsilon_0 \frac{d\Phi_E}{dt} \right) \quad \text{Ampère-Maxwell law}$$

Where i_{incl} = current through any surface bounded by C ,

Φ_E = electric flux through that *same surface bounded by curve C* , $\Phi_E = \int_S \vec{E} \cdot d\vec{a}$.

11.2 Induced Magnetic Field

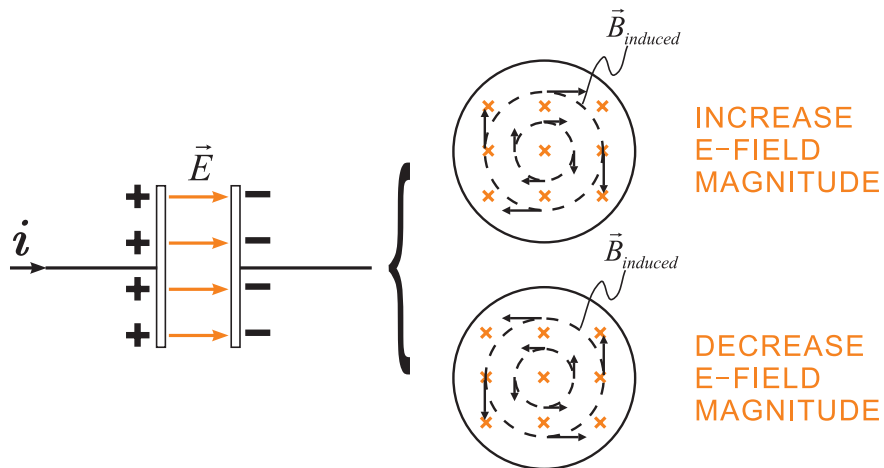
We learn earlier that electric field can be generated by

- $\left\{ \begin{array}{l} \text{charges} \\ \text{changing magnetic flux} \end{array} \right.$

We see from Ampère-Maxwell law that a magnetic field can be generated by

- $\left\{ \begin{array}{l} \text{moving charges (current)} \\ \text{changing electric flux} \end{array} \right.$

That is, a change in electric flux through a surface bounded by C can lead to an *induced magnetic field along the loop C* .



Notes The induced magnetic field is along the *same direction* as caused by the *changing electric flux*.

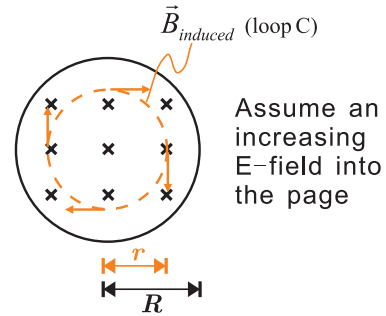
Example What is the magnetic field strength inside a circular plate capacitor of radius R with a current $I(t)$ charging it?

Answer Electric field of capacitor

$$E = \frac{Q}{\varepsilon_0 A} = \frac{Q}{\varepsilon_0 \pi R^2}$$

Electric flux inside capacitor through a loop C of radius r :

$$\Phi_E = E \cdot \pi r^2 = \frac{Qr^2}{\varepsilon_0 R^2}$$



Ampère-Maxwell Law inside capacitor:

$$\underbrace{\oint_C \vec{B} \cdot d\vec{s}}_{\because \vec{B}_{induced} \parallel d\vec{s}} = \mu_0(i_{incl} + \varepsilon_0 \frac{d\Phi_E}{dt})$$

$$\begin{aligned} \underbrace{2\pi r}_{\text{Length of loop } C} B_{induced} &= \mu_0 \varepsilon_0 \frac{d}{dt} \left(\frac{Qr^2}{\varepsilon_0 R^2} \right) \\ &= \mu_0 \frac{r^2}{R^2} \underbrace{\frac{dQ}{dt}}_{I(t)} \end{aligned}$$

$$\therefore B_{induced} = \frac{\mu_0 r}{2\pi R^2} I(t) \quad \text{for } r < R$$

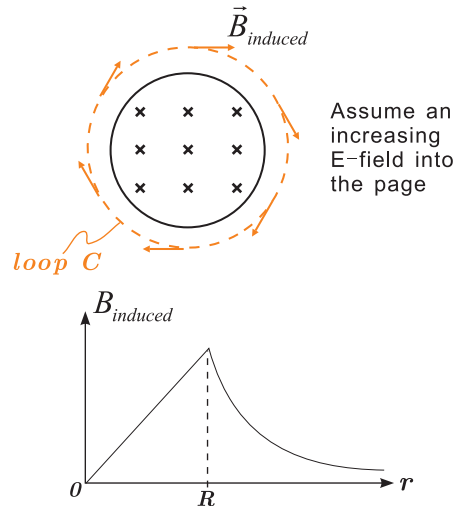
Outside the capacitor plate:

Electric flux through loop C : $\Phi_E = E \cdot \pi R^2 = \frac{Q}{\varepsilon_0}$

$$\oint_C \vec{B} \cdot d\vec{s} = \mu_0(i_{incl} + \varepsilon_0 \frac{d\Phi_E}{dt})$$

$$2\pi r B_{induced} = \mu_0 \varepsilon_0 \left(\frac{1}{\varepsilon_0} \cdot \frac{dQ}{dt} \right)$$

$$\therefore B_{induced} = \frac{\mu_0 I(t)}{2\pi r}$$



11.3 Maxwell's Equations

The four equations that *completely* describe the behaviors of electric and magnetic fields.

$$\begin{aligned}
\oint_S \vec{E} \cdot d\vec{a} &= \frac{Q_{incl}}{\epsilon_0} \\
\oint_S \vec{B} \cdot d\vec{a} &= 0 \\
\oint_C \vec{E} \cdot d\vec{s} &= -\frac{d}{dt} \int_S \vec{B} \cdot d\vec{a} \\
\oint_C \vec{B} \cdot d\vec{s} &= \mu_0 i_{incl} + \mu_0 \epsilon_0 \frac{d}{dt} \int_S \vec{E} \cdot d\vec{a}
\end{aligned}$$

The one equation that describes *how matter reacts to electric and magnetic fields*.

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

Features of Maxwell's equations:

- (1) There is a high level of *symmetry* in the equations. That's why the study of electricity and magnetism is also called **electromagnetism**.

There are *small asymmetries* though:

- i) There is *NO point "charge" of magnetism / NO magnetic monopole*.
 - ii) Direction of induced E-field *opposes to* B-flux change.
Direction of induced B-field *enhances* E-flux change.
- (2) Maxwell's equations predicted the existence of propagating waves of E-field and B-field, known as **electromagnetic waves (EM waves)**.

Examples of EM waves: visible light, radio, TV signals, mobile phone signals, X-rays, UV, Infrared, gamma-ray, microwaves...

- (3) Maxwell's equations are *entirely consistent with the special theory of relativity*. This is *not* true for Newton's laws!